

Kobon Triangle Constructions: Uniform Lower Bounds, Structural Defect Bounds, and Formal Verification

Alejandro Zarzuelo Urdiales

Comprehensive research edition, 7 October 2026

Abstract

The Kobon triangle problem asks for the largest number of bounded triangular cells determined by real affine lines. We consolidate a construction and verification program begun in March 2026, including the research of 5–6 October. An all-order simple construction gives $G(n) = \lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2)$ for $n \geq 4$. Compatible geometric seeds now instantiate the proved Bartholdi–Blanc–Loisel recursion at orders based on 32, 36, 48, and 60, alongside the earlier eleven-derived family. For $q = 60 \cdot 2^t$, actual simple arrangements give $1200 \cdot 4^t - 10$ triangles at $q+1$ and $1200 \cdot 4^t + 30 \cdot 2^t - 11$ at $q+2$, respectively nine and ten below the simple upper floors. Poola supplied the finite 61 : 1190 count; the uniform compatible refit and its 29-pair visibility certificate are the developments here. An executable portfolio retains the earlier all-order envelope. The quantitative deficit-dependent successor stated in the previous edition now has a complete actual-geometric Lean proof and indefinite iteration for both parities. We also extract nonsimple core fans, matching and components from actual arrangements, proving degree-free mixed-multiplicity defect bounds and sharper all-triple bounds with positive local contributions. Exact translation germs yield birth/loss identities and interval counterexamples to a specific desingularization loss formula. The inherited numerical families, standard axioms, finite native checks, external classifications and bounded unsuccessful searches are distinguished throughout. The unrestricted Kobon problem and the arbitrary full half-order successor remain unresolved; no worldwide numerical-priority claim is made.

Keywords. Line arrangements; triangular cells; Kobon problem; constructive lower bounds; projective transformations; exact computation; Lean formalization.

Contents

1	Introduction and scope of the results	5
1.1	The main completed all-order theorem	5
1.2	Finite enhancements and other completed components	6
1.3	History, attribution, and the structure of the paper	8
2	Literature, provenance, and the scope of comparison	8
2.1	Four distinctions that affect every comparison	8
2.2	From the puzzle to arrangement theory	8
2.3	The Füredi–Palásti all-order construction	9
2.4	Optimal subsequences and the straightening problem	10
2.5	Upper-bound refinements retain their simplicity assumptions	11
2.6	Recent computational constructions and versioned claims	11
2.7	The 2026 enumeration and its use as external input	11
2.8	OpenMath inputs and the 5–6 October developments	12
2.9	The March result, later verification, and attribution	12
2.10	What formal verification contributes	13
3	Real-line geometry and exact certificates	13
3.1	The two extremal problems	13
3.2	A polynomial sign certificate	14
3.3	What a finite lower-bound certificate establishes	15
4	A parity-sensitive all-order construction	15
4.1	Trigonometric geometry and the classical count	15
4.2	The repeated-index correction	16
4.3	Projective changes of affine chart	17
4.4	Assembly, exact gain, and asymptotic scale	19
5	Individual cases, finite enhancements, and provenance	21
5.1	The March values and their present verification	21
5.2	Materialized improvements from the phase and chart construction	23
5.3	The 44-line simple case	23
5.4	External records and independently checked inputs	23
5.5	The finite envelope and the complete tables	24
6	Exterior extensions: the March proposal and its precise replacement	25
6.1	A geometric theorem from explicitly visible pairs	26
6.2	Defect, boundary wedges, and a rule for both parities	27
6.3	What fails in indefinite full-gain exterior iteration	29
6.4	A conditional recurrence and an unconditional numerical bound	30
7	Compatible seeds, geometric doubling, and sparse families	31

7.1	The exact one-step theorem	31
7.2	The compatible eleven-line realization	33
7.3	Closing the geometric invariant	33
7.4	Even companions and the retained boundary invariant	35
7.5	The first recursive envelope at every natural order	37
7.6	Actual one-triangle optimality windows	38
7.7	Known optimal families and independent reproduction	39
7.8	The closed integration interfaces and the Forge tail	39
7.9	A uniform 49-line compatibility certificate	40
7.10	The compatible 37-line tail of the known 18-dyadic orbit	42
7.11	A compatible 61-line seed and the visibility refinement	42
7.12	The completed portfolio for every natural order	43
8	Multiplicity-sensitive upper-bound arguments	44
8.1	An exact elementary-segment identity	45
8.2	Parity on a line outside the core	46
8.3	The geometric obstruction in a cyclic fan	47
8.4	A weighted bound for an arbitrary finite core	48
8.5	A sharper estimate with at most two core points	49
8.6	A checked shared-fan example and the limits of smoothing	50
8.7	Actual arrangement extraction and simple upper theorems	51
8.8	Extremal fans and strengthened local restrictions	52
8.9	Actual fan extraction, matching and global assembly	53
8.10	Degree-free structural bounds and their local counts	54
8.11	Recipient geometry and the final all-triple gain	55
8.12	Restricted graph classes and multiplicity consequences	56
8.13	Parity tokens, equality attribution and remaining limits	58
9	Exact translation germs and local resolution losses	58
9.1	Simultaneous classification by affine coefficients	58
9.2	A four-line counterexample to a stated half-plane yield	59
9.3	Three losses and a reversed gain of two	60
10	Exact certificates, external reproduction, and unsuccessful searches	60
10.1	From a numerical proposal to an exact geometric witness	60
10.2	Finite improvements relative to the retained collection	61
10.3	Independent reproduction of external constructions	62
10.4	Projective charts, algebraic curves, and chamber searches	62
10.5	A proved algebraic obstruction to tangent-grid transfer	64
10.6	The full supplied affine-class obstruction census	66
10.7	New bounded searches and their exact outcomes	66
10.8	Cap-cone searches and the promoted 61 visibility resource	66
10.9	Bounded screens, transport diagnostics and a scoped dual	67

10.10	Reproduction and presentation audit	69
11	Formal verification and reproducibility	69
11.1	The verified snapshot	69
11.2	Three proof and computation boundaries	69
11.3	What the audit rejects	70
11.4	Theorem-level correspondence	70
11.5	Reproducing proofs, coordinates, and figures	72
12	Contributions, limitations and remaining proof gaps	73
12.1	What changes in the completed development	73
12.2	From the earlier graph plan to actual restrictions	73
12.3	The role of exact negative evidence	74
12.4	Contribution comparison	74
12.5	Remaining proof gaps	76
A	Complete saved finite table, orders 3–60	80
B	The complete 138-certificate catalog	81
C	Historical inventory of the author’s outputs	84
D	Evidence navigation and excluded drafts	86
E	Release deltas and reproducible research boundaries	87

1 Introduction and scope of the results

An arrangement of n straight lines divides the real plane into bounded and unbounded cells. The Kobon problem asks how many of the bounded cells can be triangles. The difficulty is not creating many triples of intersecting lines: there are $\binom{n}{3}$ candidate triples. The difficulty is ensuring that their triangles are not cut by any of the remaining lines and that their interiors do not overlap. The distinction between an attractive picture and a valid arrangement is therefore central to both construction and verification.

Let $K(n)$ denote the classical maximum, allowing multiple intersections and parallel lines, and let $K_s(n)$ denote the maximum over simple arrangements: no parallel pair and no three concurrent lines. All lower-bound witnesses constructed by the universal theorem below are simple, so they establish lower bounds for both quantities. Some of the strongest retained finite certificates are nonsimple. We never use an upper bound restricted to K_s as an upper bound for K without an additional argument.

This paper consolidates the author’s March 2026 extension proposal and the subsequent program of exact construction, comparison with the literature, and formal proof. The purpose is both mathematical and documentary: to state the strongest completed results, to make their proofs and provenance inspectable, and to preserve the failed approaches that determine the next research questions. The original general extension claim is reported as an unresolved claim rather than silently promoted to a theorem.

1.1 The main completed all-order theorem

Define $B(n) = \lceil n(n-3)/3 \rceil$ for $n \geq 3$, and set $B(n) = 0$ for $n < 3$. The familiar Füredi–Palásti construction supplies this affine baseline; the underlying construction and its historical priority belong to the earlier literature [6, 10]. The first completed all-order theorem is the following refinement; the recursive envelope below now strengthens its retained finite enhancement.

Theorem 1.1 (Universal affine refinement). *For every $n \in \mathbb{N}$, there is a simple arrangement of n real straight lines with at least $G(n)$ bounded triangular cells, where*

$$G(n) = \begin{cases} 0, & 0 \leq n < 3, \\ 1, & n = 3, \\ \lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2), & n \geq 4. \end{cases} \quad (1)$$

Consequently $K(n) \geq K_s(n) \geq G(n)$.

The proof, developed in Section 4, combines a shifted trigonometric arrangement with one- and two-cap changes of affine chart. The Lean declaration `Kobon.Universal.baseline_sound` proves the geometric existence statement for arbitrary n . It has no assumed extension theorem and no dependence on finite experimental verification. Its dependencies are the standard logical axioms used by Lean and Mathlib.

For $n \geq 4$, the gains over $B(n)$ are 1, 1, 0, 2, 0, 1 in residue classes 0, 1, 2, 3, 4, 5 modulo six. Thus the refinement is strict at every odd order at least five and every positive multiple of six. Both formulas have leading behavior $n^2/3 - n + O(1)$: this is a constant-term refinement, not an improved quadratic or linear asymptotic coefficient. The original Füredi–Palásti projective count table already contains related constants. The present source audit does not establish first numerical priority for the affine formula.

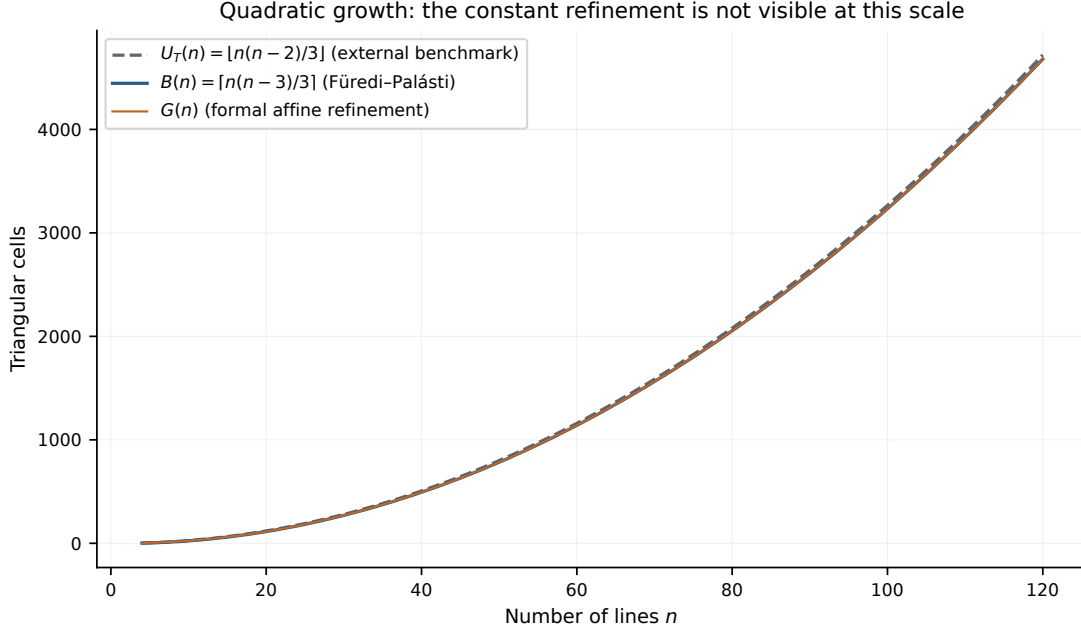


Figure 1: Growth of the classical general baseline, the proved affine refinement, and the classical polynomial upper benchmark. The close overlap on a quadratic scale is mathematically meaningful: the new uniform gain is bounded. Residue and deficit plots below resolve the smaller differences. This figure is computed from the formulas, not copied from earlier publications.

1.2 Finite enhancements and other completed components

The universal construction is supplemented by exact coordinates. Let $C(n)$ be the largest triangle count among the finite certificates in the release at order n , with $C(n) = 0$ when no such certificate is saved. Then

$$L(n) = \max\{G(n), C(n)\}, \quad K(n) \geq L(n) \quad (2)$$

is proved for all natural orders. This elementary maximum operation retains published constructions and the author’s earlier values without altering their attribution. It is not a claim that L is the largest lower bound obtainable from all published constructions by every possible interpolation or extension.

The finite catalog contains 138 distinct coordinate certificates, including 104 simple witnesses. A separate generated table has 64 orders at which a saved finite certificate strictly improves the older baseline B ; some of these improvements are already supplied by G . The catalog includes weaker checkpoints and independent reproductions. Its size is not a count of original discoveries.

The 2 October release added a recursively closed geometric family. Put $q_t = 10 \cdot 2^t$ and $A_t = (q_t^2 - 4)/3$. For every $t \geq 0$, simple arrangements realize at least A_t triangles at $q_t + 1$ lines and $A_t + q_t/2$ at $q_t + 2$ lines. Every next seed retains the sorted grid, saturation, positive central apex and visible-boundary data. Taking the maximum of these candidates and L gives the earlier total bound H of Theorem 7.7. It strictly improves the previous repository envelope at every family pair from 321/322 onward, without claiming a new literature record for the inherited numerical family.

The research of 5–6 October completes the compatible 33- and 49-line seed proofs, adds the known 37-line orbit, and realizes a further 60-dyadic orbit from Poola’s 61-line point input. Its strongest even branch uses 29 visible pairs. The resulting portfolio J retains H pointwise

Result	Mathematical content	Release status
Universal G Envelope L	Simple real-line witnesses at every order Maximum with all saved finite witnesses	General Lean proof Lean; finite native checks
28:238, 30:275, 34:357	Earlier attributed finite inequalities	Exact Lean certificates
Exterior extension Full successor recurrence 49-seed numerical target	New line and gain from a visible-pair list $K(n+1) \geq K(n) + \lfloor n/2 \rfloor$ Lower function obtained by summing the proposed recurrence	General Lean proof Unproved Independently proved
One-step BBL doubling	$(q+1, T) \mapsto (2q+1, T+q^2)$ for compatible seeds	General Lean proof
Infinite hybrid iteration	Odd and full-gain even geometric families	Complete Lean; native seed
33/37/49 orbits	Known numerical families, compatible real seeds and simple optimality	Complete Lean; native seeds
61 orbit	Uniform 1190/29 seed; constant simple gaps 9 and 10	Complete Lean; native seed
Quantitative successor	Deficit-dependent actual rule, both parities, indefinite iteration	General Lean proof
Local fan structure	Incidence, antipodal and run-length restrictions	General Lean proofs
Simple upper windows	Actual odd/even upper bounds and family existence	Complete Lean
Actual clean-line charging	Charge existence and finite fiber bounds	General Lean proof
Global multiplicity consequences	Actual fans, matching, mixed and triple defect estimates	General Lean proofs; class scope
Local surgery	Exact translation germs, retention, births and losses	General Lean proofs
Loss-rule counterexamples	Four- and six-line real parameter intervals	General Lean proofs
Tangent-grid obstruction	Explicit ten-sign algebraic impossibility	General Lean proof

Table 1: Claim status in the 7 October edition. “General Lean proof” does not imply that every proposed application has been formalized. Published construction inputs retain their original authorship.

(Theorem 7.11). These are actual straight-line witnesses at every finite depth; their historical attributions and simple-model gaps appear in Table 7.

The earlier manuscript’s quantitative boundary-defect successor now has an end-to-end Lean proof, including terminal-ray extraction and averaging, and it iterates through both parities. The original full half-order recurrence remains unproved. On the upper side, actual cyclic fans, side matching, graph components and geometric escape are now extracted from the arrangement. The strongest all-triple global estimate is

$$2\delta + B_{15} \geq 2U + 3(E_3 + P_{21}) + N_{12} + M_{22},$$

with the local counts defined in Section 8.10. Mixed-multiplicity and component estimates retain explicit corrections; they do not yield an unrestricted numerical bound depending only on n . The translation analysis of Section 9 proves exact birth/loss accounting and counterexamples to a specified loss formula. Table 1 separates these claims from the still open problem.

1.3 History, attribution, and the structure of the paper

The March archive records a proposed extension from odd to even orders, associated finite bounds, and an incomplete Lean file. That file contains eight proof placeholders, including its central extension step. Later exact certificates establish the finite inequalities independently, while the current all-order proof uses a different construction. Accordingly, a corrected account must separate the survival of numerical conclusions from the validity of the original proof strategy.

The September work broadened the program to both parities, exterior visibility, boundary-defect estimates, compatible doubling seeds, exact rational realization, and multiplicity-sensitive upper arguments. Several plausible routes failed: direct repetition of exterior gains lost the necessary boundary resources; small perturbations of nonsimple records sometimes reduced the triangle count; and large local searches failed to improve their chosen seeds. These negative findings retain their finite scope. The 2 October work closed the recursive and clean-line interfaces and added grid obstructions. The 5–6 October work closes the quantitative successor and actual nonsimple fan extraction, strengthens structural component bounds, and isolates exact surgery losses. The unrestricted full-gain successor and unrestricted numerical upper problem remain open.

Section 3 states the geometric definitions and the certificate-to-cell bridge. Section 4 proves the uniform refinement; Section 5 records finite cases and provenance. The following sections develop extension and family results, local upper-bound ingredients, and computational searches. The final sections describe the formal trust boundary and future obligations. The appendices contain the complete finite tables and a map from mathematical claims to proof sources. The code and evidence are available at <https://github.com/alejandrozu/kobon-proof> [35].

2 Literature, provenance, and the scope of comparison

2.1 Four distinctions that affect every comparison

The literature studies several related extremal functions. Their numerical tables cannot be combined without retaining their hypotheses. Throughout this paper, $K(n)$ concerns straight lines in the real affine plane and permits multiple intersections and parallel pairs, whereas $K_s(n)$ restricts to pairwise nonparallel lines with no three concurrent. Only bounded triangular cells are counted. A pseudoline arrangement is a topological relaxation; a lower bound obtained there becomes a straight-line lower bound only after realizability has been established. Finally, projective triangular faces need not remain bounded after an affine chart is chosen.

There are also two different projective-to-affine operations. Choosing a new line at infinity outside an n -line arrangement keeps n lines, but may lose triangular faces to infinity. Making one of the arrangement’s own lines the line at infinity and removing it produces an $(n-1)$ -line affine arrangement. The latter operation underlies several optimal subsequences; the former is central to the uniform construction formalized here.

2.2 From the puzzle to arrangement theory

Fujimura’s puzzle reached English readers through *The Tokyo Puzzles*, edited by Gardner and translated by Adachi [9]; Gardner also discussed it in his collected mathematical recreations [11]. The broader theory of line and pseudoline arrangements is developed in Grünbaum’s monograph [13]. The papers of Strommer and Purdy document an established

Distinction	What transfers immediately	What requires another argument
Simple / unrestricted	A simple example is an unrestricted example.	A simple upper bound need not bound a configuration with concurrent lines.
Straight / pseudoline	Every straight arrangement is a pseudoline arrangement.	A pseudoline construction requires a straight realization.
Affine / projective	Projective transformations preserve incidence and projective faces.	Boundedness depends on the line chosen as infinity.
Finite / infinite	A certified configuration proves its individual lower bound.	Iteration requires the next configuration to satisfy every input invariant.

Table 2: Logical distinctions used throughout the source audit.

extremal-triangle literature preceding the construction used in this paper [26, 27, 31]. We cite those early papers as historical context rather than silently identifying all their projective or non-simple questions with the affine function K .

The familiar comparison polynomial is

$$U_T(n) = \left\lfloor \frac{n(n-2)}{3} \right\rfloor. \quad (3)$$

The attribution to Saburo Tamura is transmitted in the Clément–Bader draft [3]; an original publication by Tamura was not located in this audit. We therefore do not manufacture an independent bibliographic item for that attribution. Clément and Bader report the stronger classical envelope $U_T(n) - 1$ for $n \equiv 0, 2 \pmod{6}$. Section 8 explains both the scope of this reported result and a local proof-reading issue. A source-reported upper bound and an upper bound proved in our Lean development are different kinds of evidence.

2.3 The Füredi–Palásti all-order construction

Füredi and Palásti gave highly triangular straight-line arrangements using trigonometric coordinates [10]. Their projective Table 1 has a residue-dependent constant term. The familiar affine consequence is commonly written

$$B(n) = \left\lceil \frac{n(n-3)}{3} \right\rceil \quad (n \geq 3). \quad (4)$$

Erickson states this formula explicitly in Section 7.2 while using arrangements in a computational-geometric lower-bound argument [6]. Felsner and Kriegel place the construction in the study of Euclidean triangular cells [7]. A later discussion by Damásdi, Martínez-Sandoval, Nagy and Nagy quotes the slightly weaker floor version while studying triangle areas [4]. Thus a general lower bound valid at every order existed well before the present work.

Our completed formula is

$$G(n) = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1 + (n \bmod 2), \quad n \geq 4. \quad (5)$$

The constructions in later sections prove this for actual simple affine lines. Table 3 separates its arithmetic improvement over (4) from the projective count in the original paper. Here $P_{FP}(n)$ denotes the latter construction’s projective count, not the projective maximum.

$n \bmod 6$	$G - B$	$P_{\text{FP}}(n), n \geq 5$	$P_{\text{FP}} - G$
0	1	$n(n-3)/3$	-1
1	1	$(n^2 - 3n + 5)/3$	0
2	0	$(n^2 - 3n + 8)/3$	2
3	2	$(n^2 - 3n + 9)/3$	1
4	0	$(n^2 - 3n + 8)/3$	2
5	1	$(n^2 - 3n + 5)/3$	0

Table 3: Exact comparison with the quoted affine baseline and with the original projective construction’s face counts. A projective surplus does not itself supply additional bounded affine triangles.

The numerical gain is bounded by two. In particular, the quadratic coefficient $1/3$ and linear coefficient -1 remain unchanged. What the completed proof adds is a phase refinement, a controlled choice of affine chart for odd orders, and an end-to-end geometric formalization. The audited sources do not provide an explicit statement of (5) with this affine proof. That observation is not an exhaustive numerical-priority theorem: old projective constructions contain related counts, and an unnoticed affine-chart argument could have been available earlier. The defensible claim is the proved improvement over the explicitly quoted formula (4), together with the formal construction; a claim of first numerical discovery is withheld.

2.4 Optimal subsequences and the straightening problem

Harborth and Roudneff developed constructions and extremal results for pseudolines [15, 28]. Their scope matters because topological realizability does not imply straight realizability. Forge and Ramírez Alfonsín constructed a projective straight-line family at orders $2 \cdot 2^t + 2$ [8]. Removing a distinguished line at infinity gives optimal affine orders $2 \cdot 2^t + 1$; in particular, this is the relevant older source for the 33-line order.

Bartholdi, Blanc and Loisel’s preprint appeared in 2007 and its proceedings publication in 2008 [1]. Their compatible doubling step takes $q + 1$ lines to $2q + 1$ and adds q^2 triangles. It requires a prescribed tangent grid and a fully used distinguished line. Their straight-line optimal series include $14 \cdot 2^t + 1$ and $6 \cdot 2^t + 1$; their $18 \cdot 2^t + 1$ series was then a pseudoline result. The general numerical recurrence is therefore prior work, not a new theorem of this project. Our contribution is the verified real-coordinate specialization, a closed recursive seed invariant and its boundary-visibility refinement. The actual odd and adjacent-even induction is complete in this release; it does not transfer authorship of the published doubling gain.

Parpalak and Utkin supplied a compatible straight 19-line seed and established the straight-line series $n = 18 \cdot 2^t + 1$ [20]. Its triangle count is $108 \cdot 4^t - 1$. Substitution into our formula gives the exact comparison

$$(108 \cdot 4^t - 1) - G(18 \cdot 2^t + 1) = 6 \cdot 2^t - 2. \quad (6)$$

Thus their series is stronger at these orders: the comparisons begin $107 > 103$ at 19, $431 > 421$ at 37, and $1727 > 1705$ at 73. Their separate even-series draft is also relevant prior construction evidence [22]. Its numerical families are not replaced by a weaker universal formula.

It would consequently be misleading to call G the strongest numerical lower bound for every n , or an unqualified state of the art among all possible all-order formulas. Taking a maximum with known special constructions already improves it. Padding a configuration by lines sufficiently far away preserves its bounded triangles and extends each special construction to later orders. The retained function in the repository likewise takes a maximum with its

finite certificate catalogue. The purpose of G is a compact uniform guarantee with a complete formal proof, not the suppression of stronger special cases.

2.5 Upper-bound refinements retain their simplicity assumptions

The even-order bound

$$\left\lfloor \frac{n(n-7/3)}{3} \right\rfloor$$

in Bartholdi–Blanc–Loisel applies to *simple affine pseudoline* arrangements. Blanc subsequently proved the sharper common even-order polynomial

$$K_s(n) \leq \left\lfloor \frac{n(n-5/2)}{3} \right\rfloor, \quad n \text{ even}, \quad (7)$$

also in the simple pseudoline setting [2]. The latter work was posted in 2008 and published in 2011. Its parity argument is the starting point for our clean-line accounting outside a multiple-point core.

The distinction is concrete. Formula (7) equals 53 at $n = 14$, whereas a nonsimple configuration with 54 triangles exists. Applying a simple upper theorem to that configuration would be invalid. Our upper-bound work therefore counts shared sides and multiple incidences explicitly rather than treating a small perturbation to general position as automatically harmless.

2.6 Recent computational constructions and versioned claims

Savchuk’s 2025 work combines arrangement tables, SAT search, and heuristic straightening [29]. It reports optimal 23- and 27-line configurations and an exclusion of a perfect 33-triangle arrangement at order 11. Its Appendix A.4 records the earlier Honma-based 11-line construction. For our purposes, an arrangement table, a floating-point fit, an exact straight-line certificate, and an exhaustive upper-bound exclusion are four different outputs. The present work independently verifies the coordinate lower bounds it imports; it does not thereby replay the external SAT exclusions.

Maiorana released fifteen exact 14-line, 54-triangle configurations [18]. We preserve their attribution and the source’s CC BY 4.0 license. The current Parpalak–Utkin gallery supplies further exact constructions, including nonsimple even-order examples [23]. The 8-line construction has earlier provenance; citing its current host does not transfer its invention to the host authors. Each retained dataset is tied to a source revision, checked independently, and distinguished from any stronger optimality language used in its original repository.

Versioning also matters for exclusions. The fourth version of Liang, Liu and Zhang’s preprint concerns symmetry restrictions for simple 28-line configurations [16]. It is not treated here as a proof of an unrestricted exact value, and its displayed numerical comparison is not substituted for Blanc’s simple upper theorem. The ProofAtlas research page similarly reports restricted upper results while keeping the unrestricted problem open [25]. We record these as current research sources, not as independently reproduced proofs.

2.7 The 2026 enumeration and its use as external input

Parpalak and Utkin’s July 2026 preprint gives a reduced-word enumeration and classification of triangle-maximal pseudoline arrangements, with Euclidean and projective equivalence kept distinct [21]. Its larger first-hit examples are pseudoline constructions; straight realizability is

a separate problem. The present tangent-grid obstruction uses their supplied 21-line classification as an external completeness input. Our exact replay checks the algebraic contradiction certificates and all supplied affine-class/support cases, not their full enumeration algorithm inside Lean. Keeping the infinity support and every Euclidean class is essential: a smaller list of projective representatives did not suffice for the affine exclusion.

The compatible 49-line branch also requires careful comparison. Its completed odd family at $n = 48 \cdot 2^t + 1$ is the tail of BBL’s older $n = 6 \cdot 2^s + 1$ series, with $s = t + 3$. The counts cannot be presented as new lower-bound records. The uniform explicit slopes, whole-interval sign certificate, boundary data and completed formal seed interface are the contributions here. The completed 33-line branch likewise verifies the inherited Forge–Ramírez Alfonsín orbit. The new 37-line seed verifies the $q = 36$ tail of the known Parpalak–Utkin/Blanc numerical orbit. Formal coverage of these families is not numerical priority.

2.8 OpenMath inputs and the 5–6 October developments

The source audit pins Poola’s and Srirangam’s OpenMath submissions and the competition judging archive by immutable repository revisions [24, 30, 33]. Judging summaries may precede later compilation receipts; they are evidence of the competition record, not mathematical authority. Poola’s 61-line point count of 1190 is retained with that attribution. The compatible reciprocal-slope refit, whole true-tangent interval, and 29-visible-pair certificate support the 60-dyadic orbit in Section 7.11. The inspected primary sources did not locate this particular straight orbit, but that bounded comparison does not prove worldwide priority.

Srirangam’s local shared-apex, wrong-ray, cap-block and clean-line arguments informed the upper investigation. Unfinished global transfer or Hall-capacity arguments from a source packet are not imported as theorems. Section 8.9 instead proves actual fan and matching extraction and quantifies its component and multiplicity consequences. Clément–Bader’s earlier perfect-implies-simple claim keeps its attribution even where its auxiliary shared-side accounting cannot supply our proof.

Section 9.2 gives a four-line real-parameter counterexample to the half-plane yield formula in Liu’s versioned 2026 desingularization preprint [17]. Its scope is that formula, rather than unrelated results or the entire numerical Kobon problem. The replacement is a global exact germ loss set, with retention requiring a separate geometric argument.

2.9 The March result, later verification, and attribution

The author’s March manuscript proposed an even-order extension and recorded the values 238, 275, and 357 at orders 28, 30, and 34[34]. These values are now supported in this project by exact simple-line certificates. The associated original Lean file, however, contained placeholders, including the general extension lemma. The later successful formalizations do not retroactively turn that unfinished universal argument into a proof.

OEIS A006066 indexes numerical data, references, and named contributions [19]. It is not a complete priority database. In particular, the 2007 Bartholdi–Blanc–Loisel table already contains a bold, hence straight-realizable, 30:275 entry. The 34:357 value follows from the older perfect 33-line family and its exterior extension. At 28 lines, Savchuk describes the analogous extension of the 27-line solution. These overlaps require a distinction between an independently supplied construction, a new exact or formal certificate, and the first known numerical lower bound. We retain the author’s recorded contribution without claiming first numerical priority solely from the OEIS attribution.

The recurrence reproduced by HandWiki [14] is likewise a record of the proposed claim

rather than a substitute for its proof. The unrestricted statement

$$K(n+1) \geq K(n) + \lfloor n/2 \rfloor$$

has not been established by the work reported here. What has been established includes a geometric exterior extension under explicit visibility hypotheses, finite successful successor steps, and a separate unconditional all-order construction. Their logical roles remain distinct.

2.10 What formal verification contributes

The proof infrastructure uses Lean 4 [5] and mathlib [32]. For the universal result, it connects analytic line equations, determinant signs, triangular cells, affine-chart control, and finite counting, rather than checking the final arithmetic formula in isolation. For finite results, it turns attributed coordinates into explicit certificates. The upper development now proves actual segment incidence, clean-line charging, classical simple bounds, cyclic-fan and matching extraction, and global structural component inequalities. The remaining numerical upper problem lies in residual local profiles and ordinary-end trade-offs, rather than an assumed extraction bridge.

The accompanying repository [35] records source revisions, negative searches, axiom reports, and the boundary between proved statements and proposed research. This separation is part of the result: a complete formal proof of a modest uniform improvement is stronger evidence than an ambitious recurrence with an unproved geometric step. It does not by itself confer numerical priority or solve the unrestricted Kobon problem.

3 Real-line geometry and exact certificates

3.1 The two extremal problems

Definition 3.1. An affine line is a set $\{(x, y) \in \mathbb{R}^2 : ax + by = c\}$ with $(a, b) \neq (0, 0)$. An arrangement is a finite set of distinct lines. A triangular cell is a bounded connected component of the complement whose closure is a nondegenerate triangle. The maximum number of such cells among arrangements of n lines is $K(n)$. The maximum among simple arrangements, with neither parallel lines nor triple concurrence, is $K_s(n)$.

The classical problem permits degeneracies that the simple problem excludes. Consequently $K_s(n) \leq K(n)$, and a simple lower-bound construction is valid for both problems. The reverse use of an upper bound is invalid without a separate reduction. Our coordinate validator permits several lines through a triangular vertex, but excludes every line that cuts the triangle’s interior.

The main formal lower-bound predicate uses witnesses with no parallel pairs. A finite projective chart theorem now transports an arbitrary distinct-line witness and its selected actual triangles to that model while preserving boundedness (Section 8.13). The arbitrary input checks nonparallelism of each triangle’s supporting pairs rather than using the polynomial predicate unmodified on parallel supports. This is a reduction for a chosen finite witness, not a claim that every original maximizer is already nonparallel. For nontrivial orders, pairwise nonzero determinants imply valid distinct lines. Orders with fewer than three lines are handled separately.

3.2 A polynomial sign certificate

Represent a line ℓ by (a_ℓ, b_ℓ, c_ℓ) , with affine evaluation $f_\ell(x, y) = a_\ell x + b_\ell y - c_\ell$. For two lines set

$$D(\ell, m) = a_\ell b_m - b_\ell a_m, \quad (8)$$

$$\begin{aligned} V(\ell, m) &= (X_{\ell m}, Y_{\ell m}, D(\ell, m)) \\ &= (c_\ell b_m - b_\ell c_m, a_\ell c_m - c_\ell a_m, D(\ell, m)). \end{aligned} \quad (9)$$

When $D(\ell, m) \neq 0$, their actual affine intersection is $p_{\ell m} = (X_{\ell m}/D(\ell, m), Y_{\ell m}/D(\ell, m))$. For a third line r define

$$E(r; \ell, m) = a_r X_{\ell m} + b_r Y_{\ell m} - c_r D(\ell, m), \quad (10)$$

$$O(r; \ell, m) = E(r; \ell, m) D(\ell, m). \quad (11)$$

The useful identity is

$$O(r; \ell, m) = f_r(p_{\ell m}) D(\ell, m)^2. \quad (12)$$

Thus O has the sign of the affine evaluation without requiring division in the certificate. All operations are polynomial over integer or rational coordinates.

Definition 3.2 (Certified supporting triple). For a pairwise nonparallel arrangement $(\ell_0, \dots, \ell_{n-1})$, an increasing triple $i < j < k < n$ is certified if

1. $E(\ell_k; \ell_i, \ell_j) \neq 0$;
2. for each line r of the arrangement, the three numbers

$$O(r; \ell_i, \ell_j), \quad O(r; \ell_i, \ell_k), \quad O(r; \ell_j, \ell_k)$$

are either all nonnegative or all nonpositive.

The first condition excludes concurrence of the three supports; the second permits zeros at vertices but forbids a strict sign change.

Theorem 3.3 (From a sign certificate to a triangular cell). *Every certified supporting triple determines a nondegenerate bounded triangular cell. Its interior is nonempty. Two distinct increasing certified triples determine disjoint interiors.*

Proof. The three pair intersections exist because their denominators in (8) are nonzero. The nonvanishing evaluation excludes concurrence and gives three noncollinear points. Their positive barycentric combinations form a nonempty open triangle Δ .

By (12), each arrangement line has a common weak sign at the three vertices. Affine linearity gives the same sign throughout the closed triangle. It is strict in Δ : otherwise all three vertex evaluations would be zero, forcing a valid line to contain three noncollinear points. Hence no arrangement line enters Δ .

Fix an interior point z . Its strict sign cell is

$$S_z = \{x \in \mathbb{R}^2 : f_r(x) f_r(z) > 0 \text{ for every arrangement line } r\}.$$

The preceding argument gives $\Delta \subseteq S_z$. Conversely, the three support inequalities alone force all three barycentric coordinates of x to be positive, so $S_z \subseteq \Delta$. The sign cell is an intersection of open half-planes and is convex; it is exactly a component of the complement. It is bounded by the three finite segments of the triangle.

If two certified interiors meet, they have the same strict sign vector and hence the same sign cell. Its three supporting lines are determined by its three open sides. Since distinct arrangement lines cannot share an open side, the sets of supports agree, and sorting their indices makes the triples equal. The contrapositive proves disjointness. Q.E.D.

The modules `Kobon.Euclidean` and `Kobon.Cells` prove the sign-cell and nonoverlap content through `interior_nonempty`, `interior_eq_signCell`, and `distinct_interiors_disjoint`. The connected-component interpretation is the elementary convexity argument above; it is not separately encoded through a topological component API. Thus the certificates have an explicit interpretation as actual Kobon cells.

3.3 What a finite lower-bound certificate establishes

Proposition 3.4 (Soundness of exact finite validation). *Let A be an array of n integer line coefficients, and let $\mathcal{T}$ be a list of at least T certified increasing triples with no duplicates. If all pair determinants are nonzero, then $K(n) \geq T$. If all increasing triples of lines additionally have nonzero concurrence determinant, then $K_s(n) \geq T$.*

Proof. The coefficient inclusion $\mathbb{Z} \hookrightarrow \mathbb{R}$ preserves polynomial equalities, strict inequalities, and weak signs. The resulting real arrangement therefore satisfies Definition 3.2. Apply Theorem 3.3 to the distinct triples. The additional concurrence test is exactly the simplicity condition. Q.E.D.

Rational coordinates are cleared of denominators line by line before the integer check. The mathematical lower bound needs only a list of T valid triangles; it need not prove that no further triangle exists in the same arrangement. For the catalog, however, two independent exact counters also enumerate the triangle count. One works from consecutive intersections along each line, while the other directly tests triples against the arrangement. Agreement is an independent check on extraction and counting, not a substitute for the general soundness theorem.

These distinctions explain the three evidential levels used later. A proved symbolic theorem applies at arbitrary order. A finite Lean certificate establishes its listed order. An exact experimental count outside the catalog remains a computationally verified witness until promoted through the same formal pipeline. No finite list, however large, proves an infinite family.

4 A parity-sensitive all-order construction

This section gives the mathematical ingredients of Theorem 1.1. The underlying trigonometric family is a version of the classical Füredi–Palásti family. Its attribution is independent of the present formalization and choice of affine chart.

4.1 Trigonometric geometry and the classical count

For $0 < \theta < \pi$, define

$$\ell(\theta) : \quad \sin \theta x + \cos \theta y = \sin(3\theta). \quad (13)$$

The normal has length one. Direct trigonometric expansion gives

$$D(\ell(x), \ell(y)) = \sin(x - y), \quad (14)$$

$$E(\ell(z); \ell(x), \ell(y)) = 4 \sin(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (15)$$

In particular, distinct angles in $(0, \pi)$ give nonparallel lines. Multiplying (15) by (14) yields

$$O(\ell(z); \ell(x), \ell(y)) = 4 \sin^2(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (16)$$

Only three factors determine the sign on the right.

For the half-phase choice $\theta_i = (i + 1/2)\pi/n$, select increasing triples with

$$i + j + k \equiv n - 1 \text{ or } n - 2 \pmod{n}. \quad (17)$$

For the shifted choice $\theta_i = (i + 1/6)\pi/n$, select instead

$$i + j + k \equiv 0 \text{ or } n - 1 \pmod{n}. \quad (18)$$

Lemma 4.1 (Triangle selection). *For $n \geq 3$, the triples in (17) certify triangles of the half-phase arrangement. The triples in (18) certify triangles of the shifted arrangement. Both arrangements are simple.*

Sign proof. The angle sum of any three lines is $(i + j + k + 3/2)\pi/n$ in the first arrangement and $(i + j + k + 1/2)\pi/n$ in the second. It is never an integer multiple of π , which, together with (15), excludes triple concurrence.

For a selected $i < j < k$, fix a fourth index r . The factors $\sin(\theta_r - \theta_j)$ and $\sin(\theta_r - \theta_i)$ change sign only when r passes the corresponding support index. For each of the pairs (i, j) , (i, k) , (j, k) , the remaining factor is $\sin((i + j + r + 3a)\pi/n)$ with $a = 1/2$ or $1/6$. Substituting the selected residue of $i + j + k$ fixes on which side of the next multiple of π this sum lies when $r < k$ and when $r > k$. Splitting the index range at i, j, k shows that the three expressions in (16) have one weak sign in each range. At a support index the appropriate evaluations are zero. Their common sign need not be the same for different r ; that is precisely the permitted sign choice in Definition 3.2.

Equivalently, this check uses only the sign of $\sin t$ on consecutive intervals $(s\pi, (s + 1)\pi)$ and the inequalities $0 \leq r < n$. No numerical trigonometric approximation is used. The complete index splits are proved in `FurediPalasti.lean` and `ShiftedFurediPalasti.lean`. Definition 3.2 and Theorem 3.3 then give the claimed cells. Q.E.D.

For any residue $b \in \mathbb{Z}/n\mathbb{Z}$, the equation $i + j + k = b$ has exactly n^2 ordered solutions. At most $3n$ have a repeated index, because each of $i = j$, $j = k$ and $i = k$ leaves at most n choices. Two distinct residue classes therefore give at least $2n^2 - 6n$ ordered distinct-index triples. Each increasing triple accounts for six permutations, so the half-phase arrangement has at least

$$\left\lceil \frac{2n^2 - 6n}{6} \right\rceil = B(n) \quad (19)$$

certified triangles. This is the classical all-order estimate, now proved directly in the real-coordinate formalization.

4.2 The repeated-index correction

Lemma 4.2 (Shifted count). *For every $n \geq 3$, the shifted arrangement has at least*

$$\left\lfloor \frac{n(n - 3)}{3} \right\rfloor + 1$$

certified triangular cells.

Proof. Use the two residues 0 and $n - 1$ in (18). In the zero residue, the three repeated-index sets all contain $(0, 0, 0)$. Their union therefore has at most $3n - 2$ members, not merely $3n$.

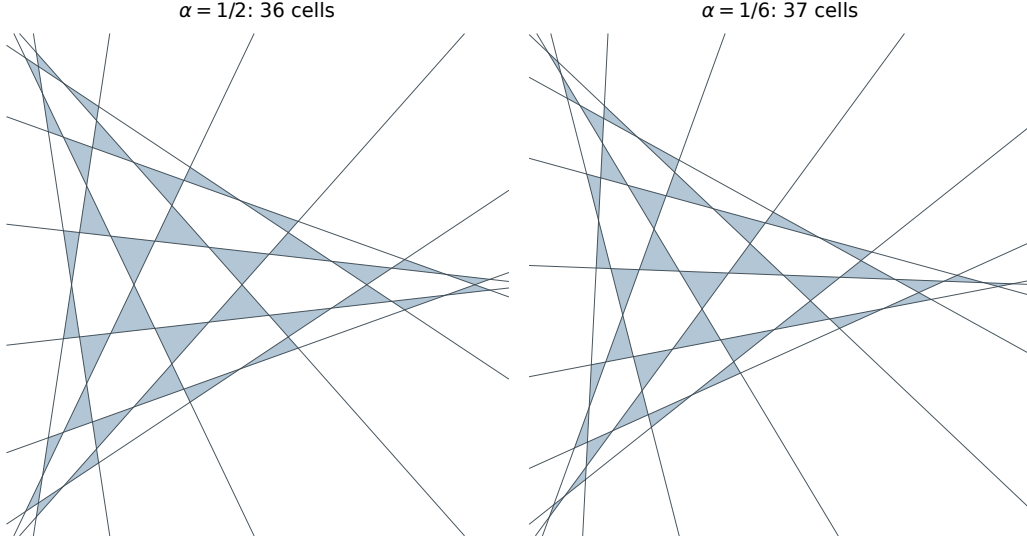


Figure 2: New rational-coordinate illustrations at $n = 12$, obtained from the half-phase and shifted trigonometric families. Two independent exact counters certify 36 and 37 triangles in these rational proxies. The axes of each panel are rescaled independently. The real-coordinate universal theorem follows from the symbolic sign and modular arguments, independently of this illustration. The underlying family is attributed to Füredi and Palásti.

In the other residue retain the bound $3n$. The number of ordered distinct-index solutions is at least $2n^2 - 6n + 2$. If T is the number of increasing selected triples, then

$$6T \geq 2n^2 - 6n + 2, \quad T \geq \left\lceil \frac{n(n-3)+1}{3} \right\rceil = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1.$$

Lemma 4.1 supplies their actual geometry. This argument uses the overlap of bad-index sets, not a numerical rounding conjecture. Q.E.D.

4.3 Projective changes of affine chart

The shifted count is enough for the even part of (1). To obtain the additional odd-order gain we use exposed vertices of the half-phase arrangement. The exact coordinate map is

$$(x, y) \mapsto \left(\frac{x}{h-x}, \frac{y}{h-x} \right), \quad h \neq 0. \quad (20)$$

It replaces the line $x = h$ by the line at infinity. The transformed coefficients of $ax + by = c$ are

$$\Phi_h(a, b, c) = (ha - c, hb, c). \quad (21)$$

Unlike an affine transformation, this change can alter which projective triangular faces appear bounded. Thus counting triangles before and after the change requires an explicit sign argument.

Lemma 4.3 (Exact covariance). *If $D(\ell, m) \neq 0$, then*

$$D(\Phi_h \ell, \Phi_h m) = h D(\ell, m) (h - x_{\ell m}), \quad (22)$$

$$E(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^2 E(r; \ell, m), \quad (23)$$

$$O(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^3 (h - x_{\ell m}) O(r; \ell, m), \quad (24)$$

where $x_{\ell m}$ is the abscissa of the old intersection. If the cut avoids every old vertex, the transformed arrangement has no parallel pairs. Triple nonconcurrency is preserved.

Proof. Substitute (21) into the polynomial expressions in Section 3, expand, and cancel. Nonzero h and avoidance of all $x_{\ell m}$ give the pair determinant claim, while multiplication by h^2 preserves a nonzero triple evaluation. Q.E.D.

The sign multiplier in (24) also gives a direct triangle-preservation rule. If $h > 0$ and all vertices of a certified triangle lie to the left of $x = h$, all three multipliers are positive. If $h < 0$ and its vertices lie to the right, they are again positive. At selected exposed vertices on the opposite side, the sign reverses. Carefully choosing which vertices are separated by the cut creates the extra triangles.

Lemma 4.4 (One exposed vertex). *For odd $n = 2m + 1 \geq 3$, a change of affine chart of the half-phase arrangement has at least $B(n) + 1$ bounded triangles.*

Proof. The intersection of $\ell(x)$ and $\ell(y)$ has abscissa

$$X(x, y) = \cos(2x) + \cos(2y) + \cos(2x + 2y). \quad (25)$$

For the half-phase angles, the vertex of lines $0, n - 1$ is uniquely rightmost, with abscissa $c_n = 1 + 2\cos(\pi/n) > 0$. Indeed, each of the first two cosine terms in (25) is at most $\cos(\pi/n)$ and the last is at most one; if the pair is not $(0, n - 1)$, at least one necessary inequality is strict. Choose $h > 0$ between all other abscissae and c_n .

No selected triangle of (17) uses the pair $(0, n - 1)$: such a triple would require its middle index to be 0 or $n - 1$ modulo n . Hence all counted triangles are preserved by the positive multiplier rule. The additional triple $(0, m, 2m)$ has, for each test line, nonnegative evaluation at the exposed vertex and nonpositive evaluations at its other two vertices before the transformation. The cut reverses exactly the exposed-vertex sign, so the transformed triple satisfies the certificate. These signs follow from (16) by splitting the test index at m . The triple is absent from the old selected list, and Lemma 4.3 preserves simplicity. This gives one additional bounded triangle. Q.E.D.

Lemma 4.5 (Two exposed vertices). *For $n = 6k + 3$ with $k \geq 1$, a change of affine chart of the half-phase arrangement has at least $B(n) + 2$ bounded triangles.*

Proof. Write $n = 3m$, where $m = 2k + 1$. The two intersections supported by $(m - 1, m)$ and $(2m - 1, 2m)$ have common abscissa $-c_n/2$. Every other intersection has strictly larger abscissa. To verify the latter assertion, rewrite (25) as

$$X(u, v) = 2zw + 2z^2 - 1, \quad z = \cos(u + v), \quad w = \cos(u - v).$$

For nonadjacent angle indices, the cosine bounds on the discrete grid, together with completing the square in z , give $X > -c_n/2$. For adjacent indices, $w = \cos(\pi/n)$ and the same quadratic attains equality precisely at $u + v = 2\pi/3$ or $4\pi/3$, giving the two listed pairs. The exceptional wrapping pair $(0, n - 1)$ is the rightmost vertex. These exposure inequalities are proved uniformly in `FurediPalastiCaps.lean`.

Choose $h < 0$ strictly above $-c_n/2$ and below all other abscissae. The modular conditions (17) show that no counted triangle uses either exposed pair. Its vertices are therefore all to the right of the cut and its signs are preserved. The two additional support triples are

$$(2k, 2k + 1, 5k + 2), \quad (k, 4k + 1, 4k + 2).$$

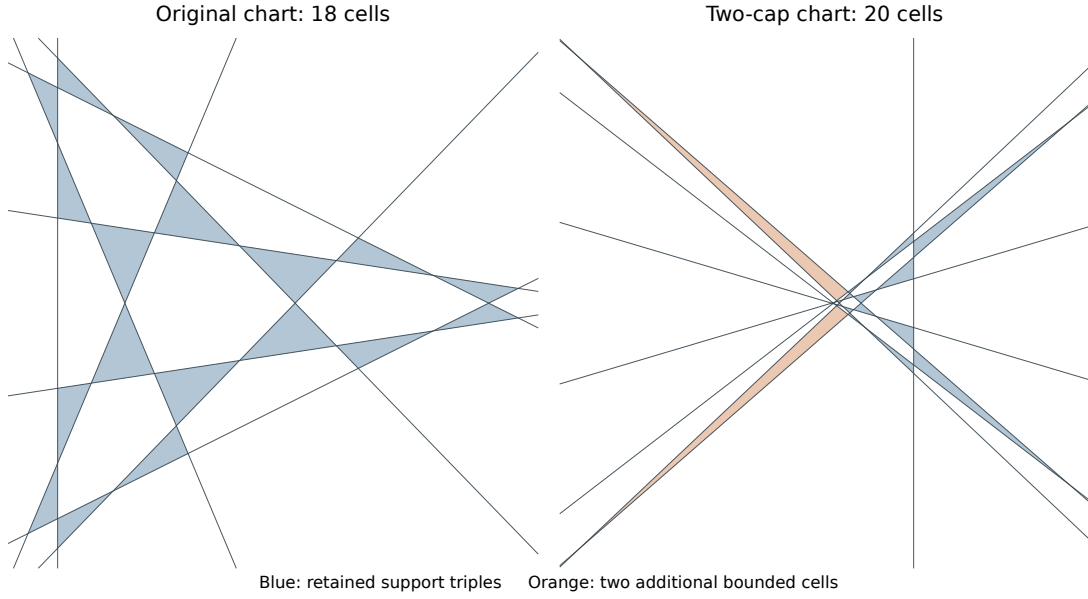


Figure 3: New exact rational illustration at $n = 9$: the change of affine chart retains all 18 old triangular cells and creates two more. The resulting 20 is below the known optimum 21; this figure illustrates the uniform mechanism. Panels are rescaled independently. Both counts are checked exactly. The general proof tracks the determinant multiplier at every vertex, independently of the drawing.

For each test index, substitution into (16) gives one sign at the exposed pair and its opposite weak sign at the other two vertices. The negative cut reverses exactly the exposed-vertex multiplier in each case, establishing the transformed certificates. The two triples are distinct and neither satisfies the old selection residues. Simplicity follows from Lemma 4.3. Appending both triples to the retained list proves the claim. Q.E.D.

4.4 Assembly, exact gain, and asymptotic scale

Proof of Theorem 1.1. Use Lemma 4.2 for even $n \geq 4$. For odd n not divisible by three, $n(n-3) \equiv 1 \pmod{3}$, so $B(n) + 1 = \lfloor n(n-3)/3 \rfloor + 2$; apply Lemma 4.4. For odd multiples of three at least nine, $B(n) = n(n-3)/3$ and Lemma 4.5 gives the same target. Three nonconcurrent lines give $G(3) = 1$, while the smaller cases have zero triangles. Each construction used is simple. Q.E.D.

Corollary 4.6 (Exact improvement over the quoted baseline). *For $n \geq 4$,*

$$G(n) - B(n) = \begin{cases} 1 + (n \bmod 2), & 3 \mid n, \\ n \bmod 2, & 3 \nmid n. \end{cases}$$

The gains in classes $0, 1, 2, 3, 4, 5 \pmod{6}$ are respectively $1, 1, 0, 2, 0, 1$.

Proof. The product $n(n-3)$ is zero modulo three when $3 \mid n$, and one modulo three otherwise. Substitute these remainders in the floor and ceiling formulas. Q.E.D.

For comparison with the classical polynomial benchmark $U(n) = \lfloor n(n-2)/3 \rfloor$, direct integer arithmetic gives, for $n \geq 4$,

$$U(n) - G(n) = \left\lfloor \frac{n}{3} \right\rfloor + \mathbf{1}_{n \equiv 2 \pmod{3}} - 1 - (n \bmod 2). \quad (26)$$

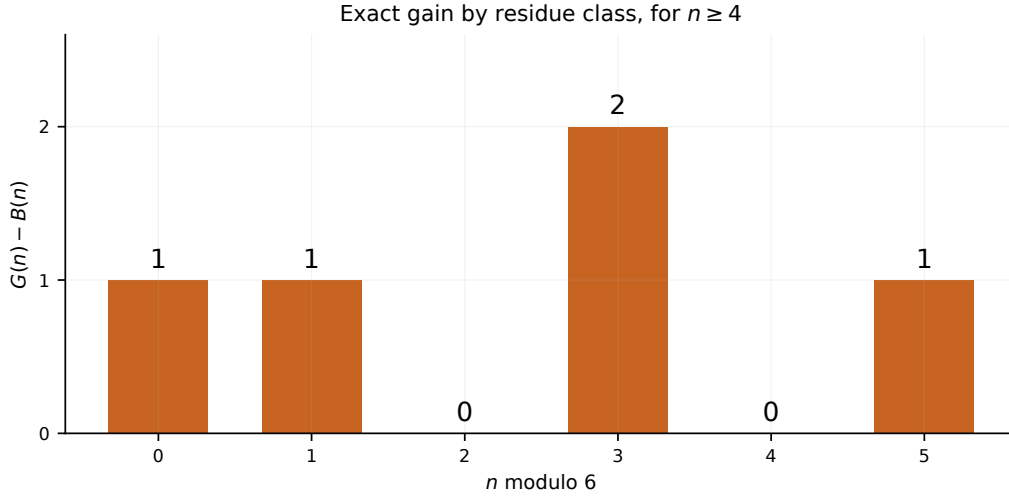


Figure 4: The exact proved gain $G - B$, resolved by residue class modulo six. The gain is strict on four classes; the universal statement includes arbitrarily large orders in each class.

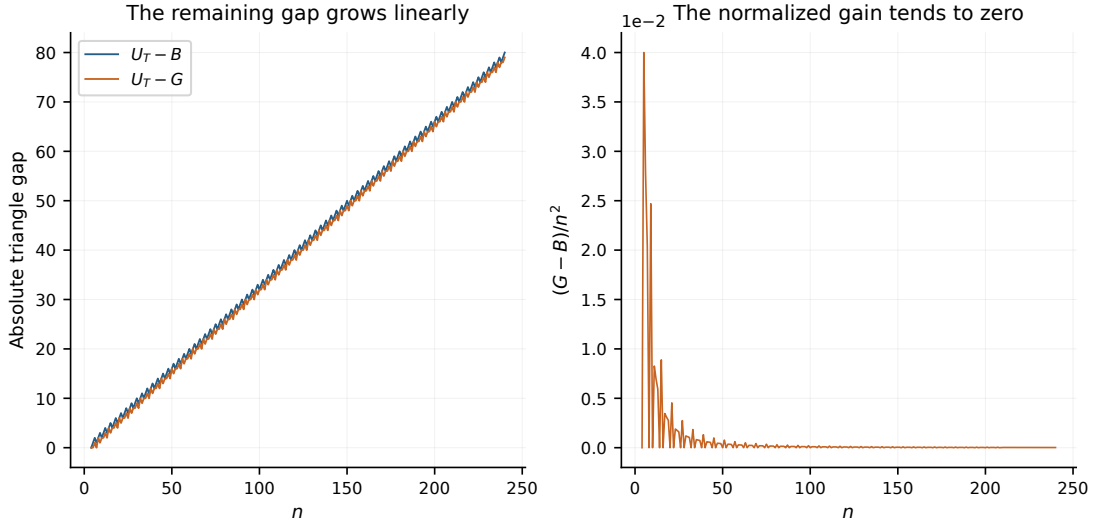


Figure 5: Deficit and normalized growth computed from the proved formulas. The constant improvement is visible in the deficit, while both general lower formulas have the same leading quadratic and linear terms. The polynomial is the classical benchmark, now also formalized for simple arrangements; its numerical upper bound is not new.

This comparison is arithmetic. Section 8.7 now formalizes the classical simple upper theorem; unrestricted attribution remains external. The deficit is still of order n , and sharper bounds or special constructions apply at some orders.

The outcome is an unconditional affine theorem, stronger than the quoted baseline at infinitely many orders. Its numerical first priority remains a separate literature question. The construction gives a witness for each n ; it does not provide an operation that adds one line to every preceding optimal witness.

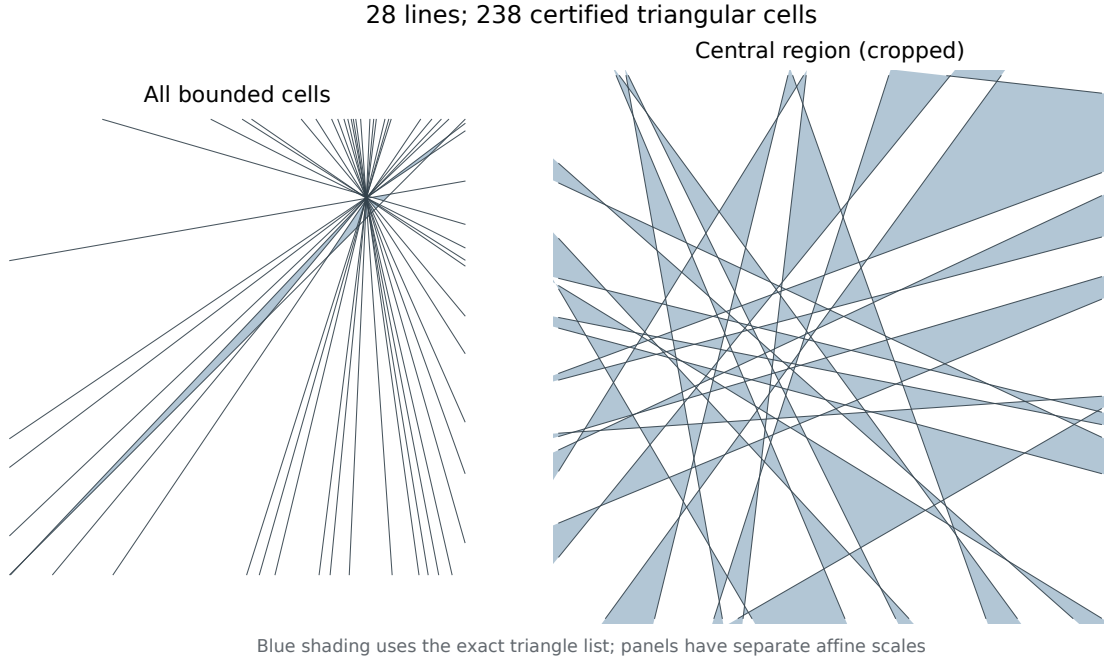


Figure 6: The retained exact 28-line witness with 238 certified triangles. This is a new rendering of the saved coordinates associated with the earlier extension program. The marked zoom, where present, is an enlargement of the same arrangement; it does not add or remove lines from the certificate. The picture illustrates a lower bound, while the exact sign tests establish it.

5 Individual cases, finite enhancements, and provenance

5.1 The March values and their present verification

The three finite inequalities associated with the author’s early extension work are

$$K(28) \geq 238, \quad K(30) \geq 275, \quad K(34) \geq 357. \quad (27)$$

Each now has an exact simple-arrangement certificate, so the same inequalities hold for K_s . Their present validity is independent of the unproved recurrence in the original March manuscript. The certificates give actual lines and complete checked triangle lists, not only numerical consequences of an extension hypothesis.

The current OEIS table associates the name Zarzuelo with these entries [19, 34]. Attribution in an index and first discovery are different questions. The earlier literature already contains matching constructions or consequences for some of these counts; the literature review records these overlaps. The claims made here are the documented attribution trail, the independently verified witnesses, and their place in the author’s extension program. No new priority claim is inferred from the absence of a value from a particular online table.

The original inventory also listed outputs for every order from 3 through 60, together with larger cases at 99 and 195. Some were direct certificates, some were numerical consequences of a boundary-defect argument, and many were weaker than an already available construction. Appendix C preserves that inventory as history. The current certificate aliases independently retain its numerical inequalities; a label inherited from the old table is not used as a present formal-proof status.

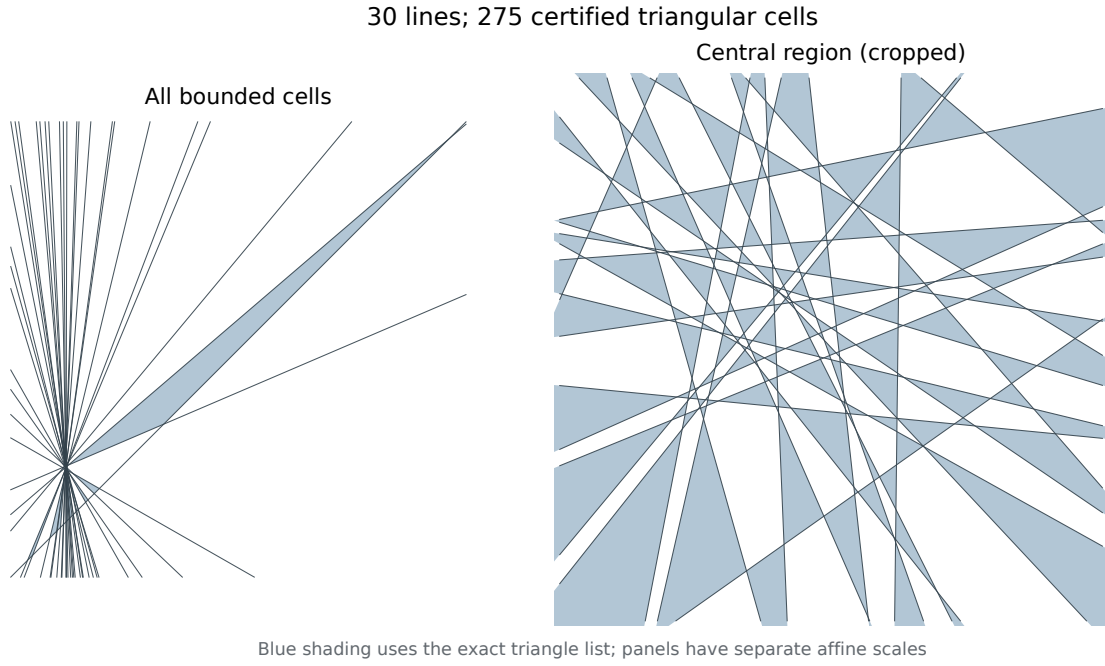


Figure 7: The retained 30-line witness with 275 certified triangles, rendered from its exact coordinates. The count has prior literature overlap and is not asserted to be a new numerical record. The complete witness remains in the catalog even though the historical general extension proof was incomplete.

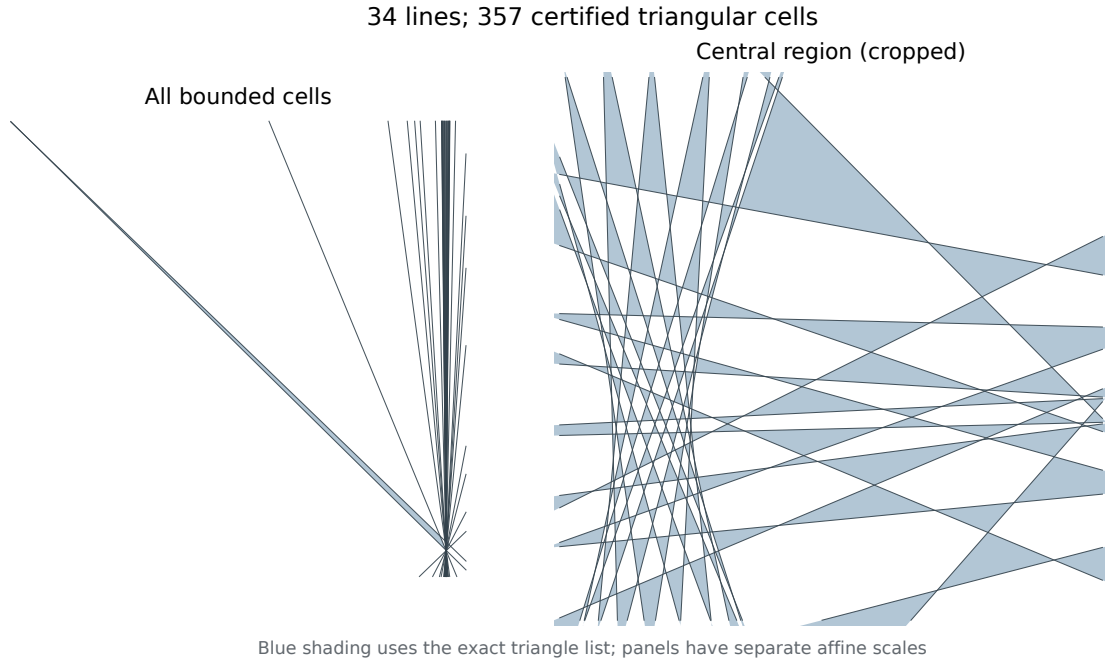


Figure 8: The retained 34-line witness with 357 certified triangles. Original plotting of the certificate makes the finite conclusion inspectable independently of the claimed universal extension. The construction input and earlier matching literature are credited separately.

n	Earlier retained count	Current simple count	Increase
39	469	470	1
47	690	691	1
48	720	721	1
51	817	818	1
53	884	885	1
54	918	919	1
55	954	955	1
59	1102	1103	1
60	1140	1141	1
99	3169	3170	1
195	12481	12482	1

Table 4: Changes relative to this project’s previous retained witnesses, not a table of first discoveries or global record improvements. “Simple” is certified for every current coordinate witness in this table.

5.2 Materialized improvements from the phase and chart construction

Table 4 records simple rational witnesses promoted during the last construction phase. Each listed new count is supplied by the universal theorem and also realized by an independently checked finite coordinate file. This double presentation is useful: the theorem proves all orders, while the coordinates support visualization and further exact searches.

In particular, the earlier additional-line values 39 : 469, 51 : 817, 99 : 3169 and 195 : 12481 are now exceeded by G itself. Their finite correctness survives, but those numbers are not distinctive of the additional-line family. Historical evidence is retained rather than deleted when a stronger construction is found.

5.3 The 44-line simple case

The exact exterior extension of the retained 43-line arrangement creates 21 triangles while preserving its 587 old triangles. Both exact counters and the Lean certificate give a simple 44-line arrangement with 608 triangles. Blanc’s classical simple upper bound at this order is also 608 [2]. Both inequalities are now formally supported, giving

$$K_s(44) = 608. \quad (28)$$

The finite Lean certificate supplies the lower inequality, and `UpperEvenSimpleOptimality` supplies the simple upper inequality. Equation (28) is their immediate numerical corollary, not a separate maximality wrapper. It neither establishes $K(44) = 608$ for unrestricted arrangements nor claims first discovery of this numerical value.

This example shows why upper-bound scope belongs next to the numerical statement. A simple witness can establish a classical lower bound, but a simple upper theorem cannot by itself close the classical problem.

5.4 External records and independently checked inputs

The updated catalog incorporates exact gallery witnesses from Parpalak and Utkin at

8 : 15, 14 : 54, 20 : 117, 26 : 204, 32 : 315, 36 : 402, 38 : 450, 42 : 553, 46 : 667, 50 : 792.

These are constructions or coordinate sources of the cited researchers, not newly discovered values in this project. The eight-line value has older priority; the modern gallery is the source of the checked coordinates. The public source snapshot and the formal certificate index preserve the provenance [19, 20].

All fifteen of Maiorana’s supplied 14-line arrangements with 54 triangles were independently checked and promoted as finite certificates [18]. Fourteen have two triple points and one has four. Their source license and immutable commit are retained. These witnesses are a particularly useful warning against conflating simple and nonsimple arrangements: removing a multiple intersection by a small perturbation need not preserve the full triangle count. Section 10 reports the exact resolution experiments.

The two source snapshots used for these imports are Maiorana commit `e47c7cfd9661e54d29befb83118bf83d5f804028` and gallery commit `feae4f5571a988a3ed45bd26a36259ebb7d7d3d2`. Their mathematical attribution is attached to each source record even when a new Lean wrapper or new figure is generated here.

5.5 The finite envelope and the complete tables

Proposition 5.1 (Retained all-order envelope). *Let $C(n)$ be the maximum count in the finite catalog at order n , or zero if that order is absent. Then $K(n) \geq L(n) = \max\{G(n), C(n)\}$ for every n . In the frozen release, $L(n) = G(n)$ for every $n > 195$.*

Proof. When $C(n) \leq G(n)$, use Theorem 1.1. Otherwise choose a certificate attaining the finite maximum and apply Proposition 3.4. No coordinate certificate in this release has order greater than 195. Larger computational witnesses discussed later have not been silently added to the Lean catalog. Q.E.D.

The formal definition is `Universal.bound = max Universal.baseline AllN.bound`. Since $G \geq B$, it equals the expression in Proposition 5.1; the older AllN layer preserves the previous finite-envelope theorem. This makes comparisons reproducible without rewriting the historical construction history.

Appendix A gives the best saved simple and classical values from 3 through 60. Appendix B lists all 138 certificate identities, including repeated orders and weaker checkpoints. Appendix C gives the older author-output inventory. Together these tables prevent three common confusions: a retained output is not necessarily the best available value; a missing OEIS row is not proof of novelty; and an exact count for one arrangement is not automatically the extremal value $K(n)$.

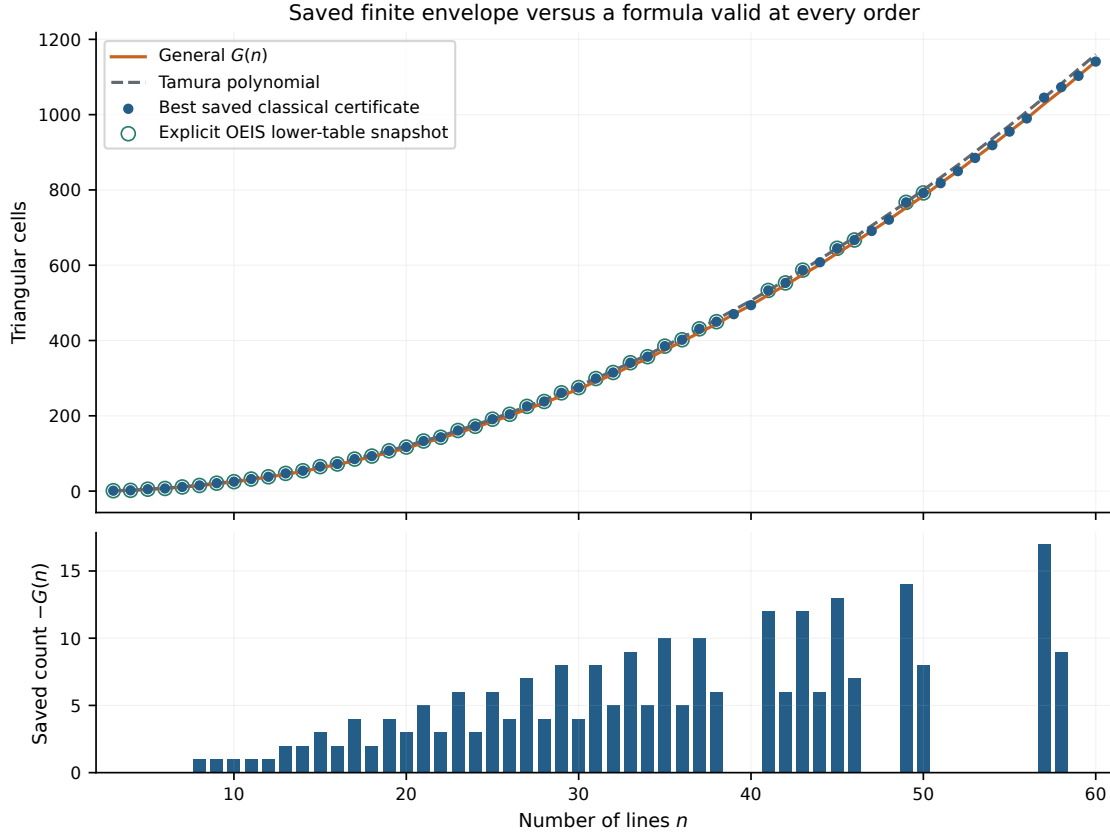


Figure 9: The strongest saved simple and classical coordinate counts, compared with the uniform formula and the polynomial upper benchmark. External witnesses retain their attribution. A marker above G is a finite enhancement of the general formula, not automatically an original record.

6 Exterior extensions: the March proposal and its precise replacement

The starting point of this project was the March 2026 proposal that an odd Kobon arrangement could always be extended with the gain

$$K(2m + 2) \geq K(2m + 1) + m. \quad (29)$$

The proposal was presented in [34] and subsequently described as an iterative rule in [14]. The preserved March source is important evidence of the development of the idea, but is not a completed proof of (29): its Lean file contains eight proof placeholders, including the extension assertion. The present work replaces the informal alternating-segment argument by an actual geometric extension theorem and identifies exactly which additional counting hypothesis yields the desired gain. Independently verified finite bounds are retained regardless of the status of that original argument.

The later OEIS attribution at orders 28, 30, and 34 is evidence of the proposal’s reception, rather than a proof of numerical priority [19]. In particular, the BBL table already records the straight-line value $30 : 275$. The perfect 33-line input belongs to the family of Forge and Ramírez Alfonsín [8]; applying the exterior extension stated by BBL gives $34 : 357$ [1]. The historical attribution, the validity of a finite count, and the novelty of a general extension theorem are therefore assessed separately.

Two distinctions are essential. A recurrence for maxima need not extend *every* arrangement attaining a given count, and a construction for one odd-to-even step need not admit repeated full-gain extensions. Throughout this section the geometric hypotheses concern simple affine arrangements; consequently their conclusions give lower bounds for both K_s and K , but nonsimple witnesses for K cannot automatically be substituted into the hypotheses.

6.1 A geometric theorem from explicitly visible pairs

Write an old line as $L_i = \{p : \ell_i(p) = 0\}$, with ℓ_i affine, and let $A = \{L_0, \dots, L_{n-1}\}$ be simple. A linear functional w is *admissible* if it is nonzero on every old line direction. At $p_{ij} = L_i \cap L_j$, choose the direction vectors u_i, u_j on the two supports so that $w(u_i) = w(u_j) = 1$. The following criterion uses only the old arrangement.

Definition 6.1 (Visible pair). The pair $\{i, j\}$ is visible from w if, for every old affine equation ℓ_k , the three real numbers

$$\ell_k(p_{ij}), \quad D\ell_k(u_i), \quad D\ell_k(u_j)$$

have one common weak sign: either all are nonnegative or all are nonpositive. Here $D\ell_k$ denotes the linear part of ℓ_k .

Geometrically, the two increasing- w rays bound an unbounded empty wedge. The algebraic definition is convenient because it gives the required half-plane inequalities directly, without assuming that a future triangle already exists.

Theorem 6.2 (Verified exterior realization). *Let A be a simple arrangement of n real affine lines, let $\mathcal{T}$ be a set of T distinct triangular cells of A , and let $\mathcal{V}$ be a set of c distinct pairs visible from an admissible w . If*

$$H > \max_{i < j} w(p_{ij}),$$

then adjoining $L = \{p : w(p) = H\}$ produces a simple arrangement containing all triangles in $\mathcal{T}$ and at least c further triangles. In particular, $K_s(n+1) \geq T + c$.

Proof. Admissibility excludes a parallel old line, and the strict height inequality excludes every old vertex from L . Thus simplicity is preserved. Every old triangle lies in the convex hull of its vertices, strictly below the level H , and survives unchanged.

For a visible pair, the new vertices on its two supports are

$$p_{ij} + (H - w(p_{ij}))u_i, \quad p_{ij} + (H - w(p_{ij}))u_j.$$

At either vertex an old affine equation has the form

$$\ell_k(p_{ij}) + (H - w(p_{ij}))D\ell_k(u).$$

The second coefficient is positive. Definition 6.1 therefore puts all three vertices in one closed half-plane of each old line. The new line is itself a supporting side, and the triangle is nondegenerate because the old intersection lies strictly below it. No arrangement line cuts its interior. Different old pairs give different support triples, and every new triple uses L , whereas no old one does. The two lists are consequently disjoint. Q.E.D.

This is the geometric content of **Exterior.extension**. The theorem is proved for arbitrary real coefficients and either parity, with no assumption about a future triangle count. An explicit formal height is

$$H = 1 + \sum_{\substack{i, j < n \\ i \neq j}} |w(p_{ij})|,$$

so existence of a sufficiently distant line is part of the construction. The proof uses the standard Lean logical axioms and no finite native evaluation. What it does *not* supply is a universal lower bound of $\lfloor n/2 \rfloor$ on the number of visible pairs.

6.2 Defect, boundary wedges, and a rule for both parities

Let $t(A) = T$ and define the bounded-segment defect

$$d(A) = n(n - 2) - 3T. \quad (30)$$

There are $n(n - 2)$ elementary bounded segments. A segment can support at most one triangular cell: its endpoints determine the two other supports, and their intersection lies on at most one side of the segment. Thus $d(A)$ counts the unused bounded segments. Let $b(A)$ count the unbounded two-sided cells, called wedges below.

Lemma 6.3 (Boundary charging). *For a simple arrangement of $n \geq 3$ lines,*

$$3 \leq b(A) \leq n, \quad b(A) + d(A) \geq n.$$

Proof. The arrangement has $2n$ free rays. A wedge pairs two such rays at their common endpoint, and each free ray belongs to at most one wedge. An unpaired ray on L , ending at $p = L \cap M$, meets an interior vertex of M . The two bounded elementary segments of M incident to p cannot both support triangles: both triangles would also use the unique bounded segment of L starting at p . Charge the unpaired ray to an unused segment of M at p . Each unused segment receives at most two charges, one at each endpoint. Hence $2n - 2b \leq 2d$. This also proves $b \leq n$. Finally, the convex hull of all finite vertices has at least three vertices. Both supports at a hull vertex have free outgoing rays, which give a wedge. Distinct hull vertices give distinct wedges. Q.E.D.

Order the $2n$ oriented line directions cyclically. The two directions of a wedge are consecutive: a line direction strictly inside that sector would force a line to enter the wedge and cross a free boundary ray. Write $B \subset \mathbb{Z}/(2n)$ for the wedge-start indices. Define

$$c_j = |B \cap \{j, j + 1, \dots, j + n - 2\}|, \quad C(A) = \max_j c_j. \quad (31)$$

The n directions on which an admissible w is positive form one consecutive block, and every such block occurs for some normal sector. An exterior line closes precisely the wedges whose two rays lie in this block. Consequently $C(A)$ is the exact largest exterior gain and

$$\sum_{j=0}^{2n-1} c_j = (n - 1)b(A), \quad \left\lceil \frac{(n - 1)b(A)}{2n} \right\rceil \leq C(A) \leq \left\lfloor \frac{n}{2} \right\rfloor. \quad (32)$$

The sum follows by counting the $n - 1$ blocks containing a fixed wedge. The upper bound follows because complete wedge pairs use disjoint rays in a block of n rays.

Theorem 6.4 (Boundary-defect extension). *For a simple n -line arrangement with $n \geq 3$ and T triangles, an exterior line preserves the old triangles and gives at least*

$$T + g(n, T), \quad g(n, T) = \left\lceil \frac{n - 1}{2n} \max\{3, 3T - n(n - 3)\} \right\rceil \quad (33)$$

triangles. Thus, for either parity,

$$K_s(n) \geq T \implies K_s(n + 1) \geq T + g(n, T).$$

Proof. Lemma 6.3 gives $b(A) \geq \max\{3, n-d(A)\}$. Substitute into (32) and apply Theorem 6.2. If T is only a lower bound, the witness has some count $T' \geq T$; the right side of (33) is nondecreasing in T , which gives the stated maximum-function implication. Q.E.D.

Remark 6.5 (Formal scope and attribution). Theorem 6.4 and its formula were already stated with a manuscript proof in the 3 October edition. The 5–6 October development completes the actual geometric Lean proof. For any finite injective certified triangle family, `OpenMathSimpleBoundary` extracts each line’s first and last real intersections. If A counts vertices terminal on exactly one line and b those terminal on both, then $A + 2b = 2n$, $A \leq 2U$, and $d = U$. A single-terminal vertex has three bounded incident segments. The incident used degree is even, so at least one is unused; its endpoint charge proves the budget. These are `certificate_double_terminal_budget` and `certificate_boundary_defect`.

Actual outward tangents have constant affine signs against every old transverse line. Their positive normal projections imply the original `Exterior.VisiblePair` predicate. Normal charts, critical-root ordering, sector samples and a finite double count prove that each double-terminal vertex is selected by exactly $n - 1$ of $2n$ normal choices. A maximum projected vertex in each normal direction supplies the minimum resource $b \geq 3$. The endpoint `successor` therefore has no ray-order, visibility or charging oracle as a premise. It uses only the standard logical axioms.

Exterior successors from perfect odd arrangements are already described by BBL, and Blanc supplies individual deficit-two examples [1, 2]. The general deficit-dependent scope and its formal extraction are distinguished from those known cases. This edition’s advance over the preceding paper is completed formal linkage and iteration, rather than first appearance of formula (33). Priority of the broader quantitative statement remains subject to review.

Proposition 6.6 (Closed gain formula and indefinite iteration). *For $n \geq 3$ and a supplied simple lower count T , write $\delta = n(n - 2) - 3T$. The gain in (33) equals*

$$g(n, T) = \min \left\{ \lfloor n/2 \rfloor, \left\lceil \frac{\max\{3, \max\{0, n - \delta\}\}}{2} \right\rceil \right\}. \quad (34)$$

Set $T_0 = T$ and $T_{k+1} = T_k + g(n + k, T_k)$. Then for every $k \geq 0$ there is a simple real arrangement of $n + k$ lines with at least T_k triangles. If $n \geq 4$, then $T_k \geq T + 2k$.

Proof. The integer ceiling in (33) simplifies to (34); the minimum accounts for the maximum possible number of disjoint ray pairs in a normal block. This exact arithmetic identity is proved in `closed_form_successor`. Apply the actual successor repeatedly: each output is again simple and has the larger order required by the next application. Induction gives the stated sequence, in `infinite_quantitative_extension`. For $n \geq 4$, (34) is at least two; summing yields the last inequality. This is an existence theorem for every finite iteration length and does not require a fixed normal direction throughout the chain. Q.E.D.

Corollary 6.7 (Full gain for an odd saturated input). *If a simple arrangement of $2m + 1$ lines has defect at most two, an exterior line creates exactly m additional triangles. In particular, this applies to every simple odd arrangement attaining $\lfloor n(n - 2)/3 \rfloor$.*

Proof. The wedge count is at least $2m - 1$. If all $4m + 2$ direction sectors had gain at most $m - 1$, then

$$2m(2m - 1) \leq 2mb(A) = \sum_j c_j \leq (4m + 2)(m - 1) = 2m(2m - 1) - 2,$$

a contradiction. Equation (32) bounds the gain above by m . The segment-count upper bound at odd n leaves defect zero or two. Q.E.D.

Input order	Triangles	Wedges	Best exterior gain	Requested gain
49	767	47	24	24
50	791	24	23	25
51	814	3	2	25
52	816	3	2	26
53	818	3	2	26

Table 5: One exact greedy exterior chain from the attributed 49-line input. Every direction sector was counted with rational arithmetic. These are capacities of the displayed arrangements, not upper bounds for K or K_s at the corresponding orders.

For an even input $n = 2m$, the sufficient condition $b(A) \geq 2m - 1$ likewise forces gain m , because $(2m - 1)^2 > 4m(m - 1)$. In either parity the exact criterion is the finite condition $C(A) = \lfloor n/2 \rfloor$. There is no claim that every even or odd maximum has this boundary profile. Iterating (33) is now fully formalized and supplies a weaker lower-bound rule from any simple seed. The full-gain corollary and its simple optimality consequence are also completed in [OpenMathOptimalSuccessor](#).

6.3 What fails in indefinite full-gain exterior iteration

The March argument made the number of new triangles follow from an alternation assertion after placing a line far from all old vertices. Distance proves preservation; it does not prove emptiness for half the new segments. For example, the five lines $y = ix + i^2$, $i = 0, \dots, 4$, have three triangles, and adding $y = x/2 + 100$ creates only one further triangle. These counts are checked by exact certificates. This refutes that unqualified step of the argument, without refuting a better choice of line or an inequality about maxima.

A stronger obstruction concerns every chain of exterior additions. Write b_r for its wedge count and c_r for the gain at step r . Closing c_r wedges consumes those wedges. No new free ray can appear at an old vertex, and a wedge at a new vertex must use one of the two free rays of the newly added line. Therefore

$$b_{r+1} + c_r \leq b_r + 2, \quad \sum_{j < r} c_j \leq b_0 + 2r - b_r \leq b_0 + 2r - 3. \quad (35)$$

The accumulated exterior gain is at most linear in the number of steps. The accumulated gains $\lfloor n/2 \rfloor$ are quadratic, so an infinite chain of such full-gain exterior steps is impossible. The numerical contradiction is formalized in `Iteration.no_infinite_full_gain_budget`. The universal interpretation of the consumed-wedge budget is a manuscript geometric argument; the completed quantitative successor does not assume that full gains or one fixed boundary profile persist.

Table 5 records the motivating test from 49 : 767. The obstruction is independent of the greedy tie break: two exterior gains 24 and 25 from this seed would imply

$$b_2 \leq 47 + 4 - 24 - 25 = 2,$$

contrary to $b_2 \geq 3$. The five finite profile maxima have a checked Lean arithmetic certificate; their identification with the saved coordinates is supplied by the exact geometry program. A reconstruction between steps or an interior insertion could evade this boundary budget, so it does not disprove the proposed recurrence for maxima.

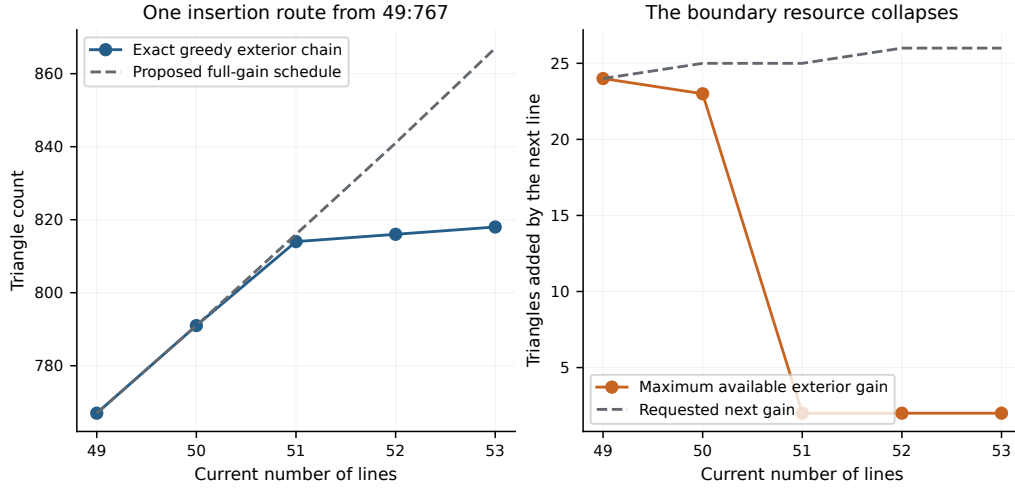


Figure 10: The boundary resource in the particular exact 49-line chain. The sharp drop in available wedges explains why its first full-gain extension cannot be repeated with the requested next gain. The plotted capacities concern this chain only.

6.4 A conditional recurrence and an unconditional numerical bound

For any integer-valued function F , assume

$$F(n+1) \geq F(n) + \lfloor n/2 \rfloor \quad (n \geq s). \quad (36)$$

Summation, including both parity cases, gives the fully checked arithmetic implication

$$F(n) \geq F(s) + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor - \left\lfloor \frac{(s-1)^2}{4} \right\rfloor \quad (n \geq s). \quad (37)$$

The hypothesis (36) remains explicit in the formal theorem. Substituting $s = 49$ and $F(49) = 767$ yields the numerical target

$$Q_{49}(n) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 191. \quad (38)$$

There is now a separate unconditional Lean theorem

$$K(n) \geq Q_{49}(n) \quad (n \geq 49), \quad (39)$$

proved using the saved witnesses at 49 and 50 and the classical all-order baseline thereafter. Indeed, for $n \geq 51$,

$$\frac{n(n-3)}{3} - \frac{(n-1)^2}{4} - 191 = \frac{n^2 - 6n - 2295}{12} \geq 0,$$

with equality at $n = 51$ and an increasing numerator beyond it. Rounding gives $B(n) \geq Q_{49}(n)$. This proves the desired numerical formula without constructing a nested full-gain chain. The stronger all-order bound G and the finite envelope elsewhere in this paper also dominate it.

For comparison, direct iteration of the defect rule $T_{n+1} = T_n + g(n, T_n)$ from $49 : 767$ guarantees $50 : 791$, $51 : 803$, $52 : 805$, and $60 : 821$. The stronger target (38) gives $51 : 816$, $52 : 841$, and $60 : 1061$. These are different constructions and different assertions. The completed numerical theorem does not retrospectively prove the March extension claim, and the failure of its proposed exterior iteration does not invalidate the completed numerical theorem.

7 Compatible seeds, geometric doubling, and sparse families

The second extension mechanism inserts a whole collection of lines near a distinguished support. It belongs to Bartholdi, Blanc, and Loisel (BBL) [1]; it is fundamentally different from the one-line exterior operation of Section 6. The present work supplies explicit compatible seeds, a recursively closed geometric Lean proof, and finite certificates at considerably larger orders. The 2 October release completed both the odd iteration and the stronger adjacent-even family; the 5–6 October work discharges further compatible seed interfaces. The numerical doubling gain retains its published attribution.

7.1 The exact one-step theorem

Let $q = 4r$ with $r \geq 5$, put $\alpha = \pi/q$, and take

$$0 < \varepsilon < \tan(\alpha/2). \quad (40)$$

Define q ordered intercepts, indexed from zero, by

$$a_j = \begin{cases} \tan((j - q/2 + 1)\alpha), & 0 \leq j < q/2 - 1, \\ -\varepsilon, & j = q/2 - 1, \\ \varepsilon, & j = q/2, \\ \tan((j - q/2)\alpha), & q/2 < j < q. \end{cases} \quad (41)$$

In particular, the extreme intercepts are $-\cot \alpha$ and $\cot \alpha$. The two central intercepts replace the repeated zero that would otherwise occur in this grid.

Definition 7.1 (Compatible tangent-grid seed). A seed at (q, ε) consists of $Y_0 : y = 0$ and the q lines $L_j : y = m_j(x - a_j)$, with the following properties: the arrangement is simple; every interval $[a_j, a_{j+1}]$ on Y_0 supports the triangular cell with supports Y_0, L_j, L_{j+1} ; and the central apex $L_{q/2-1} \cap L_{q/2}$ lies above Y_0 . A counted seed additionally carries a list of distinct actual triangular cells whose length is at least T .

Simplicity already requires all m_j to be nonzero and pairwise distinct. The distinguished-line condition is a substantial geometric restriction, not a consequence of having a large total count. It is precisely why an arbitrary optimal arrangement cannot simply be used as a repeatable BBL seed.

Theorem 7.2 (Fully verified geometric doubling). *Suppose $q = 4r \geq 20$ and (40) holds. A compatible tangent-grid seed with at least T certified triangles gives a simple real straight-line arrangement of $2q + 1$ lines with at least*

$$T + q^2 \quad (42)$$

triangles.

The precise formal theorem is `BBLDoubling.doubling`. It assumes the actual old line equations, simplicity, each distinguished triangle, the positive central height, and a duplicate-free triangle list. It does not assume a future crossing order, a list of future empty triangles, or a triangle-count recurrence. Its conclusion is an actual `SimpleLowerBound`; the generic proof uses only the standard Lean logical axioms. The numerical doubling principle is BBL's. The contribution here is its completed geometric proof for these explicit hypotheses, using the following two-parameter realization.

Geometric proof and its formal decomposition. For $0 \leq i < q$ set

$$\beta_i = -\frac{\pi}{2} + (i + \frac{1}{2})\alpha, \quad b_i = \tan \beta_i.$$

Adjoin the lines

$$M_i : \quad y = \kappa g_i(\delta)(x - b_i), \quad g_i(\delta) = \frac{2b_i}{1 + b_i^2} + \frac{\delta}{b_i} = \sin(2\beta_i) + \delta \cot \beta_i. \quad (43)$$

Choose $\delta > 0$ sufficiently small and then $\kappa > 0$ sufficiently small relative to the fixed δ and the seed. This order of choices is part of the proof. BBL's published construction uses explicit scales; the qualitative two-scale statement here is established independently by the formal crossing analysis, rather than inferred from those fixed scales.

For two new lines put $t = b_i$ and $u = b_k$. Their intersection has x -coordinate independent of κ :

$$X_{ik}(\delta) = \frac{2tu(t+u)}{2tu(1-tu) - \delta(1+t^2)(1+u^2)}. \quad (44)$$

When $tu \neq 1$, its limit is the reduced tangent $(t+u)/(1-tu) = \tan(\beta_i + \beta_k)$, and the exact displacement is

$$X_{ik}(\delta) - \frac{t+u}{1-tu} = \frac{\delta(t+u)(1+t^2)(1+u^2)}{[2tu(1-tu) - \delta(1+t^2)(1+u^2)](1-tu)}. \quad (45)$$

For sufficiently small positive δ its nonzero sign is that of $(t+u)tu$. If $tu = 1$, the crossing diverges along the specified exterior ray as $\delta \downarrow 0$; if $t+u = 0$, its coordinate is exactly zero. Thus neither exceptional case is hidden inside a continuity argument with a vanishing denominator.

The reduced tangents, the old intercepts, the new half-grid intercepts, and $-\varepsilon, 0, \varepsilon$ determine a strictly ordered finite system of cuts. Equations (44)–(45) place every new–new crossing in its prescribed cut interval. Taking δ in the intersection of the finitely many resulting positive neighborhoods also makes the new slope factors nonzero and distinct. For fixed such δ , each old–new intersection tends to the corresponding old intercept as $\kappa \rightarrow 0$. Finite continuity then realizes all the prescribed strict crossing comparisons simultaneously. Small κ separates old and new slopes. Injectivity along each new crossing row excludes new triple intersections; old simplicity excludes the remaining ones. This proves actual simplicity and crossing order.

The count can now be separated from trigonometry. Treat the q new lines and Y_0 as $q+1$ auxiliary supports. The integer row model places old support j at key $4j+2$. A pair of auxiliary supports with crossing key z yields a mixed triangle with old support j when

$$|z - (4j+2)| < 4. \quad (46)$$

Every auxiliary pair has two such old neighbors, except for at most q exterior pairs, each of which has one. Hence the number is at least

$$2 \binom{q+1}{2} - q = q^2. \quad (47)$$

The row inequalities make two sides consecutive on their supports. The two-side emptiness lemma proves that the corresponding triangle is an actual empty cell. The supporting triples are distinct. Thus (47) counts genuine triangles, not merely local crossing patterns.

It remains to preserve the seed's contribution. An old triangle not using Y_0 has a strict half-plane sign against Y_0 ; all new lines converge to Y_0 , so its three vertex signs persist

for sufficiently small κ . The old triangles with a side on Y_0 require a different argument. Saturation forces consecutive old apices to have opposite heights. The positive central apex fixes the alternating parity. The crossing order identifies a new cap support for each noncentral distinguished triangle. On both cap endpoints, every new line has the parity-correct weak sign; strict signs against other old supports persist by continuity. This replaces each such old triangle by one empty triangle with the same two old supports. The central triangle itself is retained. The finite cap and noncap requirements can again be satisfied by one common positive κ .

Finally, these T retained or replaced triangles have at least two old nonhorizontal supports, while the q^2 mixed triangles have two auxiliary supports. The supporting triples cannot coincide. Their union is a duplicate-free witness of length at least $T + q^2$. Q.E.D.

The recurrence preserves the certified segment deficit:

$$\Delta(q, T) = q^2 - 1 - 3T, \quad \Delta(2q, T + q^2) = \Delta(q, T). \quad (48)$$

Here the equality concerns the transported lower count; if the output has additional uncounted triangles, its actual defect can be smaller. Thus doubling amplifies a compatible seed to larger orders while preserving its certified constant deficit. A smaller-defect seed or a new source of additional triangles is needed to improve that deficit.

7.2 The compatible eleven-line realization

The useful eleven-line count is 32, an existing result; the search started from the Honma-based construction displayed by Savchuk [29]. The additional work here is an explicit realization compatible with the BBL grid for arbitrarily small positive central separation. Take $Y_0 : y = 0$ and

$$x - v_j y = a_j, \quad (v_0, \dots, v_9) = \frac{1}{10}(15, -2, 14, 1, 4, -4, 5, -3, 12, -5), \quad (49)$$

where

$$(a_0, \dots, a_9) = (-\tan \frac{4\pi}{10}, -\tan \frac{3\pi}{10}, -\tan \frac{2\pi}{10}, -\tan \frac{\pi}{10}, -\varepsilon, \varepsilon, \tan \frac{\pi}{10}, \tan \frac{2\pi}{10}, \tan \frac{3\pi}{10}, \tan \frac{4\pi}{10}). \quad (50)$$

For every $0 < \varepsilon \leq 10^{-5}$ the seed is simple, has at least 32 certified triangles, and has all nine distinguished triangles. Its central slopes are $5/2$ and $-5/2$ and its central height is $5\varepsilon/2 > 0$.

The stability check is finite but uniform in ε . The determinant for a triple of nonhorizontal lines is affine in the intercepts and therefore affine in ε :

$$D_{ijk} = (a_j - a_k)v_i + (a_k - a_i)v_j + (a_i - a_j)v_k.$$

Exact rational enclosures for the four tangent values and endpoint sign checks certify the whole interval. The first certificate used outward rational interval arithmetic; the subsequent Lean theorem uses proved trigonometric enclosures and the parametric certificate soundness theorem. The limit $\varepsilon = 0$ is used only to bound expressions; it is not presented as a simple arrangement. Five visible pairs for $w = (10, -13)$ are also proved uniformly, giving the actual exterior bound $12 : 37$ by Theorem 6.2.

7.3 Closing the geometric invariant

A numerical inequality after one step is insufficient for induction. The output must again satisfy every geometric hypothesis in Definition 7.1. The October construction returns precisely that information. The [seed record](#) contains actual slopes, a duplicate-free valid triangle list,

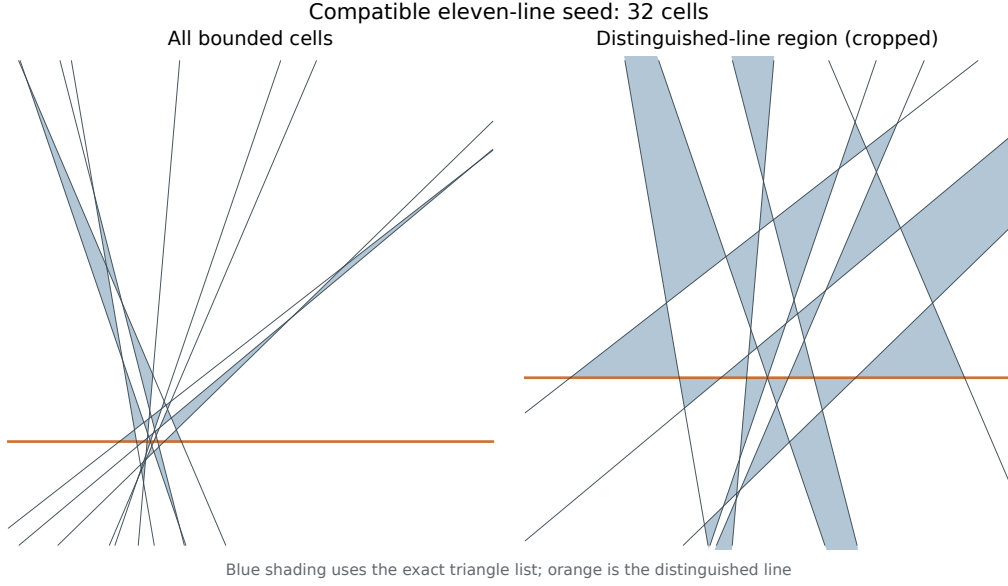


Figure 11: A finite rational realization of the compatible eleven-line seed, with global and distinguished-line views. The figure illustrates the arrangement; the all-small-parameter claim is certified by exact determinant and half-plane inequalities, not by the rendering. The count 32 is inherited from earlier work; the compatible realization and parameter certificate are the contribution recorded here.

simplicity, every consecutive distinguished triangle, and the positive central apex. The line labels are normalized by explicit inverse permutations; the transport theorem preserves actual support triples and their triangle predicates, rather than merely transporting a count.

Theorem 7.3 (Closed geometric iteration). *Fix $r \geq 5$ and a counted seed size T . Suppose that for every sufficiently small positive ε a compatible tangent-grid seed with $4r + 1$ lines and at least T triangles exists. Then for every integer $t \geq 0$ there is a simple arrangement of $4r 2^t + 1$ lines with at least T_t triangles, where*

$$T_0 = T, \quad T_{t+1} = T_t + (4r 2^t)^2, \quad 3T_t + (4r)^2 = 3T + (4r 2^t)^2. \quad (51)$$

At every finite depth the full seed invariant is available on some nonempty positive parameter interval.

Proof. The geometric construction in Theorem 7.2 is performed while retaining the next distinguished triangles and the positive central triangle. Its output intercepts, after the proved reindexing, are exactly the tangent grid for $2r$. Simplicity and the counted list are carried in the same witness. This is `seed_step`; its coefficient identities are proved in `BBLRecursiveGrid` and its unreindexed geometric witness in `BBLRecursiveWitness`.

More explicitly, let $\eta > 0$ be a valid input threshold. The next seed exists for every

$$0 < \varepsilon < \min\{\eta, \tan(\pi/(8r))\}.$$

The minimum is positive. For each such ε , the small pencil parameters are chosen as in the one-step proof. No uniform lower bound on these auxiliary parameters is needed. Thus the whole seed property, including its nonempty interval, is preserved by $r \mapsto 2r$. Induction proves the first assertion and the recurrence. Multiplying the recurrence by three and using $(2q)^2 = q^2 + 3q^2$ proves the displayed identity by induction. The formal results are `uniform_iterate`, `infinite_family`, and `triangleCount_identity`. Q.E.D.

The quantifier is $\forall t \exists \eta_t > 0 \forall \varepsilon \in (0, \eta_t)$, with a seed witness for each such parameter. It is not $\exists \varepsilon > 0 \forall t$. In particular, the theorem does not claim that a fixed drawn arrangement can be iterated with one fixed positive gap forever. This distinction closes the earlier proof gap without inserting an unproved compactness assertion.

The normalized 21-line seed supplies $r = 5$, $T = 132$ throughout $0 < \varepsilon \leq 10^{-5}$. Its nineteen distinguished triangles, actual coefficient identity, and central height $5\varepsilon/2$ are proved in `BBLSeed21Normalized`. The generic theorem starts at $r \geq 5$; the separately verified eleven-line seed supplies the initial term in the following family.

Theorem 7.4 (Verified odd family). *For every integer $t \geq 0$, set $q_t = 10 \cdot 2^t$. Then*

$$K_s(q_t + 1) \geq T_t = \frac{100 \cdot 4^t - 4}{3} = \frac{q_t^2 - 4}{3}. \quad (52)$$

Proof. For $t = 0$ use the eleven-line certificate. For $t \geq 1$ apply Theorem 7.3 to the normalized 21-line seed at depth $t - 1$. Equation (51) gives $3T_t + 4 = q_t^2$. The exact formal declaration is `eleven_odd_family`. Q.E.D.

The generic iteration uses only standard logical axioms. The concrete family additionally inherits four existing native certificate checks from its 21-line base. This is an unconditional geometric existence theorem with an explicitly reported computation trust boundary, not a conditional recurrence on unknown arrangements.

7.4 Even companions and the retained boundary invariant

The defect argument of Theorem 6.4 gives the weaker ordinary mathematical consequence

$$K_s(q + 2) \geq \frac{q^2 - 4}{3} + \frac{q}{2} - 1, \quad q = 10 \cdot 2^t. \quad (53)$$

Indeed its gain is $\lceil q(q-2)/(2(q+1)) \rceil = q/2 - 1$. The October proof establishes the additional visible pair at every depth, rather than extrapolating from finite checkpoints.

A visible seed extends the preceding seed record by an admissible normal $w = (w_x, w_y)$, a duplicate-free list of at least V visible pairs, and a rightmost old line R satisfying

$$m_R > 0, \quad w_x + w_y m_R > 0, \quad x(R \cap L_i) \leq a_R \quad (L_i \neq R).$$

These are concrete line-intersection conditions. The record is `VisibleSeed`. The generic iteration additionally assumes $w_x > 0$.

Theorem 7.5 (Closed visible-boundary iteration). *Under these uniform visible-seed hypotheses with $r \geq 5$, doubling preserves the normal and the whole visible-seed record, with*

$$(r, T, V) \mapsto (2r, T + (4r)^2, V + 2r). \quad (54)$$

Consequently, at depth t the visible count satisfies $V_t + 2r = V + 2r \cdot 2^t$. If $V = 2r$, the exterior extension yields at least $T_t + 2r \cdot 2^t$ triangles at order $4r \cdot 2^t + 2$.

Proof. Old visible pairs avoiding Y_0 persist under a sufficiently small pencil; at most one visible old pair uses Y_0 . The new construction supplies r negative complementary pairs, $r - 1$ positive almost-complementary pairs, the pair of Y_0 and the rightmost new support, and the pair of R with the support at $\beta_* = \pi/4 - \alpha/2$. This supplies $2r + 1$ pairs, compensating for the possible old loss and leaving a gain of $2r$. Pair supports and new/old tags prove that the retained list has no duplicates.

q	Odd order	Odd count	Even order	Even count	Family proof
10	11	32	12	37	Lean existence
20	21	132	22	142	Lean existence
40	41	532	42	552	Lean existence
80	81	2132	82	2172	Lean existence
160	161	8532	162	8612	Lean existence
320	321	34132	322	34292	Lean existence
640	641	136532	642	136852	Lean existence

Table 6: The completed odd and even family. Every displayed existential bound now follows from one general Lean theorem. This does not change the separate verification status of previously exported coordinate files, nor assert that these numbers exceed all literature constructions.

The last pair is controlled by

$$\cot \alpha \sin(2\beta) + \cos(2\beta) - 1 = \frac{\sin(2\beta + \alpha)}{\sin \alpha} - 1.$$

On the half-grid its unique maximum occurs at β_* , with a gap at least $2\sin \alpha$. First shrink δ to preserve this strict maximum, then shrink κ to compare the actual new crossing with all old vertices on R . The positivity of projected graph directions transfers horizontal crossing order to w -order. The corresponding rightmost clean ray is then established on the next grid. The finite requirements are intersected with the eventual one-step witness, so the same output has the old seed invariant and the new boundary invariant.

This is the geometric content of `canonical_visible_step` and `BBLEvenRecursive.visible_seed_step`; `BBLProjection`, `BBLRightmost` and `BBLNextBoundary` supply the actual order and intersection lemmas. Reindexing transports visibility and admissibility. The positive parameter-interval induction is the same as for odd seeds. Summing $V_{t+1} - V_t = 2r \cdot 2^t$ gives $V_t = V + 2r(2^t - 1)$. Finally apply the proved exterior theorem to this actual visible-pair list. Q.E.D.

Theorem 7.6 (Verified full-gain even family). *For every integer $t \geq 0$ and $q = 10 \cdot 2^t$,*

$$K_s(q + 2) \geq \frac{q^2 - 4}{3} + \frac{q}{2}. \quad (55)$$

Proof. The normalized 21-line visible seed has $r = 5$, $T = 132$, $V = 10$ and $w = (10, -13)$. Its full rightmost-boundary and visibility hypotheses hold uniformly for $0 < \varepsilon \leq 10^{-5}$. Apply Theorem 7.5; the independently verified 12 : 37 witness supplies $t = 0$. The unconditional declaration is `eleven_even_family`. Its dependencies include the four odd-seed native checks and two further finite admissibility/visibility checks. No unresolved visibility hypothesis remains in this concrete theorem. Q.E.D.

The independent adjacency and half-plane counters agree on the archived 321-, 322- and 641-line coordinate files, with 34132, 34292 and 136532 triangles. Their largest rational components have 916, 985 and 1103 decimal digits. The 642-line file passed adjacency and simplicity checks but its separate direct recount was interrupted. That coordinate artifact still has partial computational status. In contrast, existence of some 642-line simple arrangement with at least 136852 triangles is now proved by Theorem 7.6. An existential proof does not retrospectively certify the exact bytes of a saved coordinate file.

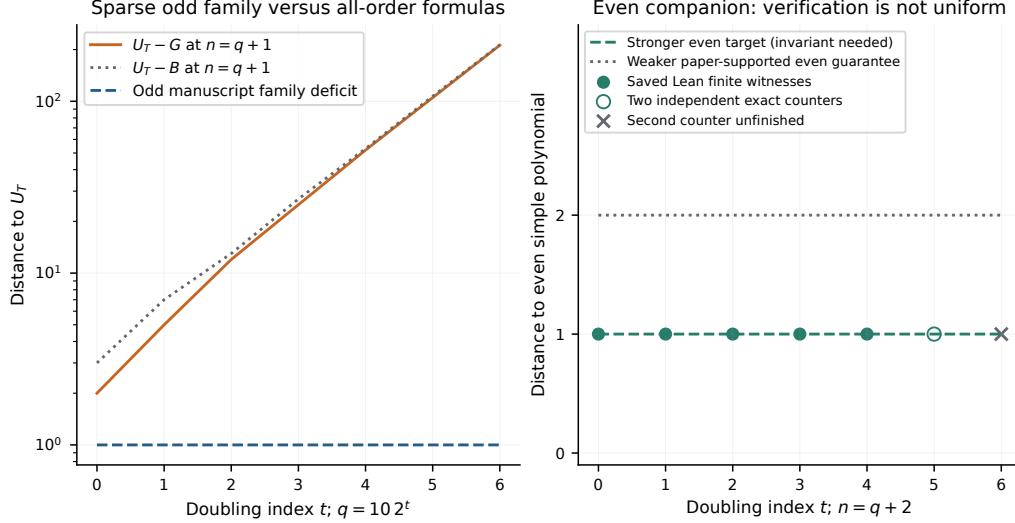


Figure 12: The original sparse-family deficit plot, retained as a historical numerical figure. Its arithmetic is unchanged. The family formulas now have completed Lean existence proofs; any finite-check status encoded in the original plot describes the September coordinate artifacts. The new figure below compares the completed envelope.

7.5 The first recursive envelope at every natural order

Recall the previous retained bound $L(n) = \max\{G(n), C(n)\}$ in (2). Define the sparse family contribution

$$R(n) = \begin{cases} (q_t^2 - 4)/3, & n = q_t + 1, \\ (q_t^2 - 4)/3 + q_t/2, & n = q_t + 2, \\ 0, & \text{otherwise,} \end{cases} \quad q_t = 10 \cdot 2^t, \quad (56)$$

and put

$$H(n) = \max\{L(n), R(n)\}. \quad (57)$$

The two cases have different parity and each determines t uniquely. The executable Lean definition uses the maximum of the candidates for $0 \leq t \leq n$, rather than an infinite search.

Theorem 7.7 (Total retained and recursive envelope). *For every $n \in \mathbb{N}$, $K(n) \geq H(n) \geq L(n)$. For every $t \geq 5$,*

$$H(q_t + 1) - L(q_t + 1) = \lfloor q_t/3 \rfloor - 2, \quad (58)$$

$$H(q_t + 2) - L(q_t + 2) = q_t/2 - \lfloor q_t/3 \rfloor - 2, \quad (59)$$

and both differences are positive.

Proof. Each nonzero candidate in (56) has a simple witness by the preceding two theorems, hence an unrestricted witness. The finite maximum chooses one such witness or a retained finite/baseline witness; it is not an addition of incompatible triangle counts. If a candidate occurs at order n , then $t < 2^t \leq n$, so the executable finite maximum includes it. This is `all_n`.

Since $q_t^2 \equiv 1 \pmod{3}$ and q_t is even, substitution in G gives

$$T_t - G(q_t + 1) = \lfloor q_t/3 \rfloor - 2, \quad T_t + q_t/2 - G(q_t + 2) = q_t/2 - \lfloor q_t/3 \rfloor - 2.$$

All saved finite enhancements end at order 195. Thus $L(n) = G(n)$ for $n > 195$, as proved by `BBLFamilyBenchmarks.previous_envelope`. For $t \geq 5$, $q_t \geq 320$ and both differences

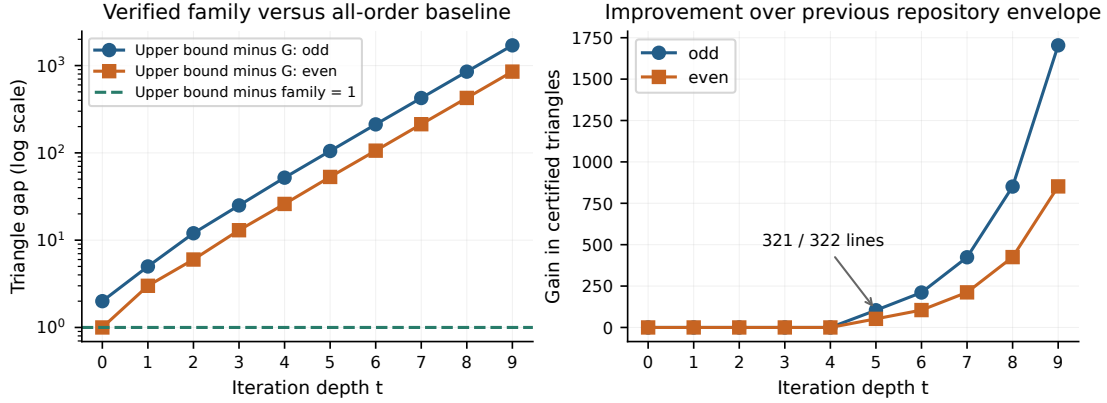


Figure 13: The completed recursive family and the retained all-order bound. The improvement is over the previous repository envelope; it is not a numerical-priority comparison against all published constructions. The upper comparisons refer to simple arrangements. Formula values are proved in Lean; the plot is newly generated from those formulas.

are positive. The formal strict-tail statement is `RecursiveEnvelope.strict_previous_tail`; uniqueness of the sparse candidate identifies H with that candidate here. Q.E.D.

At 321 and 322 lines the gains over the old verified envelope are respectively 104 and 52; at 641 and 642 they are 211 and 105. The first members do not displace stronger retained certificates: 21 : 133, 22 : 143, 41 : 533 and 42 : 553 remain in H . All 138 coordinate identities and their published attribution are retained. This theorem improves the previous verified repository envelope at infinitely many orders. It does not prove a numerical record against all published families, and its off-family values retain the preceding all-order linear deficit.

7.6 Actual one-triangle optimality windows

The October release also closes the geometric upper-bound bridge for simple arrangements, as proved in Section 8.7. This upgrades a comparison with a polynomial to a statement about the actual extremal quantity.

Corollary 7.8 (Simple optimality windows). *For $q = 10 \cdot 2^t$ and $t \geq 0$, write $A = (q^2 - 4)/3$. Then*

$$A \leq K_s(q + 1) \leq A + 1, \quad A + q/2 \leq K_s(q + 2) \leq A + q/2 + 1. \quad (60)$$

Proof. The lower halves are the constructed witnesses. For every simple arrangement, the geometric upper results give $3T \leq n(n - 2)$ and, for even $n \geq 4$, $6T \leq n(2n - 5)$. Substitution gives respectively

$$\lfloor (q + 1)(q - 1)/3 \rfloor = A + 1, \quad \lfloor (q + 2)(2q - 1)/6 \rfloor = A + q/2 + 1.$$

The exact combined declarations are `odd_window` and `even_window`.

Q.E.D.

The upper halves use standard logical axioms only; the lower halves retain their seed's native dependencies. This one-triangle interval need not be sharp at the smaller orders: the literature settles some of them more precisely. In particular it is not an upper theorem for arrangements with multiple concurrence, and it does not close the unrestricted Kobon problem.

7.7 Known optimal families and independent reproduction

The classical dyadic five-line family must be kept separate from the new compatible eleven-line realization. A convenient normalized seed uses intercepts $(-1, -\varepsilon, \varepsilon, 1)$ and

$$x - v_j y = a_j, \quad (v_0, v_1, v_2, v_3) = (-15, 10, -10, 30),$$

together with Y_0 . The complete interval $0 < \varepsilon \leq 10^{-5}$ has five certified triangles, all three distinguished triangles, and two visible pairs. Its central slopes are $1/10$ and $-1/10$; the rightmost slope is $1/30$. The normalization is deliberate: an earlier affine-equivalent numerical realization had both central slopes negative, which was inadequate as an unchecked premise for the sign convention in the published iterative proof. The corrected whole-parameter seed is verified in `TamuraSeed5`.

The known optimal odd family and its simple even companions have

$$q = 4 \cdot 2^t, \quad K_s(q+1) \geq \frac{q^2 - 1}{3}, \quad K_s(q+2) \geq \frac{q^2 - 1}{3} + \frac{q}{2}. \quad (61)$$

The current finite certificates include $5 : 5$, $6 : 7$, $9 : 21$, $10 : 25$, $17 : 85$, $18 : 93$, $33 : 341$, $34 : 357$, $65 : 1365$, $66 : 1397$, $129 : 5461$, and $130 : 5525$. These reproduce known numerical consequences of the affine family of Forge and Ramírez Alfonsín and BBL’s odd-to-even discussion [1, 8]; they are not claimed as newly discovered bounds. Theorem 7.2 does not cover the small bases $q = 4, 8, 16$. The now completed uniform 33-line seed starts its formal infinite tail at $q = 32$, while retained finite certificates supply the small cases.

Parpalak and Utkin supply a different straight-line family at $q = 18 \cdot 2^t$ [20]. Its odd count is

$$K_s(q+1) \geq \frac{q^2}{3} - 1, \quad (62)$$

including $19 : 107$ and $37 : 431$. The present work independently checked their compatible 19-line seed on $0 < \varepsilon \leq 10^{-3}$ with 1632 exact endpoint determinant checks, a rational 107-triangle witness, and all 17 distinguished triangles. The interval reproduction at 19 is separate from the now completed 37-line compatible tail below. Both the seed and the family retain Parpalak–Utkin attribution. The earlier BBL statement at $18 \cdot 2^t + 1$ concerned pseudolines; it cannot be used as a straight-line result without a realization argument. Applying Corollary 6.7 to the published straight-line family gives the attributed simple even corollary

$$K_s(q+2) \geq \frac{q^2}{3} + \frac{q}{2} - 1,$$

with $20 : 116$, $38 : 449$, $74 : 1763$, and $146 : 6983$. Stronger nonsimple published values at some of these orders concern K , and do not invalidate this simple-arrangement distinction.

7.8 The closed integration interfaces and the Forge tail

The verified 21-line base has the true tangent intercepts (41) with $q = 20$, fixed rational slopes, 132 triangles, 19 distinguished triangles, ten visible pairs, and the required rightmost-line property for every $0 < \varepsilon \leq 10^{-5}$. Its finite interval checks use native evaluation; their analytic and geometric soundness proofs are proved separately.

The September manuscript correctly identified four missing interfaces: retained geometric data, whole-list transport under the next-grid permutation, normalization of the 21-line grouped labels, and induction on a positive parameter interval. All four are now complete. The central source labels 5 and 6 have apex height $5\varepsilon/2$; normalization preserves that actual

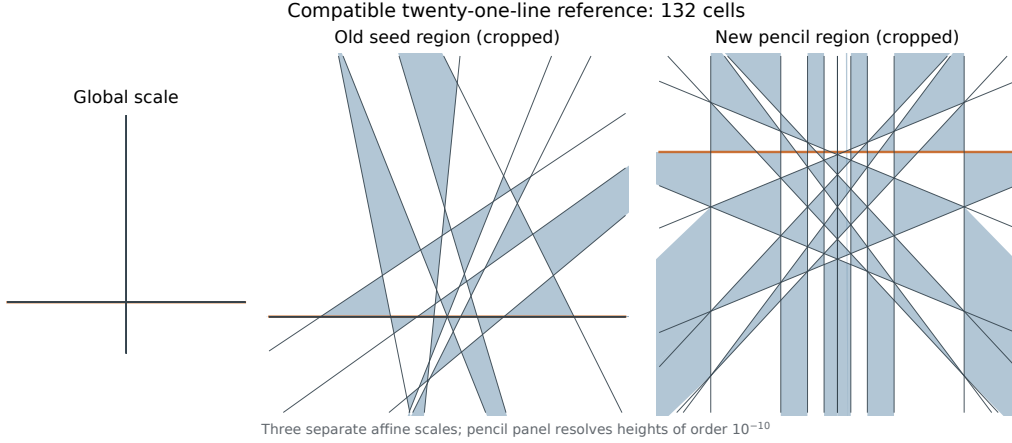


Figure 14: A 21-line hybrid checkpoint with 132 triangles. The global, core and thin-pencil views illustrate scale separation. The finite rendering is distinct from the true-tangent-grid parametric theorem in `BBLSeed21`. The normalized seed now starts the fully verified recursion; the picture is not used as its proof.

point. The next central triangle is retained independently of whether it appears in the counted list. The invariant used in the successful induction is

$$\exists \eta > 0 \quad \forall \varepsilon \in (0, \eta) \quad \exists \text{ a compatible counted seed at } (q, \varepsilon). \quad (63)$$

The former target

$$q_t = 20 \cdot 2^t, \quad K_s(q_t + 1) \geq \frac{q_t^2 - 4}{3} \quad (64)$$

is therefore a theorem, the tail of (52). The visible-boundary interfaces are also closed by the same release.

The generic Forge–Ramírez Alfonsín branch initially retained its uniform seed premises. They are now discharged by the independent `OpenMathConstructionSeed33` modules. For every $t \geq 0$,

$$K_s(32 \cdot 2^t + 1) \geq \frac{1024 \cdot 4^t - 1}{3}, \quad K_s(32 \cdot 2^t + 2) \geq \frac{1024 \cdot 4^t - 1}{3} + 16 \cdot 2^t$$

are actual simple real-line witnesses. The same rational reciprocal slopes and finite statements preserved in the earlier stopped draft are checked on $0 < \varepsilon \leq 10^{-5}$. The sound Boolean certificate engine completes the checks without changing their mathematical predicates. `odd_family` and `even_family` instantiate the proved recursion; the same file proves both simple optimality statements. These are formal completions of inherited numerical formulas, not new construction records.

7.9 A uniform 49-line compatibility certificate

The October search found a stronger compatible seed at order 49 using fixed rational reciprocal slopes of denominator 10^{10} . Its 48 intercepts are

$$-\tan \frac{23\pi}{48}, \dots, -\tan \frac{\pi}{48}, -\varepsilon, \varepsilon, \tan \frac{\pi}{48}, \dots, \tan \frac{23\pi}{48}.$$

Exact rational interval arithmetic verifies, throughout $0 < \varepsilon \leq 1/100$, a simple arrangement with 767 certified triangles, all 47 distinguished-line caps and a positive central apex. The

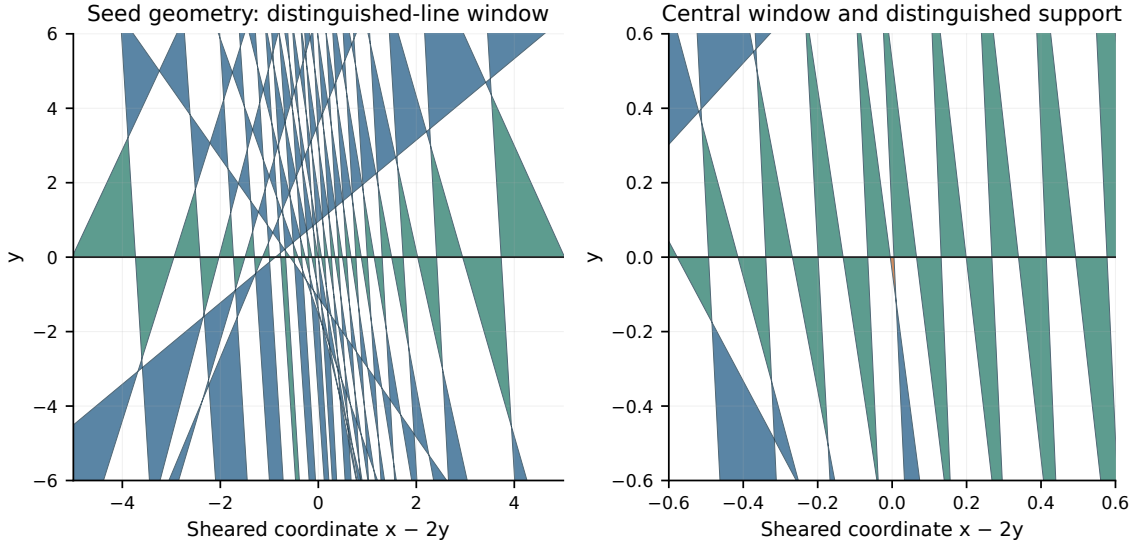


Figure 15: Two clipped windows of the rational midpoint witness for the uniform 49-line seed, with $\varepsilon = 1/200$. The invertible shear $x' = x - 2y$ makes the distinguished structure easier to see. The left window has $x' \in [-5, 5]$, $y \in [-6, 6]$; the right has both coordinates in $[-0.6, 0.6]$. Ordinary certified cells are blue, distinguished caps teal and the central cell orange. Distant cells lie outside these windows: the panels do not display all 767 triangles. This finite view illustrates the fully completed true-tangent uniform seed theorem.

fixed normal $(1, -2)$ has 24 visible pairs, including rightmost pair $(0, 48, 49)$; the last slope is $1/3$ and the required boundary inequalities hold uniformly. The full finite Lean base and its normalization and visible-boundary bridges are now completed.

The analytic part is now inside Lean: `BBLTangent48Bounds` proves rational enclosures for each of the 23 positive tangent values, using half-angle and reciprocal identities beginning with $\tan(\pi/3) = \sqrt{3}$. The endpoint widths are at most $5 \cdot 10^{-36}$. The exported finite predicates were independently replayed: 1,176 direction checks, 18,424 simplicity checks, 37,583 triangle/support checks and 1,176 visibility/support checks passed. Two independent exact cell counters also find 767 triangles at the rational midpoint witness. The finite predicates and analytic-to-geometric bridges now link the released data to the actual infinite theorem.

The completed recursive instantiation gives

$$q = 48 \cdot 2^t, \quad T_{\text{odd}} = 768 \cdot 4^t - 1, \quad T_{\text{even}} = 768 \cdot 4^t + 24 \cdot 2^t - 1 \quad (65)$$

at orders $q + 1$ and $q + 2$. The odd formula is exactly the tail $s = t + 3$ of BBL’s Theorem 1.3 for $n = 6 \cdot 2^s + 1$ [1]; these are not new numerical records. The contribution of this branch is an explicit, well-conditioned uniform compatibility certificate and its completed actual formal linkage. The declarations `odd_family` and `even_family` have all seed hypotheses discharged. Combining these witnesses with the classical simple upper bounds proves exactness in the simple model. The known 50 : 792 unrestricted witness is stronger than this orbit’s 50 : 791 simple witness, so simple exactness is not unrestricted exactness.

Earlier positive-margin fits had moved triangles across infinity. The successful fit additionally preserves the original affine sign system. This illustrates why incidence or a projective order type alone cannot certify the desired affine cell count.

A preceding search for a perfect compatible 21-line seed tested 11,547,910 floating proposals and 3,234 linear-feasibility programs without improving 132. Those bounded failures were not impossibility proofs. The exact obstruction in Section 10.5 now proves a much more

precise limitation for a prescribed grid and supplied classification; it still does not exclude other compatible families or a different geometric construction.

7.10 The compatible 37-line tail of the known 18-dyadic orbit

An independent refit of Poola’s 37-line input retains 431 triangles and 35 distinguished-line caps on the actual tangent36 box $0 < \varepsilon \leq 10^{-8}$ [24]. The 17 positive tangent enclosures are proved by analytic lemmas and rational kernel checks in `BBLTangent36Bounds`. The independent finite seed certificate and normalization supply the complete input to the existing recursion.

Theorem 7.9 (Completed 37-line orbit). *For every $t \geq 0$, put $q = 36 \cdot 2^t$. Then*

$$K_s(q+1) = 432 \cdot 4^t - 1, \quad K_s(q+2) = 432 \cdot 4^t - 1 + 18 \cdot 2^t. \quad (66)$$

Proof. The uniform compatible seed gives the first lower bound by closed doubling. Its odd certified count has defect two. The completed universal successor of Corollary 6.7 gives the second lower bound at every depth, without an additional visible-pair finite certificate. Substitution in the odd and even simple upper floors gives matching upper bounds. These are `odd_family` and `even_family`, with simple exactness proved in the same module. Q.E.D.

Numerically this is the $q = 18 \cdot 2^s$ straight-line series of Parpalak–Utkin, starting at $s = t + 1$, and its known even companions [2, 20, 22]. The compatible exact certificate and formal linkage are the additions. They raise the previous formal portfolio at 73, 74 from 1705, 1752 to 1727, 1763, and at 145, 146 from 6865, 6960 to 6911, 6983. At 38 the retained unrestricted 450-cell witness remains stronger than the simple optimum 449.

7.11 A compatible 61-line seed and the visibility refinement

Poola’s OpenMath submission supplies a 61-line point arrangement with 1190 triangles [24]. Its original fixed slopes lose eight cells as the two central tangent intercepts approach zero, so the point count did not supply the uniform premise of the recursion. We refit reciprocal slopes while preserving its affine orientation type, including the chosen infinity support, at the limiting value $\varepsilon = 0$. A floating positive LP margin proposed the rational data; exact checks then verified the whole true-tangent box $0 < \varepsilon \leq 10^{-8}$. The independent checker tests 35,990 simplicity conditions and 72,590 triangle/support conditions, retaining all 1190 triangles and all 59 axis caps. `BBLTangent60Bounds` proves the actual 29 positive tangent enclosures used by this box.

The first compatible refit has 28 visible pairs and normal $(1, -20025598529/10^{10})$. A nonlocal feasible cap-cone ray mutation, scored by the combined resource $T + V$, produces a second compatible seed with 1190 triangles and 29 visible pairs. Reflection and shear restore the canonical rightmost boundary. Its fixed normal is $(1, -8000441397/4000000000)$ and its central apex height is $(20000000000/8777037)\varepsilon > 0$. The sparse and original dense exact checkers independently verify its triangle/cap box; every Lean finite decision was replayed on the modified data. The original 28-pair seed is retained separately.

Theorem 7.10 (Actual 60-dyadic family). *For every $t \geq 0$, let $q = 60 \cdot 2^t$. Then*

$$K_s(q+1) \geq 1200 \cdot 4^t - 10, \quad (67)$$

$$K_s(q+2) \geq 1200 \cdot 4^t + 30 \cdot 2^t - 11. \quad (68)$$

The counts are nine and ten below the corresponding simple upper floors.

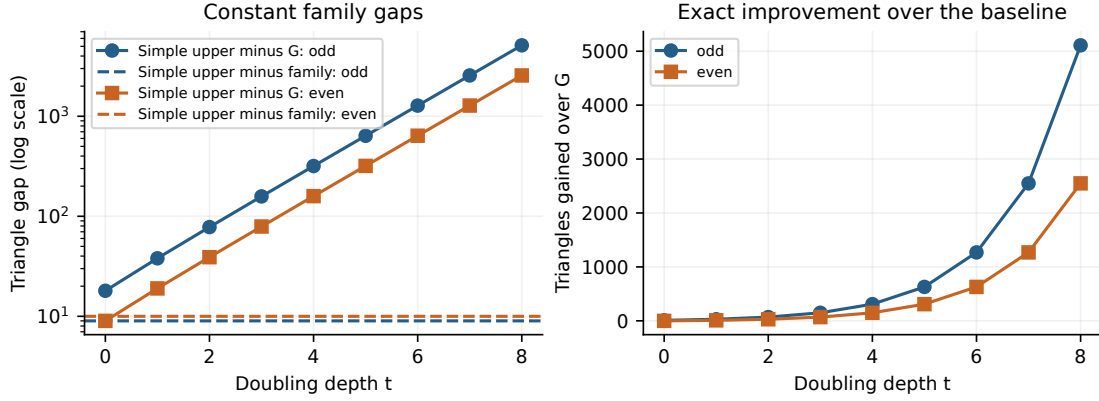


Figure 16: Exact gaps of the 60-dyadic constructions to the classical simple upper floors. The odd and strongest even gaps are constantly nine and ten. The corresponding baseline gaps grow with the grid order. At depth zero the baseline is stronger in the even case. These are formula evaluations, with no fitted counts or unrestricted upper-bound claim.

Proof. The released uniform seed has $(r, T, V) = (15, 1190, 29)$ and supplies all geometric and boundary premises of Theorems 7.3 and 7.5. Their recurrences give

$$T_t = 1200 \cdot 4^t - 10, \quad V_t = 30 \cdot 2^t - 1.$$

Exterior addition gives $T_t + V_t$. The simple upper floors at these orders are $1200 \cdot 4^t - 1$ and $1200 \cdot 4^t + 30 \cdot 2^t - 1$, respectively. The actual declarations are `odd_family` and `even_family`. The seed inherits seven explicit finite native Boolean roots; rescaling, tangent analysis, normalization, boundary transport and iteration use only standard logical axioms. Q.E.D.

The constant-gap calculation is a special case of the proved `BBLDeficitTransport`: the seed's odd triangle loss and visible-pair loss are separately conserved by doubling. The allowable positive parameter interval may shrink with depth. Neither a single fixed positive ε for all depths nor an optimum for the whole 60-dyadic class is asserted.

Relative to G , the gains are $20 \cdot 2^t - 11$ in the odd branch and $10 \cdot 2^t - 11$ in the even branch. The odd examples are 61 : 1190, 121 : 4790, 241 : 19190, and 481 : 76790. The even examples are 62 : 1219, 122 : 4849, 242 : 19309, and 482 : 77029. At 62 the retained $G(62) = 1220$ is stronger. The 29-pair improvement adds one triangle to the initial 28-pair even formula at every depth; at 122, 242, 482 this increases the newly verified retained maxima from 4848, 19308, 77028 to 4849, 19309, 77029.

The finite 1190 count remains Poola's; doubling remains BBL's. A bounded comparison of the BBL, Forge–Ramírez Alfonsín, Parpalak–Utkin and Savchuk sources did not locate this 60-dyadic straight-line orbit. This supports treating the compatible orbit as a candidate new family, but does not establish absolute numerical priority. No improved individual Kobon optimum is inferred from this source search.

7.12 The completed portfolio for every natural order

Let $F_a(n, t)$ equal the appropriate odd or even count in the row based on $a \in \{32, 36, 48, 60\}$, and zero away from those two orders. Define

$$J(n) = \max \left\{ H(n), \max_{0 \leq t \leq n} \max_{a \in \{32, 36, 48, 60\}} F_a(n, t) \right\}. \quad (69)$$

q	Count at $q + 1$	Count at $q + 2$	Attribution and simple gaps
$10 \cdot 2^t$	$(100 \cdot 4^t - 4)/3$	Odd count $+5 \cdot 2^t$	Eleven-derived; gaps 1, 1
$32 \cdot 2^t$	$(1024 \cdot 4^t - 1)/3$	Odd count $+16 \cdot 2^t$	Forge–Ramírez Alfonsín; gaps 0, 0
$36 \cdot 2^t$	$432 \cdot 4^t - 1$	Odd count $+18 \cdot 2^t$	Parpalak–Utkin/Blanc tail; gaps 0, 0
$48 \cdot 2^t$	$768 \cdot 4^t - 1$	Odd count $+24 \cdot 2^t$	BBL tail; gaps 0, 0
$60 \cdot 2^t$	$1200 \cdot 4^t - 10$	$1200 \cdot 4^t + 30 \cdot 2^t - 11$	Compatible Poola refit; gaps 9, 10

Table 7: Actual completed simple families for $t \geq 0$. Gaps compare with simple-model upper floors, not with the unrestricted maximum. The numerical formulas of the middle three rows are inherited.

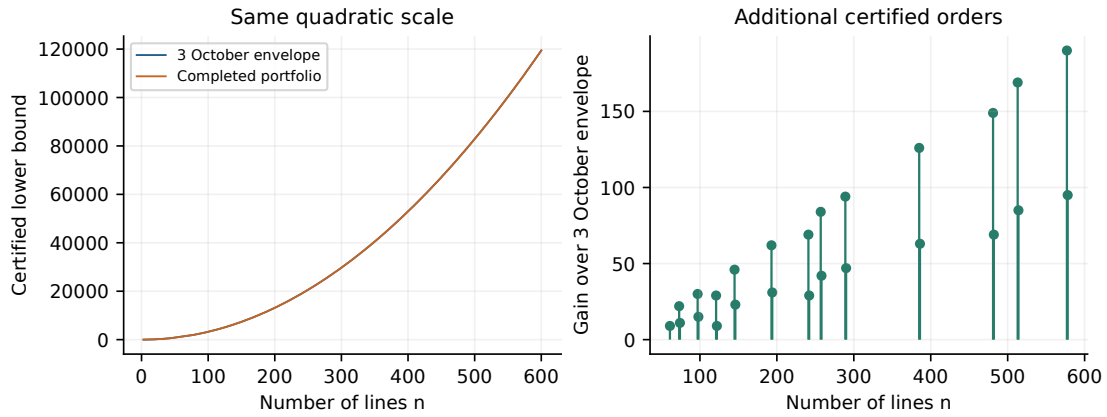


Figure 17: The completed family portfolio compared through exact formula values. Uniform certification adds coverage of known optimal orbits; the 60-dyadic orbit has the separate constant gaps in Figure 16. Taking a maximum retains stronger finite nonsimple witnesses. The figure makes no worldwide record comparison.

Theorem 7.11 (All-natural-order construction portfolio). *For every $n \in \mathbb{N}$, there is an actual real-line arrangement with at least $J(n)$ triangles. In particular $K(n) \geq J(n) \geq H(n)$ and $n^2 \leq 3(J(n) + n)$.*

Proof. Every nonzero candidate has an actual simple witness from the preceding theorems. A zero candidate has the trivial lower witness. A finite maximum selects one valid witness, and $t \leq n$ includes every matching family order. The prior envelope supplies an unrestricted witness if it is stronger; no simplicity is then asserted for that chosen maximum. This is [all_n](#), with pointwise retention in [previous_1e](#). The quality inequality follows from the inherited all-order construction and is proved at [quadratic_lower](#). It does not improve the leading asymptotic coefficient. Q.E.D.

8 Multiplicity-sensitive upper-bound arguments

This section develops upper estimates by separating ordinary crossings from finite multiple points. Actual elementary segments, incidence, clean-line charging and classical simple bounds were completed in the preceding release. The 5–6 October development additionally extracts

cyclic fans, side matching, shared-core components and geometric escape from the real arrangement. The final structural theorems therefore have no supplied fan or aggregate budget premise. Their class restrictions and residual corrections remain explicit; they are not improved unrestricted numerical bounds for K depending only on the line count.

The distinction is necessary because the even-order estimates of Bartholdi–Blanc–Loisel and Blanc require simplicity [1, 2]. The classical Clément–Bader envelope is treated as a reported external theorem [3]; the local example in Section 8.6 explains why we do not silently replace its proof by a bound on all shared edges incident to a multiple point.

8.1 An exact elementary-segment identity

Definition 8.1. For an arrangement of n distinct affine lines, let t_r be the number of finite intersection points incident to exactly r lines, and let P be the number of unordered parallel pairs. A bounded *elementary segment* is the portion of an arrangement line between two consecutive finite intersection points. Let E count these segments, U count those incident to no triangular cell, and D count those incident to two triangular cells. Let a be the number of lines containing at least one finite intersection point, and put

$$S = \sum_{r \geq 3} r(r-2)t_r, \quad \delta = n(n-2) - 3T,$$

where T is the number of bounded triangular cells.

Proposition 8.2 (Segment and defect identities). *For every such arrangement,*

$$E = n(n-1) - 2P - S - a, \tag{70}$$

$$3T = E - U + D, \tag{71}$$

$$\delta = 2P + S + a - n + U - D. \tag{72}$$

If every line meets another line, then $a = n$ and $\delta = 2P + S + U - D$.

Proof. Each nonparallel unordered pair contributes to exactly one finite point, so

$$n(n-1) - 2P = \sum_{r \geq 2} r(r-1)t_r.$$

If a line has $s > 0$ distinct finite intersection points, it contributes $s - 1$ bounded elementary segments; an isolated line contributes none. Summing point-line incidences gives

$$E = \sum_{r \geq 2} rt_r - a.$$

Subtracting these expressions yields the first identity. Each triangular cell uses three elementary segments. A segment is incident to zero, one, or two such cells, so the total number of triangle-side incidences is $E - U + D$. Substitution into the definition of δ gives the last identity. Q.E.D.

The term D cannot be dropped in a nonsimple arrangement. Multiple points reduce the available segment count by S , but sharing can compensate for some of that loss. This is precisely the interaction that a classical upper proof must control.

Lemma 8.3 (A shared side has a multiple endpoint). *An elementary segment incident to two bounded triangular cells cannot have both endpoints ordinary, where an ordinary point is incident to exactly two arrangement lines.*

Proof. At each ordinary endpoint there is a unique supporting line other than the line containing the segment. Both adjacent triangles would therefore have the same three supporting lines. Three distinct lines have at most one bounded triangular region, whereas the two incident cells lie on opposite sides of the shared segment. This is impossible. Q.E.D.

The parity estimates below assume even $n \geq 4$ and pairwise nonparallel lines. The later component statements hold for either parity at $n \geq 2$. Let the *core* consist of the finite points of multiplicity at least three. Write

$$q = \sum_{r \geq 3} t_r, \quad I = \sum_{r \geq 3} r t_r,$$

and let h count the lines meeting the core. Let D_1 count shared elementary segments with exactly one core endpoint and D_2 those with two core endpoints. By Lemma 8.3, $D = D_1 + D_2$, and hence

$$\delta = S + U - D_1 - D_2. \quad (73)$$

8.2 Parity on a line outside the core

A line is *clean* if it contains no core point. The following argument adapts the local parity method in Blanc's Section 2 [2]. Its proof is included because the transverse lines may have multiple intersections away from the clean line.

Lemma 8.4 (Clean-line charging). *For even $n \geq 4$ and pairwise nonparallel lines,*

$$n - h \leq 2U + D_1. \quad (74)$$

Proof. First fix a clean line L , place it horizontally, and number its $n - 1$ crossings from left to right. Let R_i be the transverse line at crossing i . Denote the bounded segment of L between crossings i and $i + 1$ by ℓ_i , and its two exterior rays by ℓ_0 and ℓ_{n-1} . Write p_i^+ and p_i^- for the faces above and below ℓ_i , and r_i^+, r_i^- for the elementary transverse pieces of R_i immediately above and below L . A transverse piece may be a bounded segment or an unbounded ray.

We show that some bounded $r_i^\pm$ is unused or shared. Suppose otherwise: every bounded transverse piece is incident to exactly one triangle. Each bounded ℓ_i is unshared by Lemma 8.3.

If both p_1^+ and p_1^- are nontriangular, then r_1^+ and r_1^- are unused: their other adjacent faces meet the exterior ray ℓ_0 . At least one of these transverse pieces is bounded, because R_1 meets another transverse line. This contradicts the supposition. Reflecting across L if necessary, assume p_1^+ is triangular; p_1^- is then not.

For $1 \leq i \leq n - 2$, prove the following assertions together:

- (i) if i is even, p_i^+ is nontriangular; if i is odd, p_i^- is nontriangular;
- (ii) if both faces at ℓ_{i-1} are nontriangular, then p_i^- is triangular for even i , and p_i^+ is triangular for odd i .

The base case follows from the choice of p_1^+ and the unbounded faces at ℓ_0 . Consider an even step; the odd step exchanges signs. If p_{i-1}^+ is triangular, then p_i^+ cannot be triangular, since otherwise the two triangles share the bounded piece r_i^+ , contrary to the supposition. Assertion (ii) has a false antecedent in this case.

Otherwise both p_{i-1}^+ and p_{i-1}^- are nontriangular by the preceding odd step. This cannot happen for $i = 2$, so $i \geq 4$. The earlier even step makes p_{i-2}^+ nontriangular. Hence r_{i-1}^+ is unused, and must be unbounded. The intersection of R_{i-1} and R_i must consequently lie below L . It follows that r_i^- is bounded. Its left adjacent face is nontriangular, so its right

adjacent face p_i^- must be triangular. Since ℓ_i is unshared, p_i^+ is nontriangular. This proves both assertions.

Now $n - 2$ is even, so p_{n-2}^+ is nontriangular. If p_{n-2}^- is also nontriangular, both pieces at the last crossing are unused and at least one is bounded, again a contradiction. Otherwise p_1^+ and p_{n-2}^- are triangular, while r_1^- and r_{n-1}^+ are unused. The lines R_1 and R_{n-1} intersect. If their intersection is below L , then r_1^- is bounded; if above, then r_{n-1}^+ is bounded. This is the final contradiction.

Charge each clean line to one bounded unused or shared transverse segment supplied by this argument. An unused segment can receive at most two charges, one at each ordinary endpoint. A shared segment must have a multiple endpoint, so it can receive at most one charge if counted by D_1 , and none if counted by D_2 . There are $n - h$ clean lines, giving (74). Q.E.D.

Remark 8.5 (Why parallel pairs have been excluded). Three parallel vertical lines and one horizontal line give a clean horizontal line whose transverse pieces are all unbounded. Thus simply removing parallel-participating lines from the set to be charged does not prove the lemma. The identities of Proposition 8.2 allow parallels, but the charging argument above does not. Every subsequent upper estimate in this section retains the pairwise nonparallel hypothesis.

Combining (73) and (74) gives the basic budget

$$2\delta \geq n + 2S - h - 3D_1 - 2D_2. \quad (75)$$

The remaining problem is to bound the shared-side contribution at the core.

8.3 The geometric obstruction in a cyclic fan

At an r -fold point v , the incident lines determine $2r$ rays in cyclic order, with opposite rays separated by r positions. Let $d_1(v)$ count shared elementary segments starting at v and ending at an ordinary point, and let $d_2(v)$ count those ending at another core point. A ray supports at most one such incident elementary segment.

Lemma 8.6 (No long run of ordinary-ended shared rays). *For $r \geq 3$, no $r - 1$ consecutive rays at an r -fold point can all support segments counted by $d_1(v)$.*

Proof. Such a run would border r consecutive triangular sectors. At the ordinary endpoint of each selected shared segment, the opposite supporting lines of the two neighboring triangles must coincide: besides the radial line through v , only one arrangement line passes through that endpoint. Propagating this equality makes the opposite sides of all r triangles lie on one line M .

The line M would meet $r + 1$ consecutive rays from v , which include an opposite pair. A line meeting both open opposite rays must contain their connecting segment and hence v . But the opposite side of a nondegenerate triangle with vertex v cannot lie on a line through v . This contradiction proves the lemma. Q.E.D.

This is an actual geometric statement, not an assumed combinatorial prohibition. The Lean module `FanGeometry` proves uniqueness and propagation of opposite supports for real lines and points. The module `CyclicFan` supplies the cyclic-to-geometric bridge from triangular sectors, antipodal rays and ordinary selected endpoints.

Proposition 8.7 (Local fan budget). *At every finite point of multiplicity $r \geq 3$,*

$$d_1(v) \leq 2r - 3, \quad 3d_1(v) + d_2(v) \leq 6r - 6. \quad (76)$$

Proof. Count selected rays in all $2r$ cyclic windows of length $r - 1$. Each window contains at most $r - 2$ selected rays by Lemma 8.6, and each selected ray belongs to exactly $r - 1$ windows. Therefore

$$(r - 1)d_1(v) \leq 2r(r - 2).$$

If $d_1(v) \geq 2r - 2$, the left side is at least $2(r - 1)^2 = 2r(r - 2) + 2$, a contradiction. Finally $d_1(v) + d_2(v) \leq 2r$, so

$$3d_1(v) + d_2(v) = 2d_1(v) + (d_1(v) + d_2(v)) \leq 6r - 6.$$

Q.E.D.

The cyclic double count is proved in `FanCount`. The theorem `CyclicFan.selected_card_le` combines it with the real geometric obstruction, without assuming the no-long-run conclusion as an unexplained premise.

8.4 A weighted bound for an arbitrary finite core

Proposition 8.8 (Multiplicity-weighted defect estimate). *Let $n \geq 4$ be even and let the arrangement be pairwise nonparallel. Then*

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 6)t_r. \quad (77)$$

Proof. Each D_1 segment has exactly one core endpoint and each D_2 segment has two. Summing (76) gives

$$3D_1 + 2D_2 \leq 6I - 6q.$$

Also $h \leq I$, since every core line contributes at least one point-line incidence. Inserting both inequalities into (75) yields

$$2\delta \geq n + 2S - 7I + 6q,$$

which is (77) after expanding S, I, q .

Q.E.D.

The local weights are $-9, -6, 1, 12, 27$ at multiplicities $3, 4, 5, 6, 7$, respectively. Consequently the obstruction to recovering a simple-style parity defect is concentrated at triple and quadruple points; higher multiplicity has a positive weight in this accounting.

Corollary 8.9 (Nonnegative core weight). *Under the hypotheses of Proposition 8.8, if*

$$\sum_{r \geq 3} (2r^2 - 11r + 6)t_r \geq 0,$$

then

$$T \leq \left\lfloor \frac{n(n - 5/2)}{3} \right\rfloor.$$

In particular, this holds if every finite multiple point has multiplicity at least five.

Proof. The weighted estimate implies $\delta \geq n/2$, giving the displayed upper bound by the definition of δ . For $r \geq 5$, the exact factorization

$$2r^2 - 11r + 6 = 1 + (r - 5)(2r - 1) \geq 1$$

proves the stated sufficient condition.

Q.E.D.

This extends the same polynomial to a restricted nonsimple class. It says nothing comparable for a general core consisting of many triple points. Numerical first priority for this restricted consequence has not been established.

8.5 A sharper estimate with at most two core points

Lemma 8.10 (A fan with at most one core-to-core ray). *If $d_2(v) \leq 1$ at a point of multiplicity $r \geq 3$, then $d_1(v) \leq 2r - 4$.*

Proof. Mark which of the $2r$ sectors at v are triangular. A ray supports a shared elementary segment exactly when both neighboring sectors are triangular. If every sector is triangular, every ray is shared; deleting at most one d_2 ray leaves a run of at least $2r - 1$ selected rays, contradicting Lemma 8.6. Hence a nontriangular sector exists.

Suppose $d_1(v) \geq 2r - 3$. Two separated nontriangular sectors exclude at least four shared rays, as do three consecutive nontriangular sectors. Both situations are impossible. If exactly one sector is nontriangular, the shared rays form a path of length $2r - 2$. Deleting at most one d_2 ray leaves at most two selected runs. Their total capacity is $2(r - 2)$, less than the required $2r - 3$. If exactly two adjacent sectors are nontriangular, at most $2r - 3$ shared rays remain. All would have to be selected, forming a run of length $2r - 3 > r - 2$. These cases exhaust the possibilities and prove the lemma. Q.E.D.

Theorem 8.11 (At most two finite multiple points). *An arrangement of an even number $n \geq 4$ of pairwise nonparallel affine lines with at most two finite multiple points satisfies*

$$T \leq \left\lfloor \frac{n(n - 5/2)}{3} \right\rfloor + 1. \quad (78)$$

More precisely, the defect bounds are $\delta \geq n/2$ if $q = 0$, $\delta \geq n/2 - 1$ if $q = 1$, and $\delta \geq n/2 - 3$ if $q \leq 2$.

Proof. With at most two core points, at most one ray at a core point can terminate at another core point. Lemma 8.10 gives $D_1 \leq 2I - 4q$. There is at most one core-to-core shared segment, so $D_2 \leq q(q - 1)/2 \leq 1$. Moreover,

$$h \leq I - D_2,$$

because any such segment lies on a line counted twice in the core-incidence total and once among the core lines. Substitution in (75) yields

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 12)t_r - D_2.$$

For $r \geq 3$,

$$2r^2 - 11r + 12 = -3 + (r - 3)(2r - 5) \geq -3.$$

Thus $2\delta \geq n - 3q - D_2 \geq n - 7$. Since n and 2δ are even, $2\delta \geq n - 6$. Rearranging gives (78). If $q = 1$, then $D_2 = 0$ and the inequality $2\delta \geq n - 3$ rounds to $2\delta \geq n - 2$; if $q = 0$, it gives $2\delta \geq n$ directly. Q.E.D.

The hypotheses are attained by externally attributed exact constructions. Table 8 records examples whose pairwise nonparallel status, triangle count, and two triple points were checked exactly. Their equality with (78) proves sharpness within this restricted class, not optimality over all classical arrangements.

ProofAtlas reports a stronger restriction-specific result at $n = 14$, covering up to four finite multiple points [25]. Its proof was not independently reproduced here. Our two-point estimate covers all even orders and arbitrary core multiplicities, but neither that fact nor the table establishes priority for the restricted theorem.

n	T	Restricted upper	Coordinate provenance
8	15	15	Earlier construction, retained in the current gallery
14	54	54	Maiorana, two-triple-point witnesses
26	204	204	Parpalak–Utkin gallery
50	792	792	Parpalak–Utkin gallery

Table 8: Attainment in the class covered by Theorem 8.11. Sources: Maiorana [18], Parpalak and Utkin [23]. These are not new numerical records claimed by this paper.

8.6 A checked shared-fan example and the limits of smoothing

Consider the six lines

$$x = 0, \quad y = 0, \quad x + y = 3, \quad x - y = 0, \quad 2x + y = 3, \quad x + 2y = 3. \quad (79)$$

They are the three sides of a triangle and its three medians. There are six bounded triangular cells and four triple points. At $v = (1, 1)$, six distinct nonzero elementary radial segments are shared sides, each belonging to two of those cells. Three have ordinary other endpoints and three have multiple other endpoints, giving $d_1(v) = d_2(v) = 3$; globally $D_1 = D_2 = 3$.

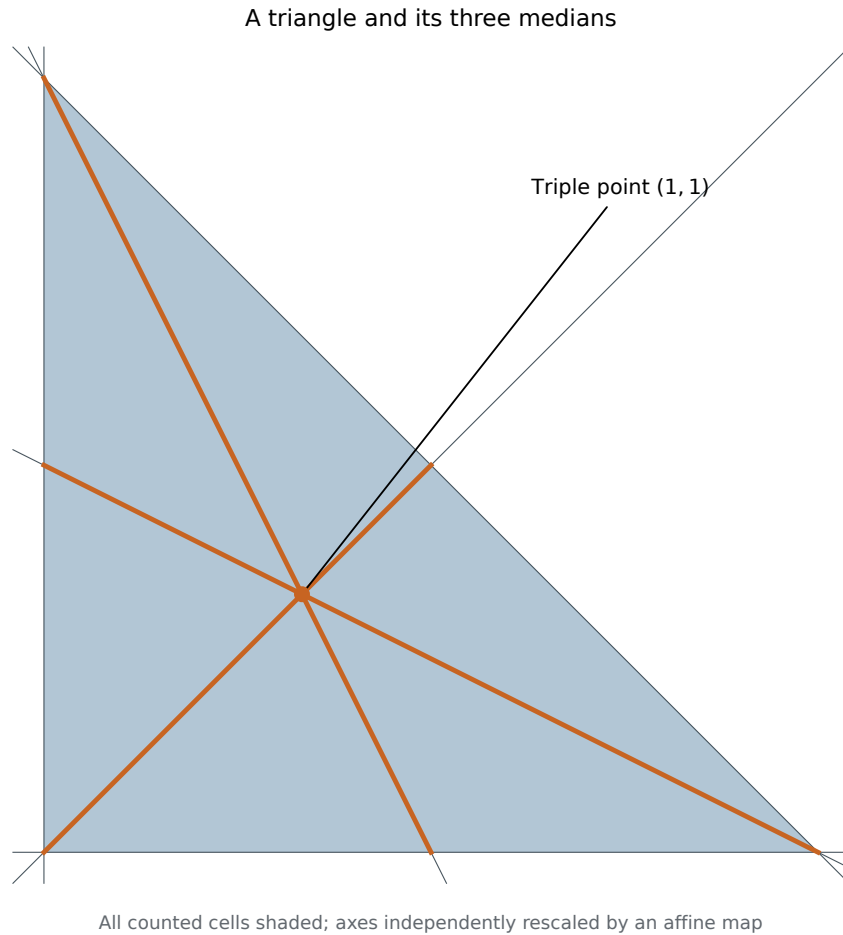


Figure 18: The exact arrangement (79). Six triangular cells meet around the central triple point, and all six radial sides are shared. This refutes only a literal local incidence reading of the shared-pair statement in Clément–Bader’s Lemma 1; it does not refute their final upper theorem. The figure is newly generated from the integer equations.

The **SharedFan** module proves the real geometric facts: each designated triangle has nonempty uncut interior, the six interiors are pairwise disjoint, the radial segments are distinct and elementary, and each is a side of two distinct triangles. The proof uses ordinary kernel checking.

This matters when interpreting the Clément–Bader proof. Its local shared-pair sentence cannot literally bound all shared segments incident to a multiple point by two. An intended assignment of each shared side to one endpoint would be a different assertion requiring global justification. The example has only six triangles, below the reported seven-triangle upper bound at order six; it is therefore not a counterexample to that upper theorem. The present paper leaves that distinction explicit and uses the separate D_1, D_2 accounting instead.

Nor can one justify a simple upper bound merely by perturbing every multiple point. For a retained two-triple-point Maiorana 14:54 witness, the four local generic resolution types have triangle counts 52, 52, 53, 52. These counts were checked with exact arithmetic; they are finite experimental evidence, not a new Lean theorem classifying all perturbations. They demonstrate why a universally nondecreasing smoothing argument needs more than the existence of a small generic perturbation. Our global upper argument never assumes such monotonicity.

8.7 Actual arrangement extraction and simple upper theorems

The formal inventory is now built from an actual finite injective family of certified triangles in $n \geq 2$ pairwise nonparallel lines. It may be a proper subset of all triangular cells. The quantities U, D_1, D_2 are defined relative to this selected family. The cell theorem proves interior disjointness, and the side-emptiness theorem proves that each side has consecutive endpoints on its supporting line.

An unordered endpoint pair represents one geometric side. At most two pairwise interior-disjoint nondegenerate triangles can share that side: three would put two on the same side of its support, forcing their interiors to overlap near an interior base point. Counting triangle-side incidences therefore gives $3T = E - U + D_1 + D_2$. Pair counting proves $\sum r(r-1)t_r = n(n-1)$ and the actual consecutive-edge inventory gives $E = n(n-2) - S$. Thus

$$\delta = S + U - D_1 - D_2 \tag{80}$$

is proved from geometry by `certificate_defect_identity`; capacity, incidence and edge cardinality are conclusions of the proof, not assumed numerical identities.

Theorem 8.12 (Actual clean-line charging and simple bounds). *Let $n \geq 4$ be even. For any pairwise nonparallel real arrangement and finite injective certified triangle family,*

$$n - h \leq 2U + D_1. \tag{81}$$

For every simple arrangement with T certified triangular cells,

$$3T \leq n(n-2) \quad (n \geq 2), \quad 6T \leq n(2n-5) \quad (n \geq 4 \text{ even}). \tag{82}$$

Proof. The parity proof of Lemma 8.4 can also be organized as an odd-cardinality matching contradiction, which is the formal route. A clean line has exactly $n-1$ distinct crossings. One common open half-plane contains a further crossing on every transverse support. Choose in that half-plane the consecutive bounded segment incident to each clean-line crossing. Existence is obtained from the nearest actual intersection, with the bounded-side and uniqueness lemmas handling both signs.

If every chosen edge had degree one in the certified triangle family, its unique triangle would have its base on the clean line: at an ordinary crossing a triangle must use one of

the two incident supports. Consecutive edge uniqueness and interior disjointness then make these bases a disjoint two-element cover of the clean-line crossings. Such a cover has even cardinality, contradicting $n - 1$ odd. Hence some incident bounded edge has degree zero or two. A degree-two edge has a multiple other endpoint by the shared-side lemma, so it is counted by D_1 .

Choose one resulting charge for each clean line. A fixed ordinary endpoint of an edge determines at most one clean transverse line. An unused edge has at most two such endpoints and receives at most two charges; a D_1 edge has exactly one ordinary endpoint and receives at most one. This proves (81). The extraction, choice and fiber bounds are all in `certificate_clean_line_budget`, using `UpperCleanHalfplane`, `UpperCleanEdges`, `UpperCleanPairing` and `UpperCleanCover`. No charge-existence or pairing premise remains.

For a simple arrangement the core is empty, so $S = D_1 = D_2 = h = 0$. Equation (80) gives $3T = n(n - 2) - U$, proving the first bound. In the even case $n \leq 2U$, so $6T = 2n(n - 2) - 2U \leq n(2n - 5)$. The `SimpleLowerBound` versions extract the injective finite family from its duplicate-free triangle list. These are `simple_lower_bound_upper` and `simple_lower_bound_even_upper`. Q.E.D.

The integer floor forms follow immediately. These classical inequalities retain their earlier attribution, including Blanc’s even parity bound. Their contribution here is a complete real-geometric formalization with standard logical axioms only. In particular, the retained simple 44 : 608 certificate attains $\lfloor 44(88 - 5)/6 \rfloor = 608$; this proves simple optimality at that order, not unrestricted optimality. Theorem 8.12 does not justify applying the even polynomial to nonsimple witnesses such as 14 : 54.

8.8 Extremal fans and strengthened local restrictions

Theorem 8.13 (Equality structure of a supplied cyclic fan). *Consider actual local sector data at an r -fold point, $r \geq 3$, satisfying the cyclic order, antipodality, triangle and ordinary-endpoint conditions of `UpperFan.Sectors`. If $d_1 = 2r - 3$, every sector is triangular and $d_2 = 3$. Consequently,*

$$d_2 \leq 2 \implies d_1 \leq 2r - 4, \quad 3d_1 + d_2 \leq 6r - 8 + 2e, \quad (83)$$

where $e = 1$ when $d_1 = 2r - 3$ and $e = 0$ otherwise.

Proof. The no-long-run property from Lemma 8.6 holds for the selected ordinary-ended shared rays. If a sector were missing, both bounding rays would be absent from that selected set. The cyclic window count with these adjacent omissions strengthens its cardinality bound to $2r - 4$. Thus the extremal case has every sector present; all $2r$ rays are then shared. Subtracting $d_1 = 2r - 3$ gives $d_2 = 3$. If $d_2 \leq 2$, equality in the old bound is therefore impossible. In the nonextremal case, $3d_1 + d_2 = 2d_1 + (d_1 + d_2) \leq 2(2r - 4) + 2r$; in the extremal case its value is $6r - 6$. These give (83). The exact proofs are `all_sectors_of_extremal`, `core_shared_card_eq_three_of_extremal` and `weighted_local_bound`. Q.E.D.

This strengthens the previous manuscript’s local hypothesis $d_2 \leq 1$ to $d_2 \leq 2$. The formerly supplied local interface is now extracted from every actual core, as described below. A supported extreme core cannot carry the full extremal fan when its core-ended rays are identified with points in the supporting closed half-plane (`UpperFanSupport` and `UpperCoreBoundary`). Also, two full extremal triple fans cannot be adjacent across the same matched elementary side with its two neighboring certified triangles (`UpperTripleFan`). The latter uses the actual coincident endpoints and opposing adjacent sectors; an arbitrary pair of abstract

fan records need not supply that matching. The completed arrangement wrapper derives it from the original selected triangle occurrences, rather than assuming compatible abstract records.

8.9 Actual fan extraction, matching and global assembly

At each core the development orders the actual supporting directions by an admissible affine chart, appends their opposite representatives, and obtains a cyclic list of $2r$ nonzero rays. Consecutive selected triangle sectors determine actual side endpoints. Simplicity of the elementary segment inventory identifies a shared ray with its unique ordinary or core endpoint, and injectivity of the selected triangle labels identifies its two neighboring cell occurrences. Thus the local ordinary and core ray counts equal $d_1(v)$ and $d_2(v)$, rather than supplied numerical surrogates. The actual local bound is `certificate_local_fan_bound`; matching along a shared elementary edge is `certificate_neighbor_matching`. The geometric no-long-run and extremal conclusions above now apply to these extracted records at every core.

Double counting their actual side incidences gives $\sum_v d_1(v) = D_1$ and $\sum_v d_2(v) = 2D_2$. In particular, the weighted sum and exceptional-fan corrections no longer require a global fan premise. The result is assembled in `certificate_weighted_fan_sum` and `certificate_defect_even`. This completes the formal geometric scope of Proposition 8.8 and its higher-multiplicity corollary. The previously conditional refined budget is now an actual theorem:

$$2\delta \geq n + 2S - 7I + 8q - 2e + D_2, \quad (84)$$

where e counts full extremal fans with ordinary degree $2r - 3$. The stronger boundary-sensitive assembly proves the two-core theorem and the three-core estimate with their actual geometry, in `certificate_three_core_upper` and `certificate_two_core_upper`. The former gives, when $q \leq 3$ and $n = 2m$,

$$\delta \geq m - 6, \quad T \leq \lfloor m(4m - 5)/3 \rfloor + 2, \quad (85)$$

equivalently $6T \leq n(2n - 5) + 12$. The retained labels record the historical conditional statements; their arrangement extraction is now complete. For $q \leq 1$, or $q \leq 2$ with at most one triple core, the sharper rounding is $6T \leq n(2n - 5) + 2$, proved by `UpperOpenMathSparseUpper`. These are restricted numerical consequences at every even order.

Several independent exact resources retain geometry that earlier budgets discarded. Let C count core endpoints of bounded nonshared elementary sides, and R_c count unbounded rays at cores. Actual first/last vertex flags give

$$D_1 + 2D_2 + C + R_c = 2I, \quad (86)$$

$$2\delta = 2 \sum_{r \geq 3} r(r - 3)t_r + 2U + C + R_c - D_1. \quad (87)$$

The source `UpperOpenMathRayResources` proves both identities. The actual shared-core graph has core vertices and D_2 edges, including isolated vertices in its component partition. Its component hull points consume at least two nonshared or ray resources each. If b_h is the sum of the numbers of supporting hull points of the individual components, then the completed hull estimate is

$$2\delta \geq 2U + 2 \sum_{r \geq 3} r(r - 4)t_r + 3q + 3b_h. \quad (88)$$

It follows from separately bounding ordinary sharing and total sharing at each supported core; see `certificate_component_hull_deficit`. If all core multiplicities are at least $R \geq 4$, this gives $6T + (2R(R - 4) + 3)q + 3b_h + 2U \leq 2n(n - 2)$. The cases $R = 4, 5$ retain penalties $3q, 13q$, respectively. These are structural estimates, not a formula for the unrestricted maximum.

8.10 Degree-free structural bounds and their local counts

All statements in this subsection concern $n \geq 2$ pairwise nonparallel real lines and an arbitrary finite injective family of certified triangular cells. All core, side and component counts are extracted from that family. At a core put $a = d_1$, $d = d_2$. A *marked port* is a shared core ray whose two neighboring shared rays end at ordinary vertices. Let m be the number of such ports. In the all-triple class each marked edge reaches a poor fan, with $a \leq 1$ and $a + d \leq 2$, and a shared edge receives at most one source marker. Define

$$\begin{aligned}\Delta &= \sum_{r \geq 4} r(r-4)t_r, \\ A_{\text{all}} &= \#\{\text{triple cores with } (a, d) = (2, 4)\}, \\ A_0 &= \#\{\text{triple cores with } (a, d, m) = (2, 4, 0)\}, \\ B_{15} &= \#\{\text{triple cores with } (a, d) = (1, 5)\}.\end{aligned}$$

Use A_s, B_s for the corresponding unmarked two-cap and one-cap counts in component s . The following component bound has no degree, size or multiplicity restriction.

Theorem 8.14 (Mixed and marked component defects). *For arbitrary core multiplicities,*

$$\delta + A_{\text{all}} + \sum_s \lceil B_s/2 \rceil \geq U + c + \Delta. \quad (89)$$

Its complementary doubled form is $2\delta + 2A_{\text{all}} + B_{15} \geq 2U + c + 2\Delta$. If every core is triple, the first formula strengthens to

$$\delta + A_0 + \sum_s \lceil B_s/2 \rceil \geq U + c. \quad (90)$$

Proof. The actual cap budgets, and in the all-triple case the marked-edge budgets, pay extremal and partial sources against poor endpoint capacities. Higher multiplicity pays through its quadratic surplus, without assuming that higher extremal fans are independent. On each nonempty closed component the adjusted paid cost is strictly positive. Equality in the finite budget restricts the local fan types: balanced two-cap fans, marked full two-cap fans and full zero-cap fans remain, with the residual full types added in the later paid equality case. A closed balanced subset forces a three-cycle by matching its actual triangle identities. The third normalized cap axis would have to contain a point beyond both endpoints of one elementary core side, which is impossible.

The remaining full-fan cluster cannot be finite. A full zero-cap center is surrounded by cluster neighbors; a marked two-cap center has an actual opposite continuation through a poor fan to another cluster point. A point maximizing squared norm in this finite cluster cannot satisfy either property. In the mixed case a finite supporting line excludes the closed higher/extremal cluster, and the normalized remaining triple mixture is handled by the same occurrence matching. Integer rounding of the strictly positive paid cost gives one unit per component. The final roots are `certificate_mixed_curvature_component_bound` and `certificate_component_half_penalty_bound`. No planar graph, supplied fan, convex-hull oracle or unproved aggregate identity is a hypothesis of these arrangement wrappers. Q.E.D.

The residual corrections cannot simply be dropped. Actual unmarked full $(2, 4)$ fans exist; full $(1, 5)$ fans may be adjacent. In particular, exact rational witnesses have an unmarked $(2, 4)$ core adjacent to a marked $(2, 4)$ core at eight lines and twelve triangles, and to a full $(1, 5)$ core at nine lines and sixteen triangles. Therefore an independence argument for all negative local types would be false.

8.11 Recipient geometry and the final all-triple gain

Let an *antipodal source* mean a core counted by A_0 . Its two ordinary shared rays are opposite. Actual triangle matching proves that two such sources cannot share a side, nor can a source meet an unmarked balanced $(2, 2)$ fan. Each source has at least two nonfull neighbors; its two cap flanks cannot both meet full triple fans. Two distinct sources have at most two common actual core neighbors. The latter theorem allows arbitrary multiplicities at the other cores: opposite-pair extraction would otherwise force an intervening vertex on an elementary side or crossing strict cevians inside such sides. This is [certificate_common_neighbor_card_le_two](#).

In the all-triple class, a recipient of total shared degree at most four has at most two source neighbors. For a one-cap $(1, 3)$ recipient, complete extraction of its five-sector chart strengthens this to at most *one*. Adjacent source tips contradict source independence; the remaining six tip placements are excluded by support exhaustion, ordinary-tip identification and opposite-ray continuation, with affine area signs handling the separated positions. The actual endpoint is [certificate_n13_no_two_anti_neighbors](#). A $(1, 2)$ recipient similarly has a three-shared-ray run: two sources would be adjacent or separated by the ordinary middle ray, whose continuation is impossible. See [certificate_n12_anti_degree_le_one](#). A partial $(2, 1)$ recipient has no source neighbor, because its only core edge reaches a poor fan while a source has ordinary shared degree two. An actual $(1, 3)$ recipient with a source neighbor also cannot maximize an affine functional injective on a finite closed shared-core set containing it. The extracted support contradiction is [certificate_n13_with_anti_not_max](#). This restricts exposed equality profiles but does not by itself prove strict positivity of every component.

Define the positive local counts

$$\begin{aligned} E_3 &= \#\{a = 3\}, & P_{21} &= \#\{(a, d) = (2, 1)\}, \\ N_{12} &= \#\{(a, d) = (1, 2)\}, & M_{22} &= \#\{(a, d) = (2, 2), m \geq 1\}. \end{aligned}$$

These definitions in this subsection range over actual triple cores.

Theorem 8.15 (Source-free triple defect with marked balanced gain). *If every actual core has multiplicity three, then*

$$2\delta + B_{15} \geq 2U + 3(E_3 + P_{21}) + N_{12} + M_{22}. \quad (91)$$

Equivalently, $6T \leq 2n(n - 2) + B_{15} - 2U - 3(E_3 + P_{21}) - N_{12} - M_{22}$. There is no degree, component-size or source-count restriction.

Proof. The finite unit discharging rule charges antipodal-source edges against the actual marked-port and recipient capacities. The $(1, 3)$ restriction removes its former double-recipient correction completely; the $(1, 2)$ restriction leaves one positive unit, and the $(2, 1)$ exclusion increases its coefficient to three. These yield the completed N12 theorem [certificate_n12_source_free_defect_bound](#).

For the late additional term, marked outgoing edges and incoming antipodal-source edges have disjoint opposite endpoint types. Both sets lie in the actual incident core-edge inventory. Thus at every core

$$m(v) + \deg(A_0, v) \leq d(v). \quad (92)$$

At a marked balanced core the local discharging weight is zero but $m \geq 1$; (92) gives $m + x \leq 2$, hence $2m - x \geq 1$. This retains the extra unit counted by M_{22} . The local extraction is [certificate_marked_plus_anti_degree_le](#), and the global actual wrapper is [certificate_m2_2_source_free_defect_bound](#). The finite weights, recipient geometry and actual wrappers all use only standard logical axioms. Q.E.D.

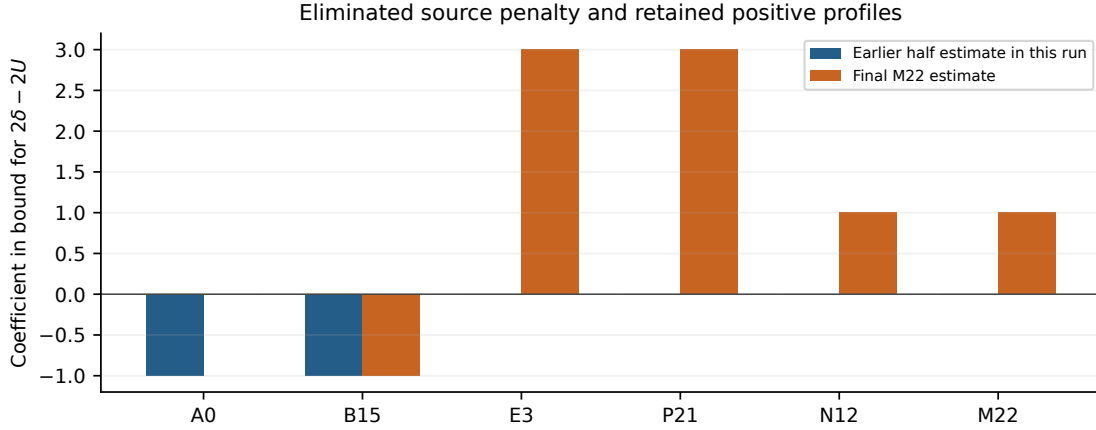


Figure 19: Coefficients in the earlier half-curvature inequality and the final all-triple global estimate developed during 5–6 October. Both are structural bounds for actual nonparallel arrangements. The bars compare algebraic charges of local types; they do not assert that an arbitrary collection of those types is geometrically realizable, and are not a comparison with an unrestricted numerical upper bound.

The estimates developed on the way are retained as independently checked intermediate resources. The half-curvature rule is $2\delta + A_0 + B_{15} \geq 2U$; its first positive refinement is $4\delta + 2A_0 + 2B_{15} \geq 4U + 6E_3 + 5P_{21}$, improved to coefficient six for P_{21} after partial-recipient exclusion. The double-recipient rule $2\delta + B_{15} + J_{13} \geq 2U + 3E_3 + 2P_{21}$ used the exact count of (1,3) double receivers; actual geometry now proves $J_{13} = 0$. The final global estimate dominates these global unit/half gains but retains B_{15} . No unrestricted upper polynomial follows by omitting it.

Theorem 8.16 (Componentwise N12 portfolio). *Under the all-triple hypothesis, let A_s, B_s, E_s, P_s, N_s be the corresponding counts in the actual component s . Then*

$$\delta \geq U + \sum_s \max \left\{ 1 - A_s - \lceil B_s/2 \rceil, \left\lceil \frac{3E_s + 3P_s + N_s - B_s}{2} \right\rceil \right\}. \quad (93)$$

This score dominates the preceding half-gain and source-free component scores. If each component satisfies $B_s + 1 \leq 3E_s + 3P_s + N_s$, then $\delta \geq U + c$.

Proof. Localize the actual discharging and strict paid-cost arguments to each shared-core-closed component. The component contribution H_s to $\delta - U$ is integral. The paid theorem gives $H_s \geq 1 - A_s - \lceil B_s/2 \rceil$; the source-free N12 theorem gives the other rounded score. Take their maximum *before* summing over components. Actual component partition and side double counting prove $\delta - U = \sum_s H_s$. The final component endpoint is `certificate_n12_source_free_hybrid_component_defect`; its dominance is proved in the same module. The sufficient positive profile is `certificate_positive_profile_component_defect`. Q.E.D.

The M_{22} strengthening in (91) is currently a *global* theorem. Equation (93) is the separately completed component portfolio; no componentwise M22 extension is claimed.

8.12 Restricted graph classes and multiplicity consequences

The cap budgets themselves require no graph restriction. In the all-triple class they give

$$D_1 + 2E_3 + P_{21} \leq 2q, \quad \delta \geq U + q - D_2 + 2E_3 + P_{21},$$

in `certificate_triple_cap_budget`. For arbitrary multiplicities, define $\mathcal{A} = \sum_{r \geq 4} (r(r-4) + 1)t_r$ and $P_{\text{poor}} = \sum_{v \text{ poor triple}} d(v)$, where poor means $a \leq 1$ and $a + d \leq 2$. The mixed weighted budget is

$$3D_1 + 3q + 3\mathcal{A} + 2P_{\text{poor}} \leq 3S, \quad 3\delta \geq 3(U + q - D_2 + \mathcal{A}) + 2P_{\text{poor}}.$$

It is `certificate_mixed_weighted_budget`. An order-four, -five or -six core contributes respectively one, six or thirteen to $\mathcal{A}$. These contributions pay triple sources without assuming that higher extremal fans are independent.

Theorem 8.17 (Component and forest classes). *For the same actual arrangement and triangle-family semantics:*

- (i) *If every core has shared-core degree at most three, then $\delta \geq U + c + \Delta$, with arbitrary multiplicities.*
- (ii) *At degree at most four, the bound is $\delta + A_{\text{all}} \geq U + c + \Delta$.*
- (iii) *If each actual component has at most five vertices, then $\delta \geq U + c + \Delta$, with arbitrary total core count and multiplicities.*
- (iv) *For an actual forest, put $\mathcal{A} = \sum_{r \geq 4} (r(r-4) + 1)t_r$ and let $P_{\text{poor}} = \sum_{v \text{ poor triple}} d(v)$. Then $3\delta \geq 3(U + c + \mathcal{A}) + 2P_{\text{poor}}$. For an all-triple forest the separate cap budget gives $\delta \geq U + c + 2E_3 + P_{21}$.*

Proof. At degree at most three, adjusted component cost is nonnegative. Its zero case leaves a closed extremal cluster or a balanced two-cap cluster; finite supporting geometry excludes the former and actual cycle geometry the latter, giving strict integer cost. The degree-four version retains full two-cap corrections. In a component of at most five vertices, a full two-cap center and its four neighbors exhaust the component. Opposite radial segments and ordinary cap tips block outer diagonals. The remaining endpoint weights pay more than the center's negative weight; mixed outer multiplicities supply their adjusted surplus. Both antipodal and non-antipodal full two-cap shapes are handled. These are `certificate_mixed_degree_three_component_bound`, `certificate_mixed_degree_four_component_bound`, and `certificate_mixed_degree_five_component_bound`.

For a forest, the actual component trees prove $D_2 + c = q$. The mixed weighted cap budget $3\delta \geq 3(U + q - D_2 + \mathcal{A}) + 2P_{\text{poor}}$ gives the result. The all-triple budget retains $D_1 + 2E_3 + P_{21} \leq 2q$ before substitution in the exact defect identity. The forest endpoints are `certificate_mixed_forest_weighted_bound` and `certificate_triple_forest_bound`. No degree bound is required for the forest statements. Q.E.D.

A closed all-triple component of at most seven vertices contains at most one antipodal source: two nonadjacent sources need two four-element neighbor sets among at most five other vertices, contradicting the at-most-two-common-neighbor theorem. The single-source charging rule gives $\delta \geq U + \sum_s \lceil (3E_s + 2P_s - B_s)/2 \rceil$ in that class; an adaptive maximum retains the stronger rule separately in each applicable component. This does not remove B_s or prove $\delta \geq U + c$ for all seven-core components. The final unrestricted N12 portfolio is retained alongside these earlier source-sensitive results.

The mixed high-multiplicity consequences also have actual formal proofs. All multiplicities at least four imply $3T \leq n(n-2)$. At even order, multiplicities at least six give $6T + 6q \leq n(2n-5)$; the sharper five-or-higher boundary theorem gives $12T + 5q + \min(q, 3) \leq 2n(2n-5)$. The proofs are in `UpperOpenMathMultiplicity`, `UpperOpenMathHighMultiplicityHull` and `UpperOpenMathBoundaryBudgets`. These restricted numerical formulas are not replacements for control of triple and quadruple cores in the unrestricted problem.

8.13 Parity tokens, equality attribution and remaining limits

The new parity extraction can charge every cut line in an even all-triple arrangement, rather than only clean lines. Ordinary crossings there have odd cardinality. The base cover either fails degree one on a transverse segment or ends a positive triangle base at a core. Separately accounting for these two modes gives the safe actual bound

$$n \leq 2U + 2D_1 + C \quad (n \geq 4 \text{ even, all cores triple}). \quad (94)$$

Its proof is in `UpperOpenMathOrdinaryGlobalCharge`. The second D_1 is necessary: a four-line pencil-plus-cap witness has $U = 0, D_1 = 1, C = 2$, so replacing $2D_1$ by D_1 would give the false $4 \leq 3$. The conjectural trade-off $n \leq 2U + D_1 + R_c + 2$, together with $C \geq 2D_1$, would yield the corrected even bound $6T \leq n(2n - 5) + 2$ in the all-triple class. Neither conjectural resource rule is assumed here. Without the correction two, the known 38 : 450 witness has token total 36; with correction two the rule still fails for mixed multiplicities, where a 14 : 52 witness has shortfall three.

Clément–Bader already stated that exact Tamura equality forces a simple configuration [3]. The quantitative penalties and restricted equality corollaries above are independently checked mechanisms in explicit scopes, not a claim of first discovery of that broad conclusion. The wrong-ray/shared-apex mechanism also has prior local arguments in Srirangam’s notes [30]; their actual fan extraction, extended partial-fan scopes and global weighted assembly are separate developments here. Source priority for the quantified component penalties requires further comparison.

An exact phase-zero Füredi–Palásti witness with eight lines has seven triple cores, fourteen triangles and fifteen shared sides, refuting the auxiliary general shortcut $D \leq 2q$. At eighteen lines its exact profile is $q = 46, D_1 = 15, D_2 = 105$. An eleven-line witness has adjacent order-five extremal full fans, so higher extremal independence is also false. These retained exact external counterexamples invalidate specified charging shortcuts, not the final numerical conjectures in which those shortcuts were used. A seven-vertex kernel-checked abstract half-charging example is sharp for its finite relaxation but unrealizable: its two sources would have the same four neighbors. It therefore identifies missing geometry rather than sharpness for real Kobon arrangements.

A finite projective chart theorem now removes parallel pairs while preserving a chosen finite set of actual bounded triangles. It requires distinct projective lines and nonparallel supporting pairs for each triangle; the old polynomial predicate alone is not used on a parallel support pair. The endpoint `arbitrary_iff_lowerBound` links the arbitrary and nonparallel lower-witness models. Class properties and local profile counts must still be checked in the transformed chart; this does not silently extend an all-triple theorem to every parallel arrangement. The remaining general upper task is mathematical control of residual profiles and the ordinary-end token trade-off, rather than unfinished extraction of the fans already used above.

9 Exact translation germs and local resolution losses

A small perturbation of a multiple point can create a tiny triangle and destroy distant old cells simultaneously. This section gives an exact description of that wall crossing. It does not assume that smoothing preserves or increases the triangle count.

9.1 Simultaneous classification by affine coefficients

Translate one indexed line parallel to itself, writing its equation as $a_px + b_py = c_p + \varepsilon$ and leaving the others fixed. Pairwise nonparallelism is preserved. Every determinant nondegeneracy test

and oriented half-plane test in the triangle certificate has the form $u + v\varepsilon$: multilinearity and the single changed constant term exclude higher powers. For positive sufficiently small ε ,

$$u + v\varepsilon \geq 0 \iff (u > 0) \text{ or } (u = 0 \text{ and } v \geq 0),$$

with analogous rules for nonpositivity and nonzero tests. A *triangle germ* is the static conjunction of these coefficient conditions for one ordered supporting triple.

Theorem 9.1 (Exact eventual translation count). *For any finite pairwise nonparallel arrangement and one translated line, there is $\eta > 0$ such that, simultaneously for every supporting triple and every $0 < \varepsilon < \eta$, the moved triangle predicate is equivalent to its germ. Let $T(0)$ be the old count, and let born and lost count germs absent from the old triangle set and old triangles absent from the germ set, respectively. Then throughout that interval,*

$$T(\varepsilon) + \text{lost} = T(0) + \text{born}. \quad (95)$$

Proof. For each affine test with nonzero constant coefficient, choose a positive threshold preserving its sign; zero constants are decided exactly by the linear coefficient. There are finitely many tests, supporting triples and transverse lines. Intersect their positive neighborhoods to obtain one simultaneous threshold. Nondegeneracy and the common half-plane tests then give exactly the original geometric certificate predicate. The finite set identity follows by partitioning both the old and germ sets into their intersection and their differences. These are [simultaneous_triangle_stability](#) and [translated_birth_loss_identity](#). The lower-bound wrappers supply actual real-line witnesses from germ counts, rather than treating a formal count as a geometric oracle. Q.E.D.

Explicit corner-incidence conditions make every old triangle's zero side tests stationary; their other signs persist. This proves no loss and a one-triangle gain when a new germ is present, in [germ_gain_one](#). If there is exactly one concurrent supporting triple and no other line through its center, translating one of its supports creates exactly one tiny-triangle germ. Other new germs would have to be old degenerate supporting triples and are excluded by uniqueness. Consequently

$$T(\varepsilon) + \text{lost} = T(0) + 1. \quad (96)$$

This is [isolated_resolution_identity](#), using the local birth theorem in `OpenMathTripleBirth`. The loss set remains the *global* germ difference. These theorems use only the standard logical axioms and permit arbitrary finite order.

9.2 A four-line counterexample to a stated half-plane yield

Liu's versioned desingularization preprint, Theorem 1.1 on printed page 2, states a formula $T(A') = T(A) + 1 - \tau_i$, with τ_i defined by old triangle interiors lying in the selected open half-plane [17]. Its hypotheses include one isolated triple point and otherwise ordinary vertices. The following example satisfies those hypotheses:

$$y = 0, \quad y = x, \quad y = -x, \quad 2x + y = 2. \quad (97)$$

Index these supports 0, 1, 2, 3. The old triangular cells have supports 013 and 023, and both interiors satisfy $x - y > 0$. Move support 1 to $y = x + \varepsilon$, so its motion is into $x - y < 0$. The stated half-plane count is zero. Yet for every $0 < \varepsilon < 1$ the moved arrangement has exactly two triangles, with supports 012 and 023. One old cell is lost and one is born; the proposed yield would instead predict three triangles.

Motion	Old count	Lost	Born	Moved count
Four-line positive resolution	2	1	1	2
Six-line upward resolution	6	0	1	7
Six-line downward resolution	6	3	1	4

Table 9: Complete real-parameter interval counts for the displayed arrangements. Birth and loss refer to supporting-triple sets, and the interval conclusions are Lean theorems.

This is an interval counterexample within the formula’s stated scope, not a criticism of unrelated results in the preprint. Completeness of the old and moved support lists, the unique triple, the whole parameter interval, and positivity throughout the old triangle interiors are proved in [halfplane_yield_fails](#). The exact germ identity (95) accounts for the loss without replacing it by the half-plane count.

9.3 Three losses and a reversed gain of two

A second arrangement is

$$y = 0, \quad y = x, \quad y = -x, \quad x + 2y = 3, \quad -2x + y = 2, \quad x + 4y = -4. \quad (98)$$

It has one concurrent triple and six old triangles. For every $0 < \varepsilon < 1/100$, replacing the first line by $y = \varepsilon$ retains all six and creates one, whereas replacing it by $y = -\varepsilon$ destroys three old triangles and creates one. The old affected cells occupy alternating radial sectors 0, 2, 4. The complete support classification is proved in [up_count](#) and [down_count](#). The reverse collapse from the downward arrangement to the singular one gains two triangles, in [reversal_gain_two](#). Thus neither a universal cap of two destroyed cells nor a one-unit reverse yield is valid in this scope. The positive direction also shows that resolution can be useful when a genuine global retention proof accompanies its local birth certificate.

10 Exact certificates, external reproduction, and unsuccessful searches

The experimental program serves two purposes: producing concrete coordinate witnesses for lower bounds, and identifying which proposed general arguments survive exact checks. These purposes require different standards of evidence. A successful coordinate witness proves an existential lower bound at its specified order. A failed search ordinarily proves neither an upper bound nor the nonexistence of a better construction. This section records both positive and negative outcomes without merging their logical scopes.

10.1 From a numerical proposal to an exact geometric witness

The stored coordinates use the convention $a_i x + b_i y = c_i$, with rational or integer coefficients. Each line is normalized to a primitive integer triple. Pair intersections are computed as homogeneous integer triples, normalized with a positive final coordinate. Consequently, collinearity, concurrence, parallelism, and comparisons along a supporting line can all be tested without floating-point tolerances.

Two independently implemented counters are used. The first sorts the distinct intersection vertices on each line and constructs the graph of elementary bounded segments; a three-cycle supported by three different lines is a triangular cell. The second enumerates support triples

and tests whether every other line has a common weak sign at their three vertices. Nonzero area and the distinction between open interiors and boundaries are checked explicitly. Where a complete independent verification is reported, both counters agree on the triangle set, not merely on a rounded area or a visual estimate. The Lean geometric soundness theorem provides a third, logically separate link from accepted coordinate predicates to real affine triangles.

Numerical optimization is used only to propose coordinates or rank candidate chambers. A proposal is rationalized and recounted before promotion. In the phase/chart experiments, progressively finer integer scales were accepted only when the entire exact triangle set and simplicity were retained. Such a finite acceptance does not prove that an arbitrary rounding of an irrational construction is valid. The real all-order theorem for G has its own determinant, trigonometric, and counting proof.

The released catalog contains 138 distinct promoted coordinate identities. Several identities can occur at the same order and triangle count, and an identity may have several repository source paths. Thus 138 is a catalog size, not a count of new records or independent discoveries. The complete catalog in Table 16 includes simple versus nonsimple status, provenance collections, and coordinate-hash prefixes; its machine-readable companion records full hashes, source paths, and theorem modules. The finite Lean certificates use native evaluation where indicated. This trust boundary is broader than that of the universal construction, whose proof uses only Lean’s standard logical axioms.

10.2 Finite improvements relative to the retained collection

The phase and affine-chart work raised the preceding saved values at eleven orders:

n	39	47	48	51	53	54	55	59	60	99	195
previous	469	690	720	817	884	918	954	1102	1140	3169	12481
current	470	691	721	818	885	919	955	1103	1141	3170	12482

Each row is an improvement over the project’s preceding retained coordinate catalog. The underlying trigonometric family is due to Füredi–Palásti [10]; these comparisons do not establish numerical first priority. The general formula G now explains all eleven displayed counts. The first projective search reached 47 : 691 after 11,264 proposals; the subsequent exposed-cap description replaced that search by a direct construction and an all-order proof.

The earlier OEIS-attributed values 28 : 238, 30 : 275, and 34 : 357 remain independently certified [19, 34]. The drawings in this paper render the currently retained exact witnesses, obtained by exterior extension of retained odd-order inputs. They are not asserted to be literal reproductions of the March diagrams. Likewise, the historical inventory in Table 17 preserves earlier outputs without upgrading every original derivation to a completed general theorem.

The current strongest saved simple and classical coordinate counts for every $3 \leq n \leq 60$ appear in Table 15. An absent entry in the explicit OEIS table is recorded as a dash. Absence from that table is not evidence that a value is absent from the literature; conversely, a coordinate certificate does not establish optimality without an upper theorem covering the same arrangement class.

10.3 Independent reproduction of external constructions

The current Parpalak–Utkin coordinate gallery [20] supplied exact witnesses at

$$\begin{array}{cccccc} 8 : 15, & 14 : 54, & 20 : 117, & 26 : 204, & 32 : 315, \\ 36 : 402, & 38 : 450, & 42 : 553, & 46 : 667, & 50 : 792. \end{array}$$

Both independent counters reproduced their triangle sets. These witnesses retain their source attribution; in particular, the eight-line numerical value has older discovery priority than the coordinate gallery. Their addition to this repository improves its verified coverage, not the discoverer’s claim to the numbers. The inspected gallery snapshot is commit `fcae4f5571a988a3ed45bd26a36259ebb7d7d3d2`.

All fifteen of Andrea Maiorana’s published 14 : 54 coordinate configurations [18] were also independently checked. Fourteen have exactly two triple points, and one has four; none has a parallel pair. The source snapshot is commit `e47c7cfd9661e54d29befb83118bf83d5f804028`. Its CC BY 4.0 licence, original coordinate files, and source attribution are retained. Added wrappers record counts and verification metadata. This reproduction imports no claim that a simple-arrangement upper theorem applies to these nonsimple witnesses.

These examples also expose a concrete limitation of smoothing. In Maiorana’s first example, the triple supports are $\{0, 4, 10\}$ and $\{0, 11, 12\}$. Independently shifting the offsets of lines 4 and 11 by $\pm 1/1000$ realizes all four local resolution choices while preserving every originally nonzero determinant sign. The resulting triangle counts are 52, 52, 53, 52, all below 54. Thus the proposed nondecreasing local smoothing strategy fails even when the local sign choices are independently realizable by straight lines. The argument identifying these sign choices with all sufficiently small local resolutions is an ordinary finite sign-stability argument, not a completed Lean theorem.

For the gallery’s collinear-core examples at orders 8, 14, 20, 26, 32, 38, 50, all 132 recorded local resolutions were realized by exact private-line offsets. Their respective best simple counts were 14, 51, 114, 203, 313, 448, 791. In particular, a universal claim that resolving such a core loses at most one triangle is also false. Figure 20 separates these finite obstructions from any general upper-bound assertion.

10.4 Projective charts, algebraic curves, and chamber searches

Projective geometry provides a useful global neighborhood: changing the line at infinity moves all lines together and may bound formerly unbounded triangular faces. It can also lose bounded cells. For the exact saved 81 : 2132 and 161 : 8532 hybrid arrangements, the projective triangular-face census already equals their bounded-cell count; a chart change alone cannot increase those particular counts.

Exact rational linear-arithmetic searches give sharper restrictions for some fixed configurations. The nine-line half-phase rational arrangement has 21 projective triangular faces, but the recorded QF_LRA solver result rules out a chart retaining all 21. Similar fixed-seed tests find no gain for the tested half-phase arrangements at 14, 20, 26, 44, and no chart retaining all 471 projective triangles of the saved 39 : 470 witness. For a nonuniform cubic 48-line seed with 724 projective triangles, 2,903 exact linear-arithmetic checks rule out 722 bounded triangles within the tested chart-retention formulation. These are exact-solver results about specified rational seeds. Their UNSAT certificates have not been independently checked in Lean, and they are not global upper bounds at those orders.

The algebraic-curve direction tested both nodal-cubic angle perturbations and torsion cosets on real elliptic curves. A larger projective face count did not automatically produce a

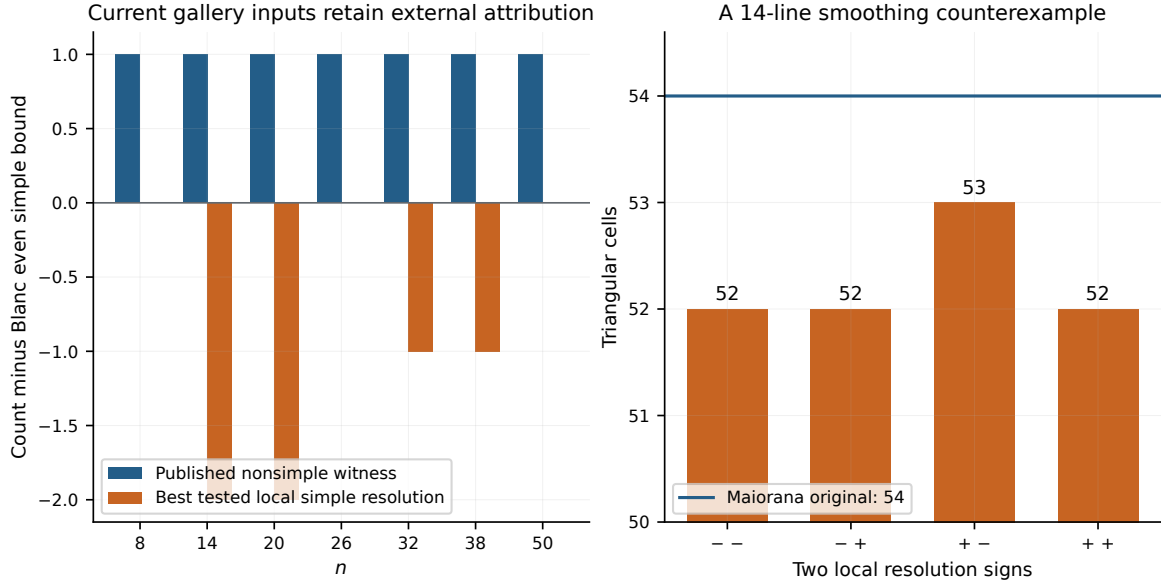


Figure 20: Exact local-resolution tests on attributed external constructions. Left: Parpalak–Utkin gallery witnesses and their best tested simple local resolutions, measured relative to Blanc’s even simple-arrangement polynomial; the eight-line value has older priority. Right: all four local resolutions of Maiorana’s two-triple-point 14 : 54 example. The original arrangements are nonsimple, so exceeding the simple polynomial is consistent. These are smoothing obstructions, not upper bounds for the classical problem.

better affine construction. For the better two-real-component elliptic examples, the projective, original affine, and best found affine counts were respectively

n	42	48	54	60
projective	550	724	922	1144
original affine	538	710	907	1127
best found affine	539	712	908	1129

All fall below the corresponding retained affine bounds. The chart scans at 54 and 60 had 90-second limits. Numerical ranking and bounded running time preclude a global nonexistence inference.

Oriented-matroid mutations and kinetic chamber sweeps addressed a different limitation: moving inside a fixed projective type cannot create new projective triangular faces. The mutation searches changed determinant signs, using linear-programming realization where appropriate. Kinetic sweeps varied many offsets, or both slopes and offsets, and visited concurrency events along selected rays. Their event counts are not counts of all realizable order types, and unsuccessful floating rays were discarded. Figure 21 records effort without treating heterogeneous search units as comparable coverage measures.

The final 39-line projective beam examined 2,122,270 candidate edges through 4,559 states without exceeding 470 affine or 471 projective triangles, even at its combinatorial scoring stage. The 48-line LP walk accepted 471 mutations out of 511 attempts and retained 721 affine and 724 projective triangles. At 44 lines, singular-boundary searches included 35,352 paired collapses; the retained beam checkpoint had 771 states and 446,422 candidate edges, without exceeding 608. An unfinished round was discarded. These results motivate a change of neighborhood or a structural argument rather than repeating the same local searches.

Other bounded approaches included Fourier deformations (2,687,600 proposals in one 39-line run), partial deletions from BBL constructions, and constrained tangent-grid transfer

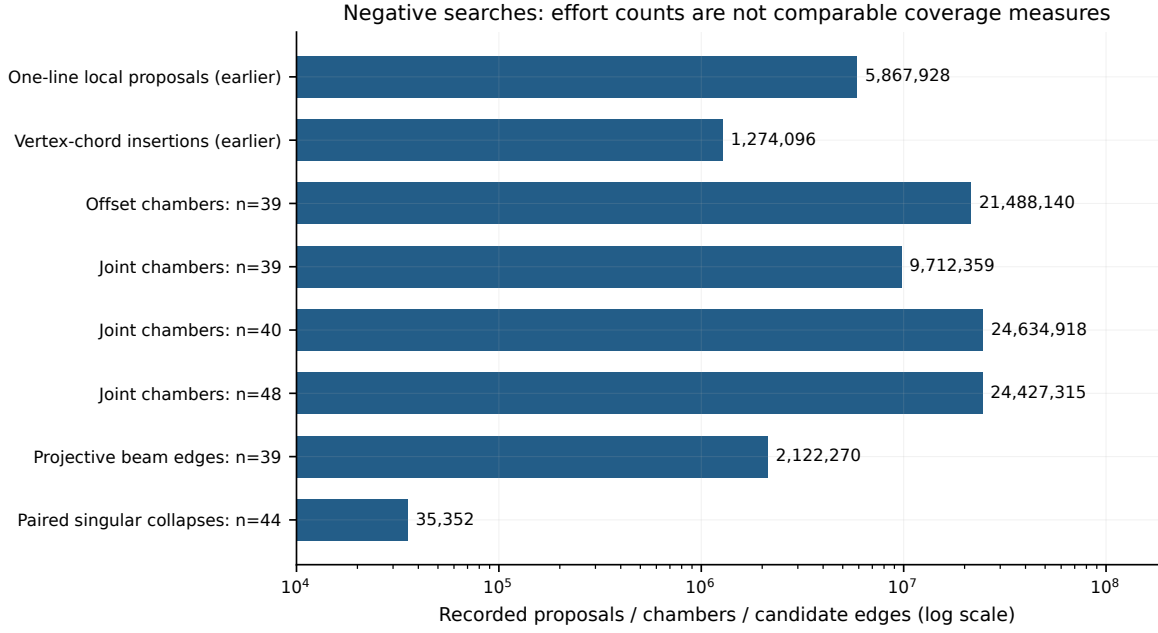


Figure 21: Selected unsuccessful searches, with their recorded units. Proposal, chamber, and candidate-edge totals describe different search neighborhoods. They do not measure exhaustive coverage, justify an upper bound, or rank the mathematical promise of the methods. All proposed improvements would require subsequent exact geometric verification.

by LP/MILP. The tested optimal 23-line chart transfers did not admit a positive-margin realization on the desired grid; a constrained near-optimal search gave 144 triangles, below a separately known compatible 145. None of these observations proves nonexistence of another compatible seed or a different successful deletion pattern.

10.5 A proved algebraic obstruction to tangent-grid transfer

The October search connects oriented-matroid sign constraints, linear programming duality and exact algebraic number arithmetic. For the reciprocal-coordinate lines $x - v_i y = a_i$, a prescribed determinant sign becomes a strict linear inequality in the unknown reciprocal slopes. Write

$$D_{ijk} = (a_j - a_k)v_i + (a_k - a_i)v_j + (a_i - a_j)v_k. \quad (99)$$

Once the intercepts are fixed, feasibility is a linear sign problem, even though the intercepts themselves are algebraic.

Lemma 10.1 (Positive-dependence obstruction). *Let $A_i \in \mathbb{R}^d$ and let $\lambda_i \geq 0$, with at least one $\lambda_i > 0$, satisfy $\sum_i \lambda_i A_i = 0$. There is no $v \in \mathbb{R}^d$ with $A_i \cdot v > 0$ for every i .*

Proof. Taking the scalar product with v makes the left side a positive weighted sum, but the assumed vector identity makes it zero. This classical strict alternative is formalized by `homogeneous_infeasible`. The formal theorem also provides an affine form for nonhomogeneous strict systems. Q.E.D.

Set $t = \tan(\pi/20)$. Its exact algebraic specification is

$$p(t) = t^4 - 4t^3 - 14t^2 - 4t + 1 = 0, \quad \frac{3}{20} < t < \frac{17}{100}. \quad (100)$$

Support triple	s_i	Positive multiplier $\lambda_i(t)$
(0, 6, 15)	-1	$-20t^3 + 120t^2 + 300t - 40$
(0, 7, 13)	+1	$-8t^3 + 24t^2 + 96t + 8$
(0, 8, 14)	-1	$-45t^3 + 205t^2 + 555t - 35$
(0, 13, 19)	+1	$-16t^3 - 6t^2 + 40t + 2$
(3, 6, 8)	+1	$20t^3 - 100t^2 - 220t + 100$
(3, 11, 15)	-1	$20t^3 - 120t^2 - 220t + 200$
(3, 13, 17)	+1	$20t^3 - 150t^2 - 300t + 130$
(7, 11, 13)	+1	$-60t^3 + 240t^2 + 860t + 320$
(8, 14, 17)	-1	$-105t^3 + 455t^2 + 1275t - 45$
(8, 15, 19)	-1	$36t^3 - 188t^2 - 452t + 124$

Table 10: An explicit positive linear-dependence certificate. Indices are zero-based intercept indices. The rows use eleven distinct intercepts and avoid the two exceptional central indices. Each column of the weighted signed coefficient matrix vanishes modulo $p(t)$.

The quintuple-angle identity with $\tan(5\pi/20) = 1$ gives $(t - 1)p(t) = 0$; the interval excludes $t = 1$. Monotonicity on the interval identifies the intended root. `BBLTangentAlgebra` proves the quartic, root interval and uniqueness; `BBLTangentValues` proves that its explicit cubic power-basis expressions are the actual nine values $\tan(k\pi/20)$ for $1 \leq k \leq 9$. Thus replacing trigonometric numbers by algebraic expressions does not insert a numerical guess.

Use the actual 20-intercept grid

$$a_j = \begin{cases} -\tan((9 - j)\pi/20), & 0 \leq j < 9, \\ -\varepsilon, & j = 9, \\ \varepsilon, & j = 10, \\ \tan((j - 10)\pi/20), & 11 \leq j < 20. \end{cases}$$

The following ten prescribed signs are impossible.

Theorem 10.2 (One exact tangent-grid obstruction). *For every real ε and every real vector $(v_0, \dots, v_{19})$, at least one row of Table 10 fails the strict inequality $s_i D_{j_i k_i \ell_i} > 0$.*

Proof. For each row define A_i to be the coefficient vector of $s_i D$ in (99). All ten displayed multipliers are positive throughout $3/20 < t < 17/100$, by rational polynomial inequalities. Using the proved power-basis expressions for the tangents, expand $\sum_i \lambda_i(t) A_i$ one column at a time and reduce modulo $p(t)$. Every column is zero. The two central columns vanish because none of the ten supports uses indices 9 or 10. Lemma 10.1 gives a contradiction. The exact source contains the polynomial multiple of p for every nontrivial column, so the computation is reproducible without a floating-point linear solver. The declarations `actual_grid_infeasible` and `actual_grid_orientation_impossible` prove this for the actual tangent grid using only standard logical axioms. Q.E.D.

This excludes one explicit orientation system with arbitrary reciprocal slopes and every real central parameter. It does not classify all 21-line arrangements. An independent exact interval Gaussian-elimination check also proves persistence under independent perturbations of radius 10^{-3} on the eleven used intercepts. That robustness calculation is external exact evidence, not part of the Lean theorem above.

10.6 The full supplied affine-class obstruction census

Parpalak–Utkin’s enumeration separates projective and Euclidean classes [21]. This distinction changed the scope of the search. The first checker verified 366 exact polynomial certificates for eighteen supplied projective representatives on $0 < \varepsilon < 1/25000$. A projective representative alone does not exhaust its affine cases: a choice of infinity support matters. That initial census was therefore insufficient for a complete affine claim.

The corrected checker expands all 236 supplied Euclidean classes and all 21 choices of distinguished support. It validates 3,765 exact polynomial dependencies and 15,060 reflected or reoriented variants, covering all $236 \cdot 21 = 4,956$ class/support cases with no missing case, for

$$0 < \varepsilon < 1/2,000,000. \quad (101)$$

Every certificate is a positive linear contradiction of the form above, so it excludes arbitrary, even parameter-dependent, reciprocal slopes. The small positive interval is a uniform bound for this finite corpus; it is not the unrestricted all-real parameter assertion proved for the single ten-sign representative.

The logical dependencies remain separate. The completeness of the classification is supplied by Parpalak–Utkin and its dataset. The present independent checker validates the listed exact certificates, the class/support coverage and the explicit symmetries. It does not rerun their full enumeration inside Lean. The resulting exclusion of a perfect 21-line seed on this prescribed saturated grid and interval is therefore computer-assisted mathematics with an external classification input. Only Theorem 10.2 is a complete representative Lean obstruction. Other grids, larger seeds, nonsimple arrangements and other construction methods remain unrestricted by this census.

10.7 New bounded searches and their exact outcomes

The 2 October work also tested finite neighborhoods selected to change features that the earlier chart-only searches preserved. None produced a new numerical lower-bound record. The exact positive and negative results are summarized below; completed enumeration of a listed neighborhood does not imply enumeration of all arrangements.

The methodological connection to recent automated mathematical search is a separation of proposal, evaluation and proof, also emphasized by Georgiev, Gómez-Serrano, Tao and Wagner [12]. Here a numerical optimizer proposes a coordinate or dual vector; exact arithmetic certifies the finite claim; the symbolic construction or obstruction explains which statements hold uniformly. Their paper is methodological context, not evidence that Kobon has already been solved. The author’s other construction projects motivated preserving an iterable invariant and exploiting symmetry orbits; none of their unrelated theorems is imported as a Kobon lemma.

10.8 Cap-cone searches and the promoted 61 visibility resource

The 5–6 October search uses a larger feasible region than the fixed orientation cell. For ordered axis intercepts a_i , write each other line as $x - u_i y = a_i$ and fix the above/below pattern of consecutive axis caps. For each adjacent pair $(j, j + 1)$ and every other support r , the determinant wall is

$$(a_{j+1} - a_r)u_j + (a_r - a_j)u_{j+1} + (a_j - a_{j+1})u_r.$$

The cap conditions are linear inequalities in the reciprocal slopes. `capRegion_iff` proves their equivalence to the actual empty nondegenerate caps with the fixed strict orientation pattern,

Search	Result	Limitation
Deletions from a 49-line seed	Best retained values 48 : 721, 47 : 677, 46 : 637, 45 : 598.	The tested deletion neighborhood only.
57-line MILP deletions	Incumbents 56 : 990, 55 : 939, 54 : 887, 53 : 837.	Timed runs; incumbents are not proved maxima.
All 9,180 tested chords of a 17-line seed	Best 18-line output has 93 triangles.	Chords of one seed.
407,253 vertex-pair insertions at 43 lines	Best 44-line output has 607 triangles, below retained 608.	The stated insertion neighborhood.
Singular 16-to-18 insertion	Best recorded count 89.	No global exclusion.
39-pseudoline straightening	Exact realizations with 353 and 358 triangles.	Neither a record nor a proof of nonstretchability.
Perfect 21-line grid fit	Exact affine-class obstruction census.	The prescribed grid, interval and supplied classification.
49-line grid fit	Uniform 767-cell certificate and full visible boundary.	Full Lean base and odd/even instantiation now completed.

Table 11: The added research directions and their actual conclusions. The 49-line positive result is compared with its known BBL numerical family in Section 7.9.

and `capRegion_convex` proves convexity. Interpolation preserves the axis caps; it need not preserve simplicity or other triangle cells while crossing concurrence or parallel facets.

Reduced-row LP walks use adjacent crossing-order constraints, initially 1,219 independent walls in place of 35,990 full simplicity conditions, and recount every realized row order. Deeper and heap walks permit temporary triangle losses. The nonlocal cap-cone ray method moves all 60 reciprocal slopes, processes concurrence and parallel events, and independently recounts selected chambers. Its promoted outcome is the 1190-triangle, 29-visible-pair seed of Theorem 7.10. More than a million explored facet crossings are discovery coverage; they are not a global 1190-triangle ceiling, and crossings may repeat. Rejected floating discrepancies remain in the reports. Every promoted coordinate proposal requires a whole true-tangent interval certificate and a fresh dependent Lean replay.

10.9 Bounded screens, transport diagnostics and a scoped dual

The unsuccessful directions are retained because each excludes a specific proposed route or suggests a different invariant. Their scope is the supplied types, charts, parameters or finite sample.

Table 12: Additional bounded research of 5–6 October. Numerical failure is not a theorem of geometric nonexistence.

Route	Recorded outcome and boundary
Central-slope scaling	Eight powers at four exact epsilon samples each were weaker than the 1182 fixed-slope small-epsilon seed. The full reciprocal-slope refit supersedes this 32-case screen.
Phase-shifted affine charts	120 sampled charts at each of 16 even orders from 12 through 90 did not improve their existing count. This does not exhaust all projective charts.

Route	Recorded outcome and boundary
Other 61 affine types	Every normal sector of the supplied 1189 type has at most 28 visible pairs. The 1187 type cannot improve 1218 even with all 30 pairs. These conclusions concern those types.
Deletion and random thinning	Averaged one-line deletion from a dense BBL family gives the preceding G value; the multi-line average has leading $n^3/(3N)$. Exact 61-certificate incidence transport through depth six gives minimum incidence $q - 7$, whose deletion count is four below $G(q)$. No arbitrary deletion theorem is asserted.
Visibility transport	Independently recounted rational diagnostics at 61, 121 and 241 give 1190, 4790 and 19190 cells and best sector resources 28, 58 and 118 for the original seed. Those finite rational proposals are separate from the true-tangent uniform theorem and the new 29-pair seed.
41/45 fixed-type and jet fits	The alternative chart runs tested 37/42 and 24/46 charts, visiting 1477 and 1059 unique sign cells. First- and second-order parameter jets tested eight source axes per order. No robust positive-margin fit was found.
Optimal31 recharting	All 29 original-infinity saturated axes of the published 299-cell input failed the floating margin screen. New infinity charts were recounted before fitting; no whole-interval uniform seed was promoted. A lower finite count is relevant only if compatibility improves the recursive successor.
Singular and retained-backbone probes	Raising existing cores, pencil slides, corridor grids, even corridor lifts and larger backbone variants did not produce a promoted improved compatible seed. Fixed-axis nesting and insertion failures concern those supplied types.
Other finite neighborhoods	Dual mutations, elliptic cosets, harmonic and offset probes, and local chamber searches retain their exact parameters and finite outcomes. Neither LP failure nor a sampled maximum certifies a general upper bound.
Phase-zero infinitesimal resolutions	The modeled old-triangle conflict graph has a perfect matching in 56 tested orders from 8–60 and at 99, 120, 195. Even independent modeled local choices retain at most half the old cells there. This is a finite combinatorial screen, not an all-order Lean perturbation obstruction.

A stronger kind of negative evidence is available for one prescribed 41-line cell. Eleven positive rational Farkas weights were verified by exact arithmetic on the exported axis0/original-infinity normalized constraints. Actual tangent intervals were independently checked using rational arctangent-quarter inequalities and the proved pi band. They exclude the required weighted budget for every positive epsilon up to approximately 0.01401555, even for epsilon-dependent reciprocal slopes. This is a metric obstruction for that cell, not an exclusion of all 41-line types. The generic coefficient-box/small-parameter separation theorem is [no_budget](#), with standard logical axioms. The concrete 41-line certificate remains exact external evidence pending its own concrete Lean replay.

The complete experiment records are under [construction evidence](#), [structural seed probes](#),

and the pinned [corpus proof map](#). The OpenMath source inventory pins the contestant repositories and judging archive [24, 30, 33]. Successful contestant inputs keep their attribution. The external 39-line 471-cell replay recorded there is separate from this repository’s inherited 138-certificate catalog and is not claimed as this author’s result.

10.10 Reproduction and presentation audit

The research registers preserve source snapshots, experiment scripts, seeds, parameters, checkpoints, and exactness limits [35]. The paper’s figures are original vector renderings, not copied source figures. The generator `paper/scripts/generate_figures.py` reads the research data without modifying it and exports PDF figures, PNG previews, LaTeX tables, and machine-readable rendering data. The figure manifest records input-file hashes and the mathematical status of each caption. Geometric views recompute their exact triangle lists before converting coordinates to floating point for display. Cropped detail panels and independent affine axis rescaling are explicitly labeled; lengths and angles are not inferred from those views.

The appropriate experimental conclusion is therefore selective. Exact coordinates have strengthened the retained lower-bound envelope and revealed specific failed construction rules. The successful universal advance comes from replacing a numerical chart search by a geometric exposure proof and an all-order count. The remaining negative searches narrow useful next steps, but they do not solve the general optimization problem.

11 Formal verification and reproducibility

11.1 The verified snapshot

The mathematical snapshot used by this edition is repository commit `f44f23ee062a255f5cad7d39188fd55c0feb83a8`. It pins Lean 4.31.0 and Mathlib commit `fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`. The source-linked proof map and accompanying release record identify the exact audit and compiler logs. Verification statements concern that collection of declarations and coordinates; they do not assert that every research claim or external dataset has been formalized.

The 5–6 October release record covers 638 active modules: 265 compiled during that run and 373 unchanged source/import closures reused from the completely verified baseline `112409e2c62b78ecff7adcad49f1e014ea5f0773`. Its fresh whole-project axiom audit inspects 9,058 theorem declarations: 8,046 use only standard logical axioms and 1,012 additionally depend on explicit finite native-evaluation roots. This incremental replay is distinguished from a clean build by the usual CI verifier. The counts describe the complete retained library, including auxiliary and generated declarations, rather than 9,058 independent new discoveries. The inherited coordinate catalog still contains 138 identities, of which 104 are simple; the independent new uniform seed modules have their own proof provenance.

11.2 Three proof and computation boundaries

General symbolic proofs. The arbitrary-order construction, generic closed BBL recursion, quantitative successor and iteration, actual structural upper bounds, translation germ theorems, interval loss-rule counterexamples and representative grid obstruction use the usual Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. No unproved geometric construction is installed as an axiom. Their conclusions are not extrapolated from a finite range of successful tests. Lean and Mathlib provide the real analysis, finite combinatorics and algebra used in these proofs [5, 32].

Finite certificates. Large coordinate checks use `native_decide`, which trusts Lean’s native compiler and runtime in addition to the ordinary logical kernel interface. The general certificate soundness theorem is proved symbolically; native evaluation verifies the finite integer sign conditions and list properties. The audit reports these dependencies rather than describing all certificate computation as kernel reduction. Small examples and many formula comparisons also have direct kernel-reduction proofs.

Computational and paper-level evidence. Exact rational searches, external interval calculations, manuscript arguments and experimental large witnesses retain separate status. They are not imported as axioms into the all-order theorem. The incomplete 642-line direct coordinate recount still does not certify those exact bytes; general recursion independently proves the existential bound. The old stopped 33- and 49-line prototypes remain historical drafts, while their independently promoted active seed modules now have complete real-parameter proofs. The full 21-line affine obstruction corpus retains an external classification input; the representative ten-sign obstruction is inside Lean. The concrete 41-line parametric dual and several exact corpus counterexamples are external evidence. The remaining zero-component analysis and conjectural token trade-offs are research arguments rather than final theorem roots.

11.3 What the audit rejects

The active-source scan strips comments and strings and rejects proof placeholders, admissions, custom axiom declarations, and unsafe definitions. The additional Lean audit recursively collects the axioms used by every theorem in the `Kobon` namespace and rejects unapproved dependencies. It permits and records the standard logical axioms and the generated native-evaluation dependencies just described.

The archived March file is deliberately outside the active library. Preserving it is evidence of the development history, not a way to conceal its eight placeholders inside an otherwise successful build. Likewise, an unfinished draft is kept under the research directory with an explicit status note rather than omitted silently from an allegedly complete theorem library.

11.4 Theorem-level correspondence

Table 13 gives the principal correspondence between the paper and the Lean development. The public repository contains more detailed logs and dependency information; the table is intended to make each major mathematical claim traceable without requiring a reader to inspect thousands of auxiliary declarations.

Table 13: Principal mathematical claims and formal sources. Names refer to the frozen mathematical snapshot.

Mathematical claim	Formal source
Exact certificate soundness	<code>Geometry.validate_sound</code> ; <code>Simple.validate_simple_sound</code>
Certificates are nonoverlapping cells	<code>Cells.interior_eq_signCell</code> ; <code>Cells.distinct_interiors_disjoint</code>
Classical affine construction	<code>FurediPalasti.lower_bound</code> ; <code>FurediPalastiCount.ordered_bound</code>
Shifted repeated-index gain	<code>ShiftedCount.ordered_bound_plus</code> ; <code>ShiftedFurediPalasti.simple_lower_bound</code>

Mathematical claim	Formal source
Actual projective transformation	<code>Projective.oriented_transform; Projective.triangle_preserved</code>
One exposed cap	<code>FurediPalastiWrap.one_cap_simple_lower_bound</code>
Two exposed caps	<code>FurediPalastiTwoCaps.two_cap_simple_lower_bound</code>
Universal G and exact gain	<code>Universal.baseline_sound; Universal.baseline_improvement</code>
Finite-enhanced all-order bound	<code>Universal.all_n; AllN.dominates_every_saved_certificate</code>
Unconditional 49-seed target	<code>AllN.from_49_unconditional; Universal.dominates_49_target</code>
Exterior addition from visible pairs	<code>Exterior.extension</code>
Real compatible seed family	<code>SeedFamily; Parametric; TangentBounds</code>
Geometric one-step BBL doubling	<code>BBLDoubling.doubling</code>
Closed seed and uniform recursion	<code>BBLRecursiveSeed.seed_step; BBLInfinite.uniform_iterate</code>
Visible-boundary recursion	<code>BBLEvenStep.canonical_visible_step; BBLEvenInfinite.infinite_even_family</code>
Concrete odd and even families	<code>BBLVerifiedFamilies.eleven_odd_family; BBLVerifiedFamilies.eleven_even_family</code>
Current all-order envelope H	<code>RecursiveEnvelope.all_n; RecursiveEnvelope.strict_previous_tail</code>
Actual simple optimality windows	<code>BBLFamilyOptimality.odd_window; BBLFamilyOptimality.even_window</code>
Next-grid saturation component	<code>BBLNextSaturation.next_saturated</code>
Actual segment and defect inventory	<code>UpperEdgeInventory.certificate_defect_identity</code>
Actual clean-line charging	<code>UpperCleanCharging.certificate_clean_line_budget</code>
Actual simple upper inequalities	<code>UpperSimpleOptimality.simple_lower_bound_upper; UpperEvenSimpleOptimality.simple_lower_bound_even_upper</code>
Extremal fan and matched triple obstruction	<code>UpperFan.Sectors.weighted_local_bound; UpperTripleFan.extremal_triple_fans_not_adjacent</code>
Actual core incidence	<code>UpperCoreLineIncidence.core_line_budget; UpperCoreCombinatorics.core_surplus</code>
Actual three-core bound	<code>certificate_three_core_upper</code>
Tangent algebra and actual obstruction	<code>BBLTangentAlgebra.tan_quartic; BBLTangentValues.value_eq_tan; BBLGridObstruction.actual_grid_orientation_impossible</code>
49-grid and actual orbit	<code>BBLTangent48Bounds; odd_family; even_family</code>
Completed Forge tail	<code>odd_family; even_family</code>
Completed 37-line orbit	<code>odd_family; even_family</code>
Strongest 61-line orbit	<code>odd_family; even_family</code>
Portfolio J	<code>all_n; previous_le</code>
Quantitative actual successor	<code>successor; closed_form_successor</code>
Indefinite quantitative iteration	<code>infinite_quantitative_extension</code>
Actual fans and matching	<code>certificate_local_fan_bound; certificate_neighborhood_matching</code>
Mixed degree-free components	<code>certificate_mixed_curvature_component_bound</code>

Mathematical claim	Formal source
All-triple N12 components	<code>certificate_n12_source_free_hybrid_component_defect</code>
Late global marked balanced gain	<code>certificate_m22_source_free_defect_bound</code>
Exact translation and birth/loss	<code>simultaneous_triangle_stability; translated_birth_loss_identity</code>
Loss-rule interval counterexamples	<code>halfplane_yield_fails; down_count</code>
Local fan geometry and budgets	<code>SharedFan; FanGeometry; FanCount; CyclicFan; CleanLineBudget</code>
Preserved historical finite bounds	<code>Results.earlier_NNN; Results.best_classical_NNN; Results.best_simple_NNN</code>

The generic exterior theorem assumes a valid visible-pair list; the new quantitative successor constructs one from actual terminal resources. The BBL theorem retains tangent-grid and saturation premises, which the concrete uniform seeds discharge. The older conditional upper arithmetic remains valid as an implication; the actual arrangement wrappers now derive its fan, matching and incidence inputs internally. They retain real restrictions such as all-triple cores or a graph class. Inspecting those hypotheses is as important as checking the absence of placeholders.

The optimized finite engine submits only closed Boolean equalities to native evaluation. Kernel soundness translates them into the original direction, simplicity, triangle, admissibility and visibility predicates. Memoization materializes coefficient and parameter arrays once per check; its proved extensional equalities preserve the mathematics. The 33-line full simplicity decision then completes in seconds rather than an interrupted long computation. This is a reproducibility improvement, not permission to weaken a predicate or reuse a modified seed's old native root. Construction proof-provenance files record the seven finite roots of the strongest 61-line even family separately from its standard-axiom analytic and recursive bridges.

11.5 Reproducing proofs, coordinates, and figures

After cloning the repository and installing the pinned Lean toolchain, the principal checks are:

```
lake exe cache get
python scripts/verify_lean.py --jobs 2
python scripts/verify_coordinates.py
python scripts/verify_seed_interval.py
python scripts/evidence_manifest.py --check
python manuscripts/2026-10-07/build.py --edition long
```

The finite envelope can be regenerated from the certificate index using `scripts/generate_all_n.py`. This generator emits theorem references; Lean checks the resulting proposition independently of the script's intentions. The generated source is not a trusted oracle for geometry.

All sixteen scientific figures retained from the initial manuscript are generated by `paper/scripts/generate_figures.py`. The script reads the saved formulas, coordinates, and experiment records. For geometric renderings it fixes the triangle combinatorics using exact rational arithmetic before converting coordinates to floating point for plotting. Vector PDF is the publication format; PNG images are previews. Rational proxies of the trigonometric examples are explicitly identified as finite illustrations, not substituted for the symbolic theorem.

The files `paper/figure_audit.md` and `paper/generated/figure-manifest.json` record the older figures' data sources, hashes and claim status. The finite tables come from the coordinate index used by verification. The 3 October figures illustrate the recursive envelope and uniform 49-line seed; this edition adds exact 61-gap, retained-portfolio and local-coefficient figures with `October-figure-manifest.json`. The upper coefficient plot illustrates algebraic weights rather than realizability of arbitrary local profiles. The shared proof map marks symbolic theorems, manuscript arguments, exact external evidence and excluded prototypes. The earlier editions and their frozen verification manifests remain preserved.

Computer and AI assistance. The development and preparation of this manuscript used computer algebra, exact arithmetic, automated search, Lean, and AI assistance for research, code, and drafting. AI-generated assertions are not treated as mathematical evidence. The reported conclusions are tied to explicit proofs, exact witnesses, or separately scoped computational records. Alejandro Zarzuelo Urdiales is the sole named author of this manuscript; cited construction authors retain their mathematical attribution.

12 Contributions, limitations and remaining proof gaps

12.1 What changes in the completed development

The total lower-bound function is now J of Theorem 7.11. It retains the all-order construction G , the inherited 138-certificate catalog and the eleven-derived recursive envelope, and adds actual compatible 33-, 37-, 49- and 61-line orbits. The all-order affine formula changes a constant term of the quoted baseline; the sparse optimal or near-optimal orbits improve particular orders without changing its leading coefficient. The 33-, 37- and 49-based numerical families are inherited. The 61-based orbit is a candidate distinct compatible family: Poola supplied its finite point count, while the uniform refit and 29-pair visibility resource support its verified iteration. The bounded source review does not establish worldwide numerical priority or a new individual optimum.

The advance over the preceding edition is also a proof-status change. The 3 October manuscript already stated the quantitative successor and its geometric argument. Its terminal charging, actual normal choices, visibility and indefinite both-parity iteration are now complete in Lean. The stopped uniform 33- and 49-line computations are replaced by successful independent active seed proofs. Actual cyclic fans, shared side matching and component assembly are also complete. The upper frontier is now the mathematical control of residual profiles and ordinary-end resources, rather than an unproved extraction interface.

The March values at 28, 30 and 34 remain supported by their later exact certificates independently of the original file's placeholders. The actual exterior construction explains successful individual steps; its consumed-wedge budget explains why one cannot simply repeat full gains. The unconditional 49-seed numerical target remains valid by a separate all-order argument. None of these conclusions proves the original unrestricted full half-order recurrence for maxima.

12.2 From the earlier graph plan to actual restrictions

The previous edition's proposed ray/edge correspondence and triangle matching are now the actual theorems of Section 8.9. In particular, cap-heavy triple sources cannot share a core edge, and their incident edges are counted injectively by `certificate_cap_heavy_degree_sum`. The older exceptional-triple graph deduction is thus no longer blocked by those interfaces. If

all exceptional fans are triple and e is their number, then

$$3e \leq D_2, \quad D_2 - 2e \geq e, \quad D_2 - 2e \geq D_2/3. \quad (102)$$

The consequences of (84) are

$$2\delta \geq n + 2S - 7I + 8q + e, \quad (103)$$

$$6\delta \geq 3n + 6S - 21I + 24q + D_2. \quad (104)$$

These retained intermediate deductions explain the historical next step; they do not remove the low-multiplicity correction or provide a new unrestricted numerical upper polynomial. The later marked-port, antipodal-recipient and component results retain finer resources. Higher-order extremal fans are not independent, as the exact corpus counterexample shows, so the triple independence statement is not broadened without its hypothesis.

The final global all-triple rule eliminates antipodal-source and $(1, 3)$ double-recipient penalties and retains positive E_3, P_{21}, N_{12} and marked balanced M_{22} terms. The separately completed N12 component portfolio chooses its strongest rounded cost inside each component. The B_{15} correction remains. Mixed multiplicities have a degree-free component bound with explicit full-triple corrections; forests, degree at most three, and components of order at most five supply useful correction-free subclasses. These statements can apply to arrangements of arbitrarily large total order without being unrestricted numerical formulas for $K(n)$.

12.3 The role of exact negative evidence

The eight-line shared-side witness, adjacent higher extremal fans, allowed antipodal-to-marked and antipodal-to-one-cap adjacencies, and parity-token counterexamples rule out specified shortcuts. Their role is to identify geometric obligations that an abstract discharging or matching proof cannot omit. A finite charging relaxation can have an equality graph that actual elementary-side geometry forbids; such a graph proves sharpness of that relaxation, not of real arrangements.

The translation-germ classifier supplies a different kind of progress: it explains all births and losses with one simultaneous real parameter threshold. The four-line interval example refutes the stated half-plane loss rule, and the six-line example exhibits three losses and a reverse gain of two. A useful surgery theorem must therefore certify the global loss set or its absence. Bounded chamber, chart, corridor, harmonic, elliptic and resolution screens narrow experimental routes but do not establish general infeasibility. The exact 41-line dual is stronger within its one prescribed cell; its concrete Lean replay remains separate from the generic proved separation theorem.

Data and code availability. The complete source, bibliography, vector figures, figure-generation scripts, proof-to-claim map, exact finite catalog, negative experiments and verification records accompany the repository [35]. This edition and its journal companion use the same immutable mathematical source revision. The preserved March, September and 3 October artifacts remain historical snapshots; their old proof-status statements are not silently rewritten. Preparation of these editions does not assert external peer review.

12.4 Contribution comparison

Table 14: Mathematical and verification contributions, with numerical attribution separated from proof completion.

Component	Mathematical content	Scope of this contribution
March finite inequalities	28:238, 30:275, 34:357 retained by exact certificates.	Original universal proof remains incomplete; related published constructions keep their priority.
Universal affine G	Actual simple construction at every order; bounded improvement over the quoted affine formula.	Geometric phase, chart and formal proof; no new leading asymptotic constant or absolute numerical-priority claim.
Eleven-derived recursion	Closed real grid, cap, orientation and visible-boundary invariant; one-triangle simple windows.	Completed in the earlier October release and retained; BBL operation is inherited.
33/37/49 orbits	Uniform true-tangent compatible seeds, actual infinite witnesses and simple optimality.	Formal completion and portfolio coverage of known numerical families.
61 orbit	Uniform 1190-cell seed; 29 visible pairs; constant odd/even simple gaps 9/10.	Poola's finite count, compatible refit and stronger visibility here; candidate distinct infinite orbit with bounded priority review.
All-order portfolio	Maximum of the previous envelope and four additional actual branches.	Pointwise retention and formal gains; not a claim to the best bound in all literature at every order.
Quantitative successor	Deficit-dependent real exterior rule and indefinite both-parity iteration.	Formula and manuscript argument already in the preceding edition; actual extraction and end-to-end Lean proof completed now.
Structural upper bounds	Mixed degree-free components, all-triple recipient charges, N12 component maxima and global M22 gain.	Actual fan/matching/graph assembly with quantified penalties; residual corrections remain and priority requires comparison.
Surgery and counterexamples	Exact simultaneous translation germs, isolated birth/loss and whole-interval loss-rule failures.	General reusable tool and targeted counterexamples; no general monotone smoothing or Kobon optimum claim.
Search and verification tools	Cap-cone convexity, parametric separation, true tangent bounds, Boolean soundness and memoized exact decisions.	Supporting reproducible infrastructure; finite search failure retains its bounded scope.

12.5 Remaining proof gaps

The unresolved tasks are concrete:

1. Prove or refute the original full half-order recurrence for the unrestricted maxima. The completed quantitative simple successor and finite exterior examples do not imply it; reconstruction may be needed.
2. Control the remaining full $(1, 5)$ profiles and strict component costs not paid by the N12 maximum. The actual $(1, 3)$ exposed-recipient restriction and late local port competition are available, but no universal strictly positive component theorem is asserted. The final zero-component analysis of matched boundary contacts remains research analysis rather than a separate Lean theorem.
3. Establish a correct global ordinary-end token trade-off. The all-triple candidates $C \geq 2D_1$ and $n \leq 2U + D_1 + R_c + 2$ remain unproved; mixed multiplicities need additional corrections. A proposed even upper formula must retain the hypotheses of every resource used.
4. Improve the compatible 61 triangle count above 1190, increase its visible resource above 29, or find other compatible primitive orders. The present observed maxima are not proved ceilings. Partial doubling, interpolation and an actual recursive even corridor family are separate construction questions.
5. Promote the concrete 41-line dual certificate and other external corpus counterexamples only after their own exact formal extraction. The external 21-line classification completeness remains an attributed input rather than a full Lean enumeration theorem.
6. Use the exact germ tool to prove global retention or controlled loss for successful surgery, and compare quantified structural statements against primary prior literature before making priority claims.

References

- [1] Nicolas Bartholdi, Jérémy Blanc, and Sébastien Loisel. On simple arrangements of lines and pseudo-lines in $\mathbb{P}^2$ and $\mathbb{R}^2$ with the maximum number of triangles. In *Surveys on Discrete and Computational Geometry*, volume 453 of *Contemporary Mathematics*, pages 105–116. American Mathematical Society, 2008. doi: 10.1090/conm/453/08797. URL <https://arxiv.org/abs/0706.0723>. Preprint first posted in 2007.
- [2] Jérémy Blanc. The best polynomial bounds for the number of triangles in a simple arrangement of n pseudo-lines. *Geombinatorics*, 21(1):5–14, 2011. URL <https://arxiv.org/abs/0801.2845>. Preprint first posted in 2008.
- [3] Gilles Clément and Johannes Bader. Tighter upper bound for the number of Kobon triangles, December 2007. URL <https://oeis.org/A006066/a006066.pdf>. Draft dated 21 December 2007.
- [4] Gábor Damásdi, Leonardo Martínez-Sandoval, Dániel T. Nagy, and Zoltán Lóránt Nagy. Triangle areas in line arrangements. *Discrete Mathematics*, 343:112105, 2020. doi: 10.1016/j.disc.2020.112105. URL <https://real.mtak.hu/114794/7/1-s2.0-S0012365X20302910-main.pdf>.
- [5] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, Lecture Notes in Computer Science, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

- [6] Jeff Erickson. New lower bounds for convex hull problems in odd dimensions. *SIAM Journal on Computing*, 28(4):1198–1214, 1999. doi: 10.1137/S0097539797315410. URL <https://jeffe.cs.illinois.edu/pubs/pdf/convex.pdf>.
- [7] Stefan Felsner and Klaus Kriegel. Triangles in Euclidean arrangements. *Discrete & Computational Geometry*, 22(3):429–438, 1999. doi: 10.1007/PL00009471. URL <https://page.math.tu-berlin.de/~felsner/Paper/tri.pdf>.
- [8] David Forge and Jorge Luis Ramírez Alfonsín. Straight line arrangements in the real projective plane. *Discrete & Computational Geometry*, 20(2):155–161, 1998. doi: 10.1007/PL00009373.
- [9] Kobon Fujimura. *The Tokyo Puzzles*. Charles Scribner’s Sons, New York, 1978. ISBN 0684155362. Edited by Martin Gardner; translated by Fumie Adachi.
- [10] Zoltán Füredi and Ilona Palásti. Arrangements of lines with a large number of triangles. *Proceedings of the American Mathematical Society*, 92(4):561–566, 1984. doi: 10.1090/S0002-9939-1984-0760946-2. Also indexed by JSTOR DOI 10.2307/2045427.
- [11] Martin Gardner. *Wheels, Life, and Other Mathematical Amusements*. W. H. Freeman, New York, 1983. Kobon discussion cited in Clément–Bader at pp. 170–171 and 178.
- [12] Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale, 2025. URL <https://arxiv.org/abs/2511.02864v3>. Version 3, 22 December 2025; cited for methodology, not as a solution of Kobon.
- [13] Branko Grünbaum. *Arrangements and Spreads*, volume 10 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI, 1972. doi: 10.1090/cbms/010.
- [14] HandWiki contributors. Kobon triangle problem, 2026. URL https://handwiki.org/wiki/Kobon_triangle_problem. Consulted 22 September 2026; a secondary record of the extension claim, not its proof.
- [15] Heiko Harborth. Some simple arrangements of pseudolines with a maximum number of triangles. *Annals of the New York Academy of Sciences*, 440(1):31–33, 1985. doi: 10.1111/j.1749-6632.1985.tb14536.x.
- [16] Ke Liang, Youming Liu, and Yurui Zhang. On the symmetries of optimal simple Kobon arrangements for $n = 28$. Zenodo preprint, version 4, July 2026. URL <https://zenodo.org/records/21181834>. Version-specific record, 3 July 2026; not an unrestricted exact-value theorem.
- [17] Youming Liu. Triangular cells and the classical Kobon count: a desingularization calculus. Zenodo preprint, version 1, July 2026. URL <https://zenodo.org/records/21181452>. Version-specific record, 3 July 2026; PDF dated 29 June; the present audit concerns only Theorem 1.1’s half-plane loss formula.
- [18] Andrea Maiorana. Exact fourteen-line Kobon configurations with 54 triangles. Public coordinate repository, 2026. URL https://github.com/rufio72/kobon_triangles_k14/tree/e47c7cfd9661e54d29befb83118bf83d5f804028. Descriptive title; commit e47c7cfd9661e54d29befb83118bf83d5f804028, 12 August 2026; CC BY 4.0.

- [19] OEIS Foundation Inc. Entry A006066: Kobon triangles. The On-Line Encyclopedia of Integer Sequences, 2026. URL <https://oeis.org/A006066>. Consulted 22 September 2026; revision 245 and its attribution comments audited.
- [20] Roman Parpalak and Denis Utkin. The $18 \cdot 2^t + 1$ triangle-maximal series of straight lines, 2026. URL <https://arxiv.org/abs/2604.22035>. Version 1, 23 April 2026.
- [21] Roman Parpalak and Denis Utkin. Enumeration and classification of triangle-maximal pseudoline arrangements, 2026. URL <https://arxiv.org/abs/2607.29236v1>. Version 1, 31 July 2026; classification is external input to the exact grid-obstruction corpus.
- [22] Roman Parpalak and Denis Utkin. The 8–14–26 even series. Public research draft and construction repository, 2026. URL <https://github.com/parpalak/kobon-even-draft/blob/4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5/8-14-26-even-series.md>. Snapshot at commit 4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5; descriptive repository title.
- [23] Roman Parpalak and Denis Utkin. Kobon solutions: coordinate gallery. Public data repository, 2026. URL <https://github.com/ud1/kobon-solutions/tree/feae4f5571a988a3ed45bd26a36259ebb7d7d3d2>. Snapshot at commit feae4f5571a988a3ed45bd26a36259ebb7d7d3d2, 21 September 2026; original individual credits retained.
- [24] Rohith Poola. A 39-line arrangement with 470 Kobon triangles. Open Math Challenge manuscript, code and exact coordinate data, 2026. URL <https://github.com/Rohith18p/rsi-kobon-triangles/blob/f462d8e18aea2a458376c523c9b6c2980237071f/paper/kobon39.typ>. Immutable source commit f462d8e18aea2a458376c523c9b6c2980237071f; repository includes the original 61-line, 1190-triangle seed; no Lean proof claimed in the submission.
- [25] ProofAtlas. Kobon triangle problem: collaboration research page, 2026. URL <https://www.proofatlas.ai/collaboration/kobon-triangle-problem/>. Consulted 22 September 2026; reported restricted results were not independently replayed.
- [26] G. B. Purdy. Triangles in arrangements of lines. *Discrete Mathematics*, 25(2):157–163, 1979. doi: 10.1016/0012-365X(79)90018-9.
- [27] George Purdy. Triangles in arrangements of lines. II. *Proceedings of the American Mathematical Society*, 79(1):77–81, 1980. doi: 10.1090/S0002-9939-1980-0560588-4.
- [28] Jean-Pierre Roudneff. The maximum number of triangles in arrangements of pseudolines. *Journal of Combinatorial Theory, Series B*, 66(1):44–74, 1996. doi: 10.1006/jctb.1996.0006.
- [29] Pavlo Savchuk. Constructing optimal Kobon triangle arrangements via table encoding, SAT solving, and heuristic straightening, 2025. URL <https://arxiv.org/abs/2507.07951>. Version 1 inspected.
- [30] Raj Harshit Srirangam. The Kobon triangle problem for 18 lines: an upper bound of 94 for all arrangements, and a reduction of $K(18) = 93$ to a single local configuration. Open Math Challenge preprint and exact proof certificates, October 2026. URL https://github.com/srirangam-r/kobon_triangles/blob/eed14659a9b2f4ca12777da5d557f2b620b966f6/paper/kobon18.tex. Dated 2 October 2026; immutable source commit eed14659a9b2f4ca12777da5d557f2b620b966f6; computer-assisted claims are distinguished from the present Lean theorems.

- [31] Thomas O. Strommer. Triangles in arrangements of lines. *Journal of Combinatorial Theory, Series A*, 23(3):314–320, 1977. doi: 10.1016/0097-3165(77)90022-X.
- [32] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824. URL <https://leanprover-community.github.io/papers/mathlib-paper.pdf>.
- [33] Alejandro Zarzuelo Urdiales. Open math challenge 2026: judging and publication-review archive. Source archive and reproducibility records, 2026. URL <https://github.com/alejandroz/alejandrozu/openmath-2026-judging/tree/3dee18bdf290e70b2f9a6bd98b889d1b1b60b51c>. Immutable commit 3dee18bdf290e70b2f9a6bd98b889d1b1b60b51c; an archive of submissions and review evidence, not a blanket validation of all contestant claims.
- [34] Alejandro Zarzuelo Urdiales. New lower bounds for even Kobon numbers. Archivara preprint, March 2026. URL <https://archivara.org/paper/48b411c9-0e03-4592-931e-179b9a1c2312>. Original March result; the archived general extension proof was incomplete.
- [35] Alejandro Zarzuelo Urdiales. Verified Kobon constructions: research and Lean proof repository. Source code, exact certificates, and research records, 2026. URL <https://github.com/alejandroz/kobon-proof>. Research state reviewed through 7 October 2026; immutable mathematical snapshot and verification scope are recorded in the accompanying proof map.

A Complete saved finite table, orders 3–60

The entries below reproduce the strongest saved coordinate lower bounds in the inherited finite catalog. The additional uniform seed families are tabulated separately in Table 7; this appendix is not an exhaustive current literature table. The classical column may use multiple intersections; the simple column may not. The OEIS column is a dated explicit-table comparison, not an exhaustive priority audit. Upper columns reproduce attributed September benchmarks; simple polynomial bounds also have actual geometric Lean proofs, while the reported Clément–Bader values retain their external-source status. A remaining gap is not a proof that intermediate values are unattainable.

Table 15: Saved finite lower bounds, separated by arrangement class, and attributed comparison upper values. OEIS is the explicit table snapshot of 20 September 2026; a dash means omitted. Upper values retain their external attribution; the simple polynomial bounds are also formalized in this release.

n	$G(n)$	Saved simple	Saved classical	OEIS lower	Blanc simple	U_T
3	1	1	1	1	1	1
4	2	2	2	2	2	2
5	5	5	5	5	5	5
6	7	7	7	7	7	8
7	11	11	11	11	11	11
8	14	14	15	15	14	16
9	20	21	21	21	21	21
10	24	25	25	25	25	26
11	31	32	32	32	33	33
12	37	37	38	38	38	40
13	45	47	47	47	47	47
14	52	53	54	54	53	56
15	62	65	65	65	65	65
16	70	72	72	72	72	74
17	81	85	85	85	85	85
18	91	93	93	93	93	96
19	103	107	107	107	107	107
20	114	116	117	117	116	120
21	128	133	133	133	133	133
22	140	143	143	143	143	146
23	155	161	161	161	161	161
24	169	172	172	172	172	176
25	185	191	191	191	191	191
26	200	203	204	204	203	208
27	218	225	225	225	225	225
28	234	238	238	238	238	242
29	253	261	261	261	261	261
30	271	275	275	275	275	280
31	291	299	299	299	299	299
32	310	314	315	315	314	320
33	332	341	341	341	341	341
34	352	357	357	357	357	362
35	375	385	385	385	385	385
36	397	402	402	402	402	408
37	421	431	431	431	431	431
38	444	449	450	450	449	456
39	470	470	470	–	481	481
40	494	494	494	–	500	506

Continued from previous page

n	$G(n)$	Saved simple	Saved classical	OEIS lower	Blanc simple	U_T
41	521	533	533	533	533	533
42	547	553	553	553	553	560
43	575	587	587	587	587	587
44	602	608	608	—	608	616
45	632	645	645	645	645	645
46	660	667	667	667	667	674
47	691	691	691	—	705	705
48	721	721	721	—	728	736
49	753	767	767	767	767	767
50	784	791	792	792	791	800
51	818	818	818	—	833	833
52	850	850	850	—	858	866
53	885	885	885	—	901	901
54	919	919	919	—	927	936
55	955	955	955	—	971	971
56	990	990	990	—	998	1008
57	1028	1045	1045	—	1045	1045
58	1064	1073	1073	—	1073	1082
59	1103	1103	1103	—	1121	1121
60	1141	1141	1141	—	1150	1160

B The complete 138-certificate catalog

Each row identifies a finite arrangement and its certified triangle count. A hash distinguishes coordinate identities even when n and T coincide. The complete coordinates, support triples, source paths, original metadata, and Lean theorem names are machine-readable in the accompanying catalog. Printing hundreds of large rational coefficients would obscure rather than improve the mathematical presentation; the hash-indexed files provide their exact reproducible specification.

The catalog contains 138 distinct promoted coordinate identities, not 138 new numerical records. S denotes a simple arrangement; M denotes a verified arrangement with multiple intersections. Every row has a classical lower-bound theorem; S rows also have simplicity certificates. The hash is a prefix of the canonical coordinate SHA-256 identity, not of the source file. Full source paths, source-file hashes, theorem modules, and complete identities are supplied in `paper/generated/certificate-catalog.json`.

Source codes identify repository collections, not discovery authors: FT, retained finite table (mixed published and derived sources); EX, exact extension; HI, historical work inventory; HY, compatible-seed/hybrid work; SC, current session (phase/chart or classical dyadic construction, distinguished in the JSON metadata); MA, Andrea Maiorana’s external certificates; PU, Parpalak–Utkin gallery inputs; KR, earlier research collection; OT, other documented source. Published constructions retain their original attribution. Native-evaluation trust boundaries are discussed in the text.

Table 16: Complete promoted coordinate catalog. Repeated orders and counts can have distinct coordinate witnesses; no numerical priority follows from inclusion.

ID	n	T	Class	Collections	Coordinate identity
1	3	1	S	FT	05c82de62571
2	4	2	S	FT	0e541bd5bbe7

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
3	5	5	S	FT	948a0bdf2dd3
4	6	7	S	FT,EX,HI	6ea9ebaa7bf2
5	7	11	S	FT	e6b9a626a163
6	8	14	S	FT	2a63f4d9308d
7	9	21	S	FT	6ebbed82d063
8	10	25	S	FT	b767c427fa30
9	11	32	S	FT,HY	b955e7c9923a
10	12	38	M	FT	38f69f380097
11	13	47	S	FT	ddec1f788550
12	14	53	S	FT	45157b2f20fe
13	15	65	S	FT	4d3bf3b46db5
14	16	72	S	FT	391cf79afe24
15	17	85	S	FT	7cf3c061a447
16	18	93	S	FT	edb6246a6de7
17	19	107	S	FT	a2f32bdbcd1e
18	20	116	S	FT,EX,HI	adc03eed024d
19	21	133	S	FT	36bae754f198
20	22	143	S	FT	54396c646a7b
21	23	161	S	FT	969ad7d3adaf
22	24	172	S	FT	e63e906b9903
23	25	191	S	FT	b568bfffb5a19
24	26	203	S	FT	e61b423c7c3d
25	27	225	S	FT	7d6ae9a04c46
26	28	238	S	FT	56ef2acb9edd
27	29	261	S	FT	d4805c25a121
28	30	275	S	FT	dcfe649a79cf
29	31	299	S	FT	70f23b9463cb
30	32	314	S	FT	9ab78b00c233
31	33	341	S	FT	95f3b770119e
32	34	357	S	FT	fbde0a760c6e
33	35	385	S	FT	9ee99af6b980
34	36	402	S	FT	3fc01b7d2a1c
35	37	431	S	FT	11f0bde7bac3
36	38	449	S	FT	46ad5a32af83
37	39	469	M	FT,HI	cf24d7d33c00
38	40	494	S	FT	785374ca691b
39	41	533	S	FT	01e2ef66f8fd
40	42	553	S	FT	2bdc9d67365a
41	43	587	S	FT	861d8be27dee
42	44	608	S	FT,HI	b5277675e356
43	45	645	S	FT	3dc6e882e9b8
44	46	667	S	FT,PU	32b8ca2f231d
45	47	690	S	FT	d8af27fc13d1
46	48	720	S	FT	5e9767e73b31
47	49	767	S	FT	957e6f151a57
48	50	791	S	FT	2ea22b558bf8
49	51	817	M	FT,HI,KR	bd215f67ab10
50	52	850	S	FT	4b0d823a1fa8
51	53	884	S	FT	723d7eff8d45
52	54	918	S	FT	915abb8ed17c
53	55	954	S	FT	ea7675d9b8a3
54	56	990	S	FT	de8a415c9b29
55	57	1045	S	FT	9a1d337c75ba
56	58	1073	S	FT	d988db552530

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
57	59	1102	S	FT	fd3b4280376e
58	60	1140	S	FT	16af6d0b260e
59	12	37	S	FT,HY	8c718a334b7b
60	39	468	S	FT	5c5b0b26ee22
61	51	816	S	FT	6fa6b7d91d28
62	5	3	S	EX	6265d6a7c557
63	6	4	S	EX	7e060e9a0ac1
64	19	107	S	EX	5d2a7588c1e2
65	21	126	S	EX	f18756b72791
66	5	5	S	EX,HI	dd0748b7cdfb
67	7	10	S	EX,HI	4b5bb6e99d62
68	21	132	S	HY	3d8bae2fa729
69	22	142	S	HY	e683945e66cd
70	41	532	S	HY	6d48a3411f41
71	42	552	S	HY	7b29a6bd1191
72	81	2132	S	HY	7bc6b303744f
73	82	2172	S	HY	d75c9101c8a5
74	161	8532	S	HY	230691d59eee
75	162	8612	S	HY	1512e6966074
76	9	19	M	HI	e16410b887ee
77	15	61	M	HI	7c38d54f9e52
78	21	127	M	HI	b8dcff224cdb
79	27	217	M	HI	6670a2b14284
80	33	331	M	HI	25fd8f0c3311
81	48	715	M	HI	ba6f5baae76d
82	99	3169	M	HI,KR	4fef38d87d0c
83	195	12481	M	HI,KR	cd3497dbf846
84	195	12480	S	HI	1b9230fb738a
85	44	602	S	HI	0b0250bf6213
86	99	3168	S	HI	a0ea30825a73
87	39	470	S	SC	30caa6f7f894
88	47	691	S	SC	fac44af18c55
89	48	721	S	SC	2fb5ef0df4cf
90	51	818	S	SC	c18e1baf25f5
91	53	885	S	SC	6bc5b14d0b68
92	54	919	S	SC	a254c1b4f05b
93	55	955	S	SC	202625a39085
94	59	1103	S	SC	ba482375103b
95	60	1141	S	SC	9bc5096cc3fd
96	65	1365	S	SC	b1dcbeb4ff39
97	66	1397	S	SC	327ecbb9b7ba
98	99	3170	S	SC	ea507e2de137
99	129	5461	S	SC	6dae0012aa49
100	130	5525	S	SC	e0559b54aee5
101	195	12482	S	SC	5413dd7b60ee
102	14	54	M	MA	d47aea63afc1
103	14	54	M	MA	18bbcff4f433
104	14	54	M	MA	78f750c071a5
105	14	54	M	MA	912da8ec9bc7
106	14	54	M	MA	9134999072a9
107	14	54	M	MA	9c871e103a46
108	14	54	M	MA	b3dcae0bc1d8
109	14	54	M	MA	9219cdaa6f90
110	14	54	M	MA	d2babcb1b9431

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
111	14	54	M	MA	2b34ca22dc89
112	14	54	M	MA	fa427b3d27ef
113	14	54	M	MA	1ec1fd5cb2ee
114	14	54	M	MA	3e5d182f8408
115	14	54	M	MA	17f8c0524eb1
116	14	54	M	MA	c472f97237f1
117	14	52	S	MA	2eb83dc242a1
118	14	53	S	MA	5e6068f5003e
119	14	52	S	MA	b3f4749f6dee
120	14	52	S	MA	c6615430ae3d
121	8	14	S	PU	ebff58687458
122	14	51	S	PU	d67e7a6b3b08
123	20	114	S	PU	bc312fc85ea5
124	26	203	S	PU	5452b4ffa1a9
125	32	313	S	PU	018f9fc6b73f
126	38	448	S	PU	7d59f504aba8
127	50	791	S	PU	6bde86a3dcdb
128	8	15	M	PU	a138081033c3
129	14	54	M	PU	105765145381
130	20	117	M	PU	382171ac62cc
131	26	204	M	PU	f71379c4db03
132	32	315	M	PU	5914f2bb88d1
133	36	402	S	PU	ce43a1077597
134	38	450	M	PU	8ba2bdb14e85
135	42	553	S	PU	9849e255d5da
136	50	792	M	PU	3d40cb582616
137	19	107	S	OT	6e8ca6f56a85
138	6	6	M	OT	af8f599eb8a1

C Historical inventory of the author's outputs

This table preserves the earlier research inventory, including bounds later superseded by G , by a stronger retained certificate, or by known literature constructions. Its original labels describe how a number entered that inventory. They do not retroactively establish a missing general geometric argument. The active result aliases independently verify the retained numerical inequalities. Published inputs and older matching results keep their original attribution.

This is a historical numerical inventory, not a list of original discoveries. B means an elementary base case; C means an explicit coordinate construction in that inventory; E means the then-evaluated defect-extension consequence (called G in the historical file, renamed here to avoid confusion with the present function G). The numerical inequalities are now retained independently by the compiled `Results.earlier_NNN` aliases. This does not formalize all premises of the former general defect argument or establish priority. The right columns display current, separately proved bounds.

Table 17: Earlier work inventory compared with the present unconditional formula and saved certificate envelope.

n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
3	1	B	1	1
4	2	E	2	2

Continued from previous page

n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
5	5	B/C	5	5
6	7	C	7	7
7	10	C	11	11
8	14	E	14	15
9	19	C	20	21
10	25	E	24	25
11	28	E	31	32
12	36	E	37	38
13	39	E	45	47
14	53	E	52	54
15	61	C	62	65
16	72	E	70	72
17	76	E	81	85
18	93	E	91	93
19	98	E	103	107
20	116	C	114	117
21	127	C	128	133
22	143	E	140	143
23	149	E	155	161
24	172	E	169	172
25	178	E	185	191
26	203	E	200	204
27	217	C	218	225
28	238	E	234	238
29	245	E	253	261
30	275	E	271	275
31	283	E	291	299
32	314	E	310	315
33	331	C	332	341
34	357	E	352	357
35	366	E	375	385
36	402	E	397	402
37	411	E	421	431
38	449	E	444	450
39	469	C	470	470
40	470	E	494	494
41	496	E	521	533
42	553	E	547	553
43	564	E	575	587
44	608	C	602	608
45	618	E	632	645
46	667	E	660	667
47	679	E	691	691
48	715	C	721	721
49	722	E	753	767
50	791	E	784	792
51	817	C	818	818
52	818	E	850	850
53	852	E	885	885
54	886	E	919	919
55	920	E	955	955
56	956	E	990	990
57	992	E	1028	1045
58	1073	E	1064	1073

Continued from previous page

n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
59	1088	E	1103	1103
60	1104	E	1141	1141
99	3169	C	3170	3170
195	12481	C	12482	12482

D Evidence navigation and excluded drafts

The following stable repository paths locate the complete evidence. This appendix is a navigation aid, not an additional source of axioms.

Path or collection	Contents and role
<code>archive/2026-03-original/</code>	Original March sources, preserved as historical evidence; incomplete and excluded from the active build.
<code>Kobon/</code>	Active Lean geometry, counts, transformations, certificates and formal proof audit.
<code>research/kobon-extension/</code>	Both-parity extension manuscript, boundary-defect analysis and examples.
<code>research/successor-formalization/</code>	Recurrence status, 49-seed numerical target, and exterior-resource obstruction.
<code>research/kobon-hybrid/</code>	Compatible eleven-line seed, interval argument and sparse family manuscript.
<code>research/kobon-research/</code>	Additional-line family draft and finite witnesses.
<code>research/kobon-own-results/</code>	Historical inventory, candidate comparisons and priority audit.
<code>research/all-n-formalization/</code>	Earlier and current all-order formulas, finite enhancement table and interpretation.
<code>research/six-hour-2026-09-21/</code>	Three-hour research release, source audits, local upper arguments, family components and cumulative review. The directory keeps its original allocation name.
<code>experiments/2026-09-20/ and experiments/2026-09-21/ verification/</code>	Frozen search scripts, parameters, exact outputs, failed approaches and interrupted-run status. Build logs, theorem audit, coordinate recounts, source hashes and the interval summary.
<code>evidence/sources.json</code>	Bibliographic roles, versions, immutable commits and provenance.
<code>paper/</code>	Preserved September comprehensive and journal manuscripts, with their corrected rendering and frozen proof manifests.
<code>research/three-hour-2026-10-02/</code>	Closed odd/even recursion, exact obstruction corpus, actual upper extraction, new searches, seed drafts and continuation plan.
<code>manuscripts/2026-10-03/</code>	Preserved 3 October comprehensive and journal editions with their frozen proof and figure manifests.

Path or collection	Contents and role
<code>research/openmath-seven-hour-2026-10-05/</code>	Compatible 33/37/49/61 proofs, quantitative successor, structural upper bounds, surgery, corpus provenance and bounded search evidence.
<code>experiments/2026-10-05/</code>	Reproducible cap-cone, chart, corridor, germ and geometric profile discovery scripts.
<code>manuscripts/2026-10-07/</code>	This comprehensive edition, journal companion, updated scientific figures, shared proof map and reproducible build instructions.

The old stopped real-parameter 33- and 49-line prototypes remain historical excluded drafts. Independent active sources now complete both seed interfaces and infinite instantiations. Other abandoned dense 61-line and sparse prototypes retain their unpromoted labels; they are not theorem roots. The large 321-, 322-, and 641-line coordinate files remain outside the inherited 138 finite identities, and the interrupted direct recount at 642 is still incomplete. Existence at these four orders follows independently from the verified family. No unfinished coordinate artifact is an assumption of the all-order portfolio.

E Release deltas and reproducible research boundaries

Table 19: Cumulative changes from the preserved September release through the research of 5–6 October.

Topic	Previous status	Current status
All-order affine G Finite coordinates	Complete Lean proof. 138 identities, 104 simple.	Unchanged and retained. Same catalog and attributions; no new numerical record added.
BBL output invariant	One-step geometry; recursive integration open.	Closed grid, seed and visible-boundary records; all finite depths proved.
Eleven-derived families	Finite checkpoints and published-theorem consequences.	Unconditional actual odd/even Lean existence.
All-order envelope	$L = \max(G, C)$.	The first recursive H and current portfolio J retain old bounds pointwise and add verified 33/37/49/61 branches.
Simple upper comparisons	Polynomial benchmarks.	Actual general simple upper proofs and one-triangle family windows.
Global upper extraction	Local fans and conditional arithmetic.	Actual rays, triangle occurrences, fan counts, matching and component assembly complete; structural corrections remain explicit.

Topic	Previous status	Current status
Perfect 21-grid transfer	Bounded failed searches.	One explicit Lean obstruction; complete exact corpus for supplied affine classes, with external classification input.
33/49 compatible seeds	Uniform checks uncompleted in the earlier October release.	Both active real-parameter bases and odd/even family instantiations complete; their numerical orbits are inherited.
37/61 compatible seeds	Not in the preceding formal portfolio.	Known 37 numerical orbit formally integrated; compatible 61:1190/29 gives constant simple gaps 9/10.
Quantitative successor	Deficit formula and manuscript proof; universal ray extraction not yet linked.	Actual both-parity successor and indefinite iteration complete, distinct from full-gain recurrence.
Upper components	Proposed extraction and finite budget interfaces.	Mixed degree-free, restricted graph classes, all-triple N12 component maxima and late global M22 gain proved.
Translation surgery	Finite perturbation diagnostics.	General simultaneous germs, exact births/losses, and four-/six-line interval counterexamples proved.
Active Lean audit	220 targets, 325 sources, 4,227 theorems in the September record.	October research record covers 638 active modules and 9,058 declarations; native trust is separately reported.
March universal extension	Incomplete archived argument.	Still unproved; finite numerical consequences independently supported.

The preserved [2 October proof index](#) records its earlier uncompleted branches. The current [research release record](#), construction provenance and corpus proof map identify promoted active proofs, independent external checks and excluded historical prototypes. They retain the distinction between complete symbolic geometry, finite native decisions, attributed classifications and bounded search.